FSM PLATFORM

SYSTEM MANUAL

# FSM User Manual

Office · Technician · Administrator

A practical guide for office staff, technicians and administrators.

| VERSION | DATE | AUDIENCE |
| --- | --- | --- |
| 0.8.11 | 2026-11-08 | Office · Technician · Administrator |

# Contents


## Introduction

What FSM Platform is
What's new in 0.8.11
How the system is put together
End-to-end flows
Navigation by role
How to use this manual

## Office

Dashboard
Work orders
Customers and contacts
Equipment and assets
Calendar and dispatch
Agreements, SLA and price lists
Bookings
Inventory, spare parts and purchasing
Mileage, expenses and payroll basis
Profitability
Reports
Documents
Map, search, notifications and messages
Customer portal from the office
Daily and periodic routines
Customer transfer (administrator only)
Interface with the technician and administrator parts
Common mistakes and troubleshooting

## Technician

First time: sign-in, 2FA and language
My work orders / Today
Work order details
The report form
Submit and sign
Mileage, expenses and spare parts
Offline, drafts and the outbox
Safety and quality in the field
Common field problems
How technician work connects to the rest

## Administrator

1. The administrator role and delegation
2. Departments and department scoping
Modules
3. Users, invitations and sessions
4. Roles and permissions
5. Customers and customer portal users
6. Checklists and templates
7. Work orders and reports
8. Service agreements, SLA and price lists
9. Booking, SMS, tracking and reviews
10. Branding, map, language and notifications
11. API keys and the REST API
Automatic email and the outbox
Push notifications
12. Activity log, privacy and data
13. Operations and health
Storage, Redis and high availability
Compliance, ISO 27001 and SOC 2
14. Administrator routines
15. Common problems
16. Relationship to the office and technician chapters

## SSO and SCIM

1. Before you begin
2. Register the application in the IdP

3. Set environment variables
4. Test sign-in
5. Auto-provisioning
6. SCIM token
7. Set up SCIM in the IdP
8. Break-glass and troubleshooting

## Quotes, invoicing and accounting

Quotes
Invoicing
Payment links (Stripe and Vipps)
Accounting sync (Fiken, Tripletex, PowerOffice Go and CSV)
Customer portal
Permissions, audit and routines
Common problems
Customer and contact

## Reference and troubleshooting

Roles and permissions in practice
Status glossary
How features talk together
Common cross-role problems
Norwegian/English glossary
Support information to collect

# Introduction and system overview

*Applies to FSM Platform version 0.8.11, 8 November 2026.*

## What FSM Platform is

FSM Platform is a service and maintenance system for work carried out at the customer's site: service, maintenance, inspection and repair. It brings planning, work orders, visits, checklists, reports, equipment, service contracts, documents and the customer portal together in one service.

The platform is designed so that the office can plan without parallel spreadsheets, the technician can work from a phone or tablet even with a poor connection, and the customer can see contracts, visits, reports and equipment without calling the office. The core documents and plans work and records time and travel per workday. Quotes, invoicing, payments, accounting sync, inventory, expenses and profitability are optional modules that an administrator must enable. Without those modules, price lists and contract amounts are reference data and invoicing happens outside the platform.

### The four roles

This manual uses four standard roles. An administrator can create custom roles and give individual users tailored permissions, but day-to-day work generally follows these four.

| Role | Who | Main task | They see |
|---|---|---|---|
| **Administrator** | System or operations manager | Sets up departments, roles, templates, branding, API access and security | Everything, across all departments |
| **Office** | Service manager, planner or case handler | Plans and dispatches work, follows up customers, contracts, bookings and reports | Own department for work orders, contracts and reports; customers, equipment and documents are shared |
| **Technician** | Field service technician | Carries out visits, completes checklists, takes photos and signs | Own department for work orders and reports; customers, equipment and documents are available |
| **Customer portal user** | Customer or contact | Books service, follows visits, reads reports and views their own equipment | Their own customer, with the portal access granted |

> **Note:** "Office" is a role, not a physical place. A service manager can work from a phone and still have the office role.

## What's new in 0.8.11

- **Quotes, invoices and payment links.** Quotes have versions, printing and a one-time approval link. Invoices have partial payments, credit notes, overdue handling and CSV. The customer can pay with

Stripe Checkout or Vipps MobilePay where the provider is configured. See quotes and invoicing.

- **Accounting sync.** Sent invoices can be synced manually to Fiken, Tripletex or PowerOffice Go, with CSV as an explicit export fallback. Sync is idempotent and audit-logged. See accounting.
- **Inventory, suppliers and purchasing.** Item list, locations, stock, reservations, consumption, supplier register, purchase orders and goods receipt. See inventory.
- **Mileage, expenses and payroll basis.** The technician records mileage and expenses with receipts; the office approves and exports a locked payroll CSV per period. See expenses.
- **Profitability.** Estimated and actual revenue, time, first-time fix and utilisation by department, customer and technician, with CSV export. See profitability.
- **Automatic email and push.** Transactional email uses a durable outbox with retry, per-event switches and test send. Push uses FCM and APNs with deep links and device registration. See administrator and technician.
- **Routing with traffic and overtime.** Routing uses OpenStreetMap/OSRM, with optional Google Routes API for traffic-aware ETA. Visits that cannot meet the time window are planned as overtime instead of staying unassigned. See route plan.
- **Manual order.** Both the route planner and **Days and time** on the work order have number fields 1/2/3/4. Manual order overrides automatic optimisation and is audited per visit.
- **SAML 2.0, SCIM groups and field-level RBAC.** SAML can be used alongside or instead of OIDC. SCIM Groups grant role and department through explicit mapping. Sensitive fields are masked on the server for technicians and the customer portal. See SSO and SCIM.
- **API for quotes, external orders and inventory.** `/api/v1/quotes`, `/api/v1/orders` and `/api/v1/inventory/**` have dedicated scopes. External order systems use `source` + `externalId` and `Idempotency-Key`, and receive the same order on replay. See administrator.
- **S3, Redis and HA/PITR.** Uploads can live in S3/MinIO, the job queue and rate limits can use Redis, and the HA profile has multiple app replicas, WAL archiving and tested PITR restore. See administrator.
- **Compliance.** The ISO/IEC 27001 and SOC 2 package is a **readiness basis**, not a certification or audit report. See administrator.

## How the system is put together

You meet the system through a few surfaces. They share data and permission checks but are adapted to different tasks.

| Surface | What you do there |
|---|---|
| **Browser and PWA** | Opens the system on a PC, tablet or phone. The field surface can be installed as an app and used with limited connectivity. |
| **Office surface** | Plans in lists, boards, calendars and dispatch. Handles customers, contracts, bookings, reports and settings. |
| **Field surface** | Shows today's jobs, the next action, map links, checklists, photos and signatures, with large touch targets and a dark theme. |
| **Customer portal** | Gives the customer an overview of their own bookings, reports, documents, equipment, contracts, visits and messages. |

| Surface | What you do there |
|---|---|
| **REST API** | Lets other systems read and write data with a dedicated API key and the same permission checks. |

Behind the surfaces is a shared application that enforces workflow and access. It is built with Next.js and React, using Prisma against PostgreSQL as the source of truth. The field app is a PWA with a service worker and IndexedDB for offline drafts. PDFs and uploaded files live in a separate file volume. Report PDFs render in-process with a concurrency limit; OCR and document jobs run through a job queue.

| Layer | Content |
|---|---|
| **User surfaces** | Customer portal, office surface, field surface and REST API clients in a browser or installed PWA. |
| **Application** | Access control, workflow, notifications, search and PDF rendering. |
| **Data and jobs** | PostgreSQL with data, revisions and audit log; file volume for PDFs and images; job queue for OCR, document jobs and notifications; email/SMS where configured. |

## Modules

The platform has a core and a catalogue of modules that can be switched on and off under **Settings → Modules**. The core (customers, work orders, workdays, time, revisions and settings) works without optional modules. A module may have dependencies, and the module guard returns `404` for pages and APIs when the module is off.

| Module | Key | Provides |
|---|---|---|
| **Quotes** | `quotes` | Quotes, versions, approval link and conversion to work order |
| **Invoicing** | `invoicing` | Invoice, payment, credit note, overdue and CSV |
| **Inventory** | `inventory` | Items, locations, stock, reservations and purchasing |
| **Expenses** | `fieldExpenses` | Mileage, expenses and payroll CSV |
| **Profitability** | `bi` | Estimated and actual revenue, time and CSV |
| **Customer portal** | `portal` | The customer's own surface |
| **API** | `api` | The REST API under `/api/v1` |

Existing installations get every delivered module enabled. Turning a module off removes navigation, settings, search hits and background jobs for that area, but keeps the data. Turn it on again to restore access.

The field surface also has a local buffer in the browser:

1. **Offline:** the form is stored in IndexedDB as a draft and outbox item.
2. **Online:** the draft is sent to the server, which confirms receipt.
3. **Always:** the server decides whether a report has actually been received; "sent" appears only after confirmation.

A draft can be saved locally, but a submitted report is not considered sent until the server confirms it. This is why the field surface shows "draft saved locally" and "waiting to send" as two different states.

## Core entities

| Entity | What it is | Key relationships |
|---|---|---|
| **Department** | Organisational unit that determines who sees what | Staff, work orders, customers, contracts, business hours |
| **User and role** | Employee or customer portal user with permissions | Department, customer, contact, API keys |
| **Customer** | The company receiving service | Contacts, sites, equipment, contracts, work orders |
| **Contact** | The person the office and technician deal with | Customer, work orders, contracts, portal user |
| **Site** | Physical address or facility at the customer | Customer, work orders, equipment, bookings |
| **Equipment/unit** | Machine or system being serviced | Customer, site, meter readings, maintenance plans, history |
| **Work order** | The service job itself, with status and responsibility | Customer, contact, site, equipment, templates, visits, reports |
| **Visit** | Time block within a work order, with technicians and status | Work order, technicians, tracking, meter readings |
| **Checklist template** | Reusable template for service or HMS | Work orders, contracts, reports, maintenance plans |
| **Report, unit and answers** | Completed checklist with units, photos and signatures | Work order, template, equipment, PDF, revisions, acknowledgement |
| **Service contract and SLA** | Agreement for recurring service and deadlines | Customer, department, template, SLA policy, work orders |
| **Booking** | Customer's request or confirmed appointment | Customer, site, equipment, work order, visit, events |
| **Quote** | Price proposal with lines, versions and approval | Customer, contact, work order, contract, equipment |
| **Invoice and payment** | Invoiced amount, payments and credit note | Customer, work order, report, quote |
| **Item and stock** | Spare part with stock per location | Location, reservation, work order |
| **Purchase order** | Supplier order with goods receipt | Supplier, receiving location, item lines |
| **Mileage and expense** | Mileage allowance or expense with receipt | User, work order, approver, payroll CSV |

| Entity | What it is | Key relationships |
|---|---|---|
| **Document** | File in the library or an attachment | Folder, tags, versions, work order, customer, contract |
| **Notification** | Message to a user about an event | User, type, source, link, email status |
| **Audit log** | Traceable history of actions | Entity, actor, time, metadata |
| **API key** | Secret for integrations | User, permissions, expiry, revocation |

## End-to-end flows

The role chapters describe each step in detail. Here you can see how the roles and data fit together.

### 1. Service contract → recurring work order → report → PDF and history

1. **Office or administrator** creates a service contract on the customer card with an interval (3, 6, 12 or 24 months), start and end dates, department, template, optional SLA and renewal mode.
2. **The system** creates a new work order when `Next service` is due. The work order inherits the customer, contact, template and department; the previous report is attached as a PDF.
3. **The office** sees the work order in "Work orders" and "Calendar", creates or checks the visit and assigns a technician.
4. **The technician** completes the checklist per unit, takes photos, comments on deviations and signs if desired. The customer can sign at the same time.
5. **The system** locks the report, saves a revision, generates a PDF and notifies the office and customer portal. If there are deviations, it creates one undated remediation work order.
6. **The customer portal user** sees the submitted report, downloads the PDF and acknowledges receipt. The acknowledgement is visible to the office.
7. **The office** follows up deviations. The equipment on the customer card shows service history from the report's units.

> **Note:** The report is frozen at submission. If the template or equipment changes later, the submitted report keeps the information that applied when it was submitted.

### 2. Customer portal booking → work order and notification → dispatch → technician

1. **The customer portal user** selects a site, service and requested date under "Bookings". The desired time is written in the description; the customer cannot select a time block.
2. **The system** creates a new work order with status `OPEN` and an unconfirmed booking request. No visit or confirmed time is created, and the office is notified.
3. **The office or technician** plans the work order and sets the status to **Planned**, or **Waiting** if the time does not fit. Only then are the date and time confirmed.
4. **The technician** receives an assignment notification and sees the visit under "Today". Cancellation triggers a new notification.

> **Tip:** A customer booking is a request, not a confirmed appointment. The office must plan it before the customer gets a final time.

### 3. Office creates a work order → calendar, board and assignment → field

1. **The office** selects "New work order" and enters a title, customer, contact, site, date, template and optionally tags and technician.
2. **The system** creates the work order and a visit in the same transaction when a date is set. New work orders start as `PENDING`; undated work orders have no visit.
3. **The office** distributes the work through "Assign technician", the calendar, the dashboard board or dispatch with suggestions and automatic assignment.
4. **The technician** sees the work order under "Today" or "Work orders" and can claim available work with "Take work order" if the role allows it.
5. **The technician** starts driving, marks "On site", completes the form and finishes. Status, comments and the report flow back to the office.
6. **The office** follows up overdue work orders, deviations, missing signatures and reports that have not been acknowledged.

> **Note:** On the board, a drop assigns a technician; it does not move the date. Date and time are changed in the calendar or in the visit details.

### 4. Tracking and reviews → customer portal and office follow-up

1. **The administrator** enables live tracking for the department. It is off by default and applies only to visits in that department.
2. **The technician** sets the visit to "On the way" and can share position. The customer opens the tracking link from "Visits" or receives it separately.
3. **The customer portal user** sees the first name, status and estimated arrival. The position is never shown as a map or coordinates, and the link expires when the visit is finished or after two hours.
4. **The customer** can leave an internal review after the visit is completed. Reviews are internal and never published; they are not shown as a separate dashboard in 0.8.11.

### 5. Permissions, notifications and audit across all flows

1. **Every** action is authorised on the server. Staff see their own work orders, contracts and reports, while customers, equipment and documents are shared. Administrators see everything, and customer portal users see only their own customer and granted portal areas.
2. **The system** notifies on assignment, date change, cancellation, completed report, booking, contract due, overdue work orders, SLA events and contact requests. Notifications go to the bell, "Notifications" and optionally email.
3. **The audit log** records who created, changed, moved, cancelled, deleted, unlocked or applied a legal hold, with time and actor. The log cannot be edited.
4. **The administrator** can change roles and permissions. Old sessions are rejected, and API keys inherit the user's current permissions.

5. **Legal hold** prevents deletion and changes. A submitted report can only be unlocked by an administrator, with a reason, and the unlock is logged.

## Navigation by role

The menu is adapted to role and permissions, so you will not necessarily see every item.

| Role | Home | Main menu | Hamburger menu and settings |
|---|---|---|---|
| **Administrator** | "Dashboard" | Dashboard, Work orders, Calendar, Customers, Messages, Reports | Bookings, Contracts, Units, Search, HMS, Documents, Map, New work order, Notifications, Quotes, Invoices, Inventory, Expenses, Profitability, Settings |
| **Office** | "Dashboard" | Dashboard, Work orders, Calendar, Customers, Messages, Reports | Bookings, Contracts, Units, Search, HMS, Documents, Map, New work order, Notifications, Quotes, Invoices, Inventory, Expenses, Profitability, Settings |
| **Technician** | "Today" | Today, Work orders, Calendar, Customers, Messages | Reports, Bookings, Contracts, Units, Search, HMS, Documents, Map, New work order, Notifications, Customers & equipment, Settings |
| **Customer portal user** | "Customer portal" | Overview, Bookings, Reports, Documents, Equipment, Contracts, Visits, Messages, Contact | Account & security |

On mobile, the bottom bar shows the most important areas: "Dashboard", "Work orders" and "Calendar" for office and administrator, and "Today", "Work orders" and "Notifications" for technicians. "More" opens the rest.

### Where do I find …?

| What you are looking for | Go to |
|---|---|
| Today's work and overdue work orders | "Dashboard" for the office, "Today" for the technician |
| Planning or moving a visit | "Calendar" or the work order's visit section |
| Assigning technicians quickly | "Dashboard" → board, "Calendar" → Dispatch, or "Assign technician" |
| Customers, contacts and portal users | "Customers" → customer card |
| Equipment and service history | "Units" or the customer card → "Equipment" |
| Service contracts and next service | "Contracts" or the customer card → "Contracts" |
| Bookings from the customer portal | "Bookings" |
| Reports and PDFs | "Reports", or the work order → "Checklists" |

| What you are looking for | Go to |
|---|---|
| Checklists and HMS templates | "Settings" → "Checklists / Templates", or "HMS" |
| Documents, folders and tags | "Documents" |
| Notifications and notification settings | The bell at the top, or "Notifications" → "Settings" |
| Audit log for one work order | The work order → "Activity log" |
| Global audit log | "Settings" → "Activity log" (administrator) |
| Employees, roles and permissions | "Settings" → "Employees & Roles" or "Roles & permissions" |
| API keys | "Settings" → "API keys" (administrator) |
| Quotes and approval links | "Quotes" |
| Invoices, payments and credit notes | "Invoices" |
| Spare parts, stock and purchasing | "Inventory" |
| Mileage and expenses for approval | "Expenses" → "Office" |
| Revenue, time and profitability | "Profitability" |
| Automatic email and push | "Settings" → "Automatic email" / "Push notifications" |
| Anything across customers and work orders | "Search" |

## How to use this manual

The manual has an introduction, three role chapters, a chapter on SSO and SCIM, a chapter on quotes, invoicing and accounting and a reference section. You do not need to read everything in order. Start with your role, and use the reference section when you need a term, a status or a troubleshooting check.

- **Role chapters:** Part 1 for office, Part 2 for technician and Part 3 for administrator. The customer portal is described alongside the customer card and the roles that administer it.
- **Specialist chapters:** SSO and SCIM and quotes, invoicing and accounting cut across the roles.
- **Reference:** the reference part contains the status glossary, permission matrix, terms and troubleshooting.

### Conventions

- Buttons, tabs, menu items and field names appear in "quotes", as they appear in the interface.
- Paths, file names, codes and technical values appear in `monospace`.
- Menu paths are shown as **Settings → API keys**.
- **Note** explains something that is easy to misunderstand. **Tip** is a shortcut or good practice. **Warning** means the action can lead to data loss or errors.
- Dates and times are shown in Europe/Oslo with a 24-hour clock. "Today" means the day in that time zone.

- When a time or date is missing, the system shows "Time not set" or "No date". That does not mean the value is midnight or that the job has been forgotten.
- Service report PDFs use Norwegian labels in 0.8.11, even when the interface is in English. Contract PDFs can be generated in the selected language.

> **Warning:** A submitted and locked report should not be changed without a good reason. Unlocking requires an administrator and a reason, and every change is recorded in the audit log.

On first sign-in you arrive at your role's home surface. Check the department, notification settings and any 2FA. Technicians should install the field surface as an app, so drafts and recent data are available even with an unstable connection.

PART 1

# Office: planning, dispatch and follow-up

This chapter is for office staff: you receive enquiries, create and plan work orders, assign technicians, follow up reports and keep customer, equipment and agreement data up to date. It applies to FSM Platform 0.8.11 (8 November 2026).

Work orders and service agreements belong to a department: you normally see only your own department, while an administrator sees everything. Customers, equipment and documents are shared, but every action requires the right permission. Work time, travel and materials are recorded on the work order. Quotes, invoicing, payments, inventory, expenses and profitability are optional modules that must be enabled; without them, price lists are reference data and invoicing happens outside the platform.

## Dashboard

The dashboard is at `/` and is titled **Planning**.

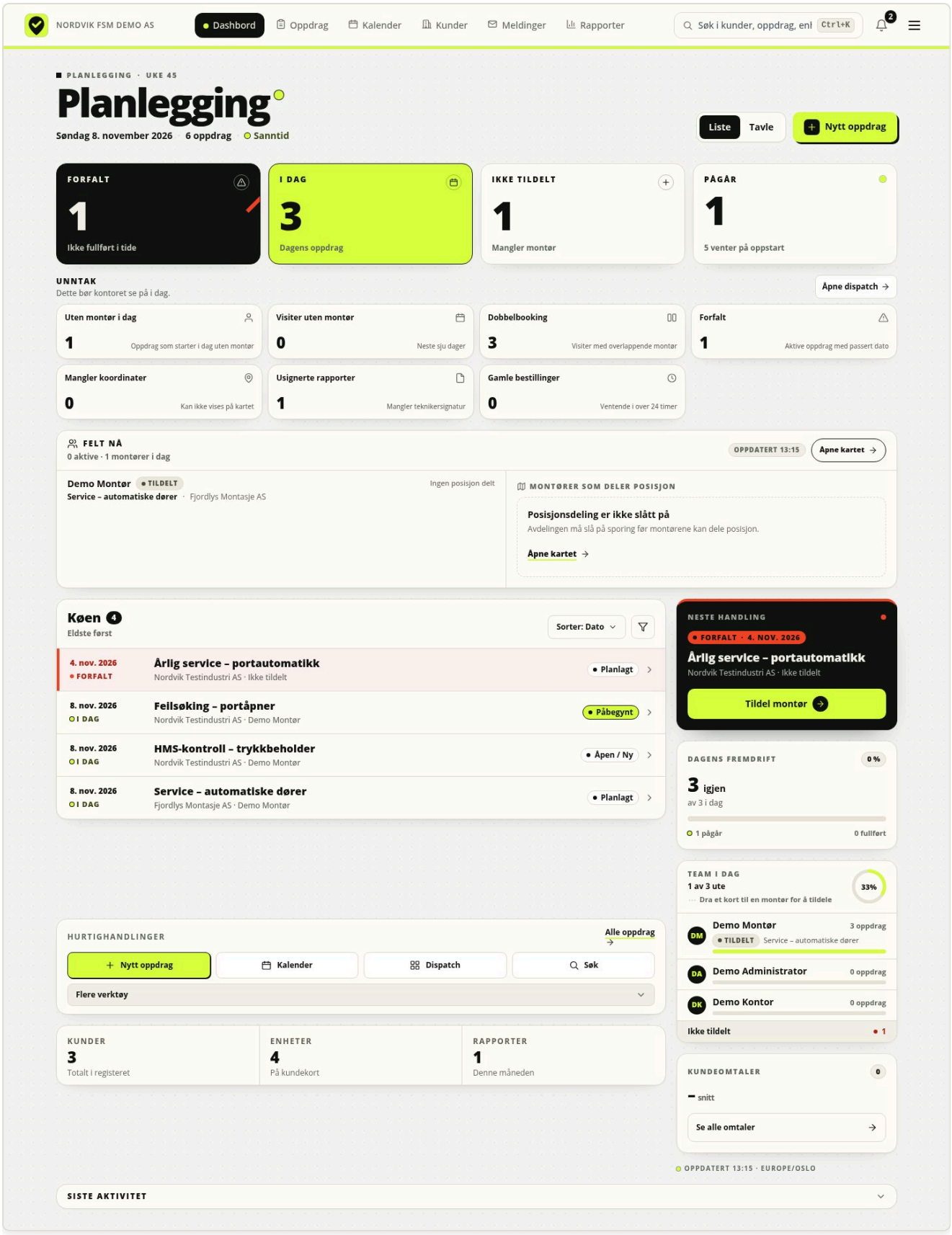



*Figure: The dashboard is the office's starting point for the day.*

Notice that **Exceptions needing action** sits above the queue itself. That is where you first see whether anything is overdue, double booked or missing a technician. The queue below shows overdue, today and unassigned in priority order.

## List and board

**List** ( `/?vis=liste` ) is the default and shows one prioritised queue with **Overdue**, **Today** and **Unassigned**.

**Board** ( `/?vis=tavle` ) has the columns **Overdue**, **Today**, **In progress** and **Unassigned**. These are priority buckets, not statuses; the status is a pill on the card.

| Card | Meaning | Click |
|---|---|---|
| **Overdue** | Scheduled before today, not completed | `/oppdrag?vis=forfalte` |
| **Today** | Scheduled today, including multi-day work orders | `/oppdrag?vis=idag` |
| **Unassigned** | Active work order without a technician | `/oppdrag?vis=ikke-tildelt` |
| **In progress** | Status **In progress** | `/oppdrag?vis=pagaaende` |

The **Queue** can be sorted by **Date**, **Customer**, **Status** or **Title**; the default is **Oldest first**. The right-hand rail shows **Next action** with **Assign technician** or **Open work order**, **Today's progress** and **Team today**.

## Field status and live map

Below the exceptions, **Field now** shows today's status for field staff. Each technician has one status:

| Status | Meaning |
|---|---|
| **On site** | A visit has started and the technician is on site |
| **En route** | The technician has started driving to a visit |
| **Assigned** | The technician has a visit today that has not started |
| **Done for the day** | All of today's visits are completed |
| **Available** | No visits today |
| **Absent** | Absent all day with no active visit |

Status precedence is **On site** → **En route** → **Assigned** → **Done for the day** → **Available**. If the technician has several jobs today, the active visit is shown; otherwise the visit starting nearest to now. **Absent** is shown only when the absence covers the whole day and the technician is not en route or on site. The row shows the work order title and customer, plus **Last shared** when the technician has shared a position. The same status badges appear in **Team today**.

The **Technicians sharing location** map panel shows only technicians who have actively started location sharing on a visit that is **En route** or **On site**. Positions are short-lived points, not history: pings older than 24 hours are not used, and the map shows the last known point. If tracking is not enabled for the department, the panel says **Location sharing is turned off**. If nobody is sharing right now, it says **No technicians are sharing their location right now**. Both states link to **Map** ( `/kart` ), and every marker has a matching link in the list below the map. The list keeps the map usable with a keyboard and screen reader.

You only see technicians, work orders and positions in your own department; an administrator sees all departments. Names and coordinates are never shown to customer portal users or anyone without work

order access. Location sharing is voluntary and controlled by the technician on the job; the department must also enable tracking.

### Exceptions, quick actions and activity

At the top of the dashboard, **Exceptions needing action** shows the cards **Unassigned today**, **Visits without technician**, **Double booked**, **Overdue**, **Missing coordinates**, **Unsigned reports** and **Stale bookings**. Each card shows a count and opens the relevant list or dispatch. **Quick actions** gives **New work order**, **Calendar**, **Dispatch**, **Search**, **My work orders**, **All work orders**, **Customers**, **Documents**, **Checklists**, **Statistics** and **Map**. **Recent activity** shows the latest events with **View all activity**.

### Quick assignment

1. Go to **Work orders** (`/oppdrag`).
2. Find the row that says **No technician assigned**.
3. Choose a technician in **Choose technician** and press **Assign**.

The technician appears in the row. The assignment applies to the work order; if only one visit needs a different technician, do it in **Planning**.

### Drag and drop

Drag a card from the board to a technician in **Team today**, or from **Resource overview** to the technician's column. Drag and drop assigns a technician, **not** a date; use the calendar or **Planning** to move time.

### Global search

**Search** in the top bar searches customers, work orders, documents, units and reports. Results show their type, and the search tolerates typos. Document hits open the document in the library. Use search when you only have a name, an order number or a keyword.

## Work orders

The work order list is at `/oppdrag` with the views **Active**, **New**, **Not started**, **In progress**, **Today**, **Overdue**, **Unassigned**, **Completed** and **Cancelled**.

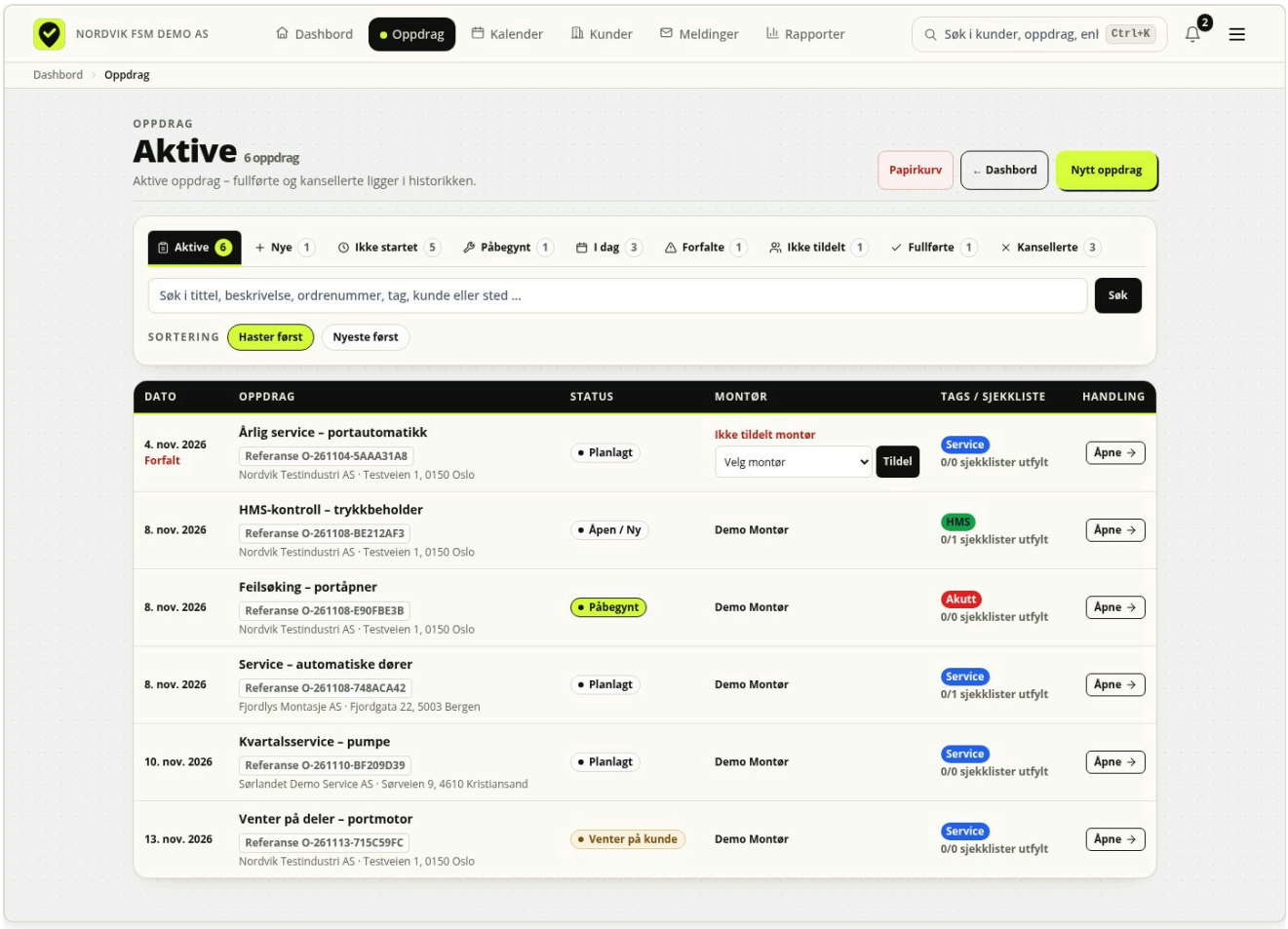



*Figure: The work order list is the queue across customers and dates.*

Use the tabs to narrow the list before searching. **Unassigned** and **Overdue** are the two most important entry points for daily follow-up; the rest is history and lookup.

## Create

1. Press **New work order** (`/oppdrag/ny`). The department is fixed to your own; only an administrator can choose another. Enter **Work order title** and optionally **Order number** (unique in the department).
2. Enter **Description / Instructions**.
3. Choose **Customer** and **Contact person**, or leave the customer empty for an **Internal work order**. **Address**, **Postal code** and **City** override the customer address.
4. Set **Date of execution** and optionally **End date (optional)**. Each date becomes a separate workday. An empty end date means a one-day job, and a multi-day work order continues on the next day.
5. Choose technicians under **Assign technicians** and templates under **Add checklists / forms**, and add **Tags** if needed.
6. Press **Save work order**. You return to **Calendar**.

**Edit work order** changes title, customer, contact, date, technicians and description.

## Priority and status

There is no separate priority field; priority comes from the date, the status and whether a technician is missing. The statuses are **Open / New**, **Waiting to be planned**, **Planned**, **In progress**, **Waiting for**

**customer**, **Completed** and **Cancelled**. Only an administrator can reopen a completed or cancelled work order.

## Checklists, technicians and visits

On the create page, templates are grouped under **Service / checklists** and **HMS documentation**. Each template produces its own report, which is always new and empty.

You can assign several technicians; **Remove** takes one off. The assignment applies to the whole work order. The office only plans the dates; each date becomes one workday in the field app.

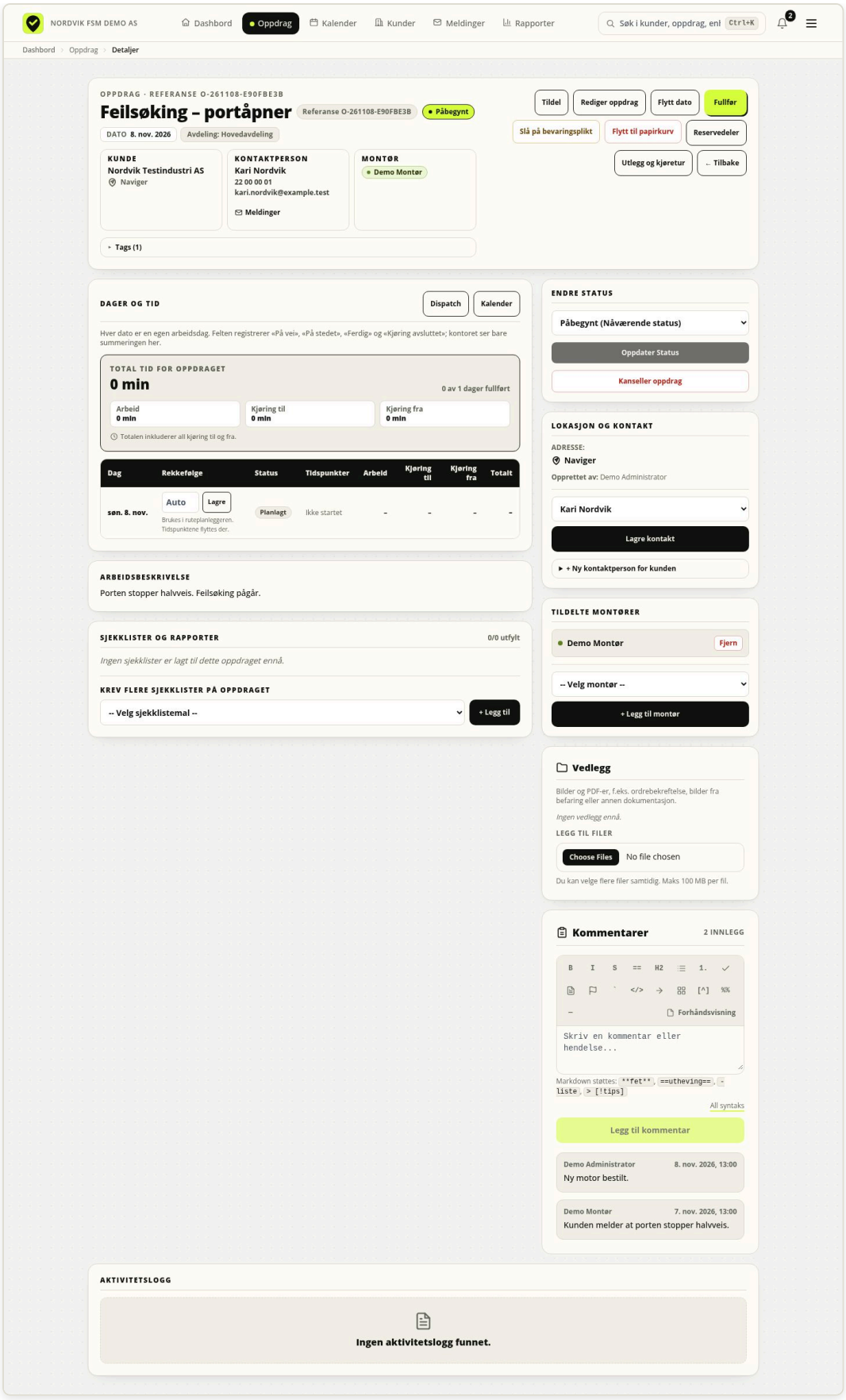



*Figure: The work order detail gathers status, contact, days and reports.*

Use **Days and time** to check actual time, and **Checklists and reports** to see report status. Assignment and status buttons are at the top, not inside the day list.

On the work order page, **Days and time** shows one row per workday with status, **En route**, **On site**, **Done** and **Travel finished**, plus work, travel to, travel from and total time. The top of the panel shows total time for the work order with the breakdown. If the work order has no workdays, the panel shows the date range and **Change dates**.

Each workday has an **order number** (1, 2, 3, 4 …). The number is stored on the visit without moving the times. The route planner uses it on the next view and moves the times when you approve the order there. An empty field or 0 means automatic order. As soon as at least one visit for the technician and day has a manual number, the planner sorts by number and skips nearest-neighbour/2-opt.

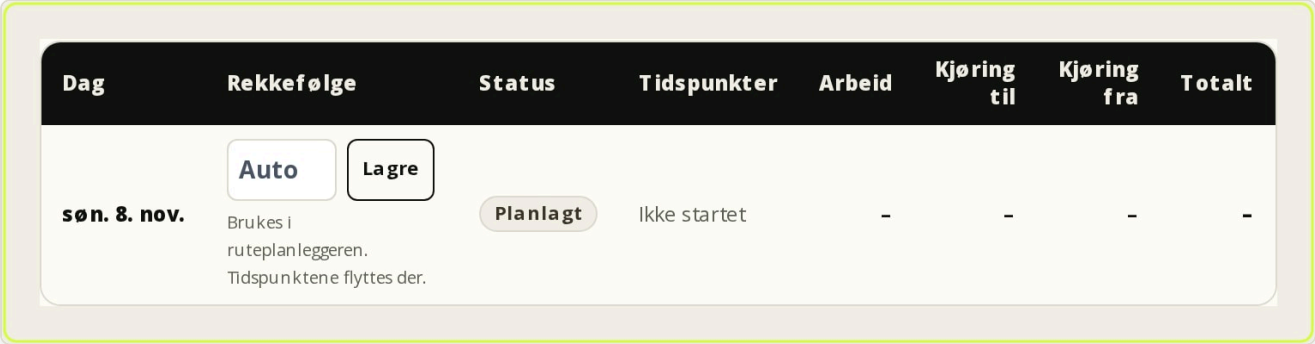



*Figure: The order number on each workday controls the route without changing the time.*

Look at the number field on the right of each day row. It is not a move of the day; it is an instruction to the route planner. Use it when the customer requires a specific order, for example main plant first and then the gate.

The office does not create, move or delete individual days here. Change the work order's date range and the days are reconciled automatically. A day that has started, is completed or has recorded time cannot be removed without deliberate cleanup. **Dispatch** and **Calendar** open the planning views for assigning technicians per day and seeing actual time.

## HMS library and QR labels

**HMS** (`/hms`) collects HMS forms by type: **Safe Job Analysis (SJA)**, **Hot work**, **Internal control**, **Deviations**, **Safety round**, **Written instruction** and **General**. Under **Link HMS form to work order**, choose an **HMS form** and an active **Work order**, then select **Link form to work order**. The form is added to the work order and filled in there. **Recent HMS documentation** shows completed HMS forms you have access to.

QR label: open the unit at `/enheter/<id>`, select **Print label**, and use **Print** on the label page (`/enheter/<id>/etikett`). Scanning the label opens the unit. **Rotate QR token** if the label is lost or shared; the old label stops working immediately.

**Existing QR codes:** If the customer already has labels from another system, register the value under **Existing QR codes** on the asset page. Paste or scan the value, optionally give it a name (for example “Old QR – gate 3”), and save. The label stays physically on the equipment and is never changed. Each value can be linked to only one asset. Deleting removes only the mapping in FSM, not the physical label, and is recorded in the audit log.

### Attachments, comments and completion

**Attachments** accepts images and PDFs, up to 15 MB per image and 50 MB per PDF, and is internal. A rolling 24-hour quota of 2 GiB per user and 10 GiB total storage per customer also applies; adjust it with `UPLOAD_QUOTA_USER_MB` and `UPLOAD_QUOTA_CUSTOMER_MB`. **Comments** are the dialogue between office and technician; write in the field and press **Add comment**.

**Complete** sets the status to **Completed** after the confirmation **Complete this work order? The work order and report are locked for editing. Cancel work order** requires a **Reason for cancellation**; the customer is notified, and only an administrator can reopen it.

### Clone, related data and trash

**Clone work order** copies the details, customer, equipment, technicians, templates, tags and visits. The clone gets the status **Waiting to be planned**, and the **order number is cleared**. Reports and attachments are not copied.

| Data | Where | Note |
|---|---|---|
| Customer/contact | Header and **Location and contact** | Locked when a report or booking exists |
| Equipment | Customer card → **Equipment** | The report remembers the units checked |
| Reports | **Checklists and reports** | Locked when submitted |
| Workdays | **Days and time** | Each date on the work order becomes one workday |
| Agreement/booking | The agreement's latest work order / booking queue | One booking per work order |

**Move to trash** is administrator-only by default; the button is shown only to roles with `workorders.delete` and only for work orders that are not closed. An administrator can restore it from **Trash** (`/oppdrag/slettede`). Work orders without reports can be removed after 365 days; work orders with reports are retained. **Legal hold** stops all automatic deletion.

## Customers and contacts

The customer directory is at `/kunder`. Search by name, org. number, city or email, or press **New customer**.

The customer card has the tabs **Overview**, **Work orders**, **Agreements**, **Equipment** and **More**. At the top you find **New work order**, **Messages**, **Edit** and, for an administrator, **Delete customer**. **Overview** shows company information, contacts, portal users, overdue and upcoming work orders.

### Contacts

1. Open the customer card → **More** → **Contacts**.
2. Press **Add contact**.
3. Enter **Name**, **Role (optional)**, **Email (optional)** and **Phone (optional)**, and save.

A portal user must always be linked to a contact. If you delete the contact, any portal users linked to it are deactivated.

### Department and responsible technician

**Edit customer** has **Department (optional)** and **Responsible technician (optional)**. The responsible technician is shown as the customer's direct contact.

### Portal users

Customer card → **More** → **Portal users**.

1. Press **Create and send invitation**.
2. Enter **Name**, **Email** and choose **Contact**.
3. Send the invitation; the user sets a password via the link.
4. Press **Activate** once the password is set.

If the email cannot be sent, the system shows **Invitation ready for {name}** with **Copy link**; send it manually. **Deactivate** closes the portal and signs the user out.

### Equipment, documents and agreements

**Equipment** shows units and service history. **More** → **Attachments** holds documents, agreements and images belonging to the customer. **Agreements** shows service agreements, coverage lines, next service, SLA and price lists.

**Sites** and **Equipment** on the customer card have their own search fields. The search runs on the server and is limited to the open customer, so you can search by name, address, postal code, city, model or serial number without other customers' rows appearing. **Reset** clears the search, and the counter shows how many matches the customer has.

## Equipment and assets

The asset register is at `/enheter`. Search by model, serial number, location or customer, filter by status and category, or turn on **Show hierarchy**.

### Create and edit

1. Press **New asset** (`/enheter/ny`).
2. Choose **Customer**, and enter **Model** and **Serial number**.
3. Enter **Location**, **Category**, optional **Parent asset**, **Status** and **Criticality**.
4. Set **Installation date**, **Warranty until** and **Meter unit** if relevant.
5. Press **Register asset**.

Once a **Customer** is selected, only that customer's **Sites** and **Parent asset** are loaded from the server. Changing customer clears and reloads the lists, so you cannot select a site or asset belonging to another customer. The same applies to **New work order** and **Edit work order**.

**Edit asset** changes the same fields.

The statuses are **Active**, **Out of service** and **Retired**; criticality is **Critical**, **Normal** or **Low**. **Parent asset** builds a hierarchy, and **Move in hierarchy** changes placement.

## Move a unit to another customer

When equipment changes owner or customer, the whole unit can be moved with its service history. The move requires **Manage equipment** and **Manage customers**, or an administrator.

1. Open the unit at `/enheter/<id>` and select **Move to another customer**.
2. Search for the target customer and optionally choose a target site. If you choose no site, the unit stays without a site at the new customer.
3. Review **What moves** and the preview. It shows the number of work orders, visits, reports, revisions, readings, plans, documents, attachments and QR codes that will follow.
4. Type the target customer's name to confirm, or tick the confirmation, and select **Move the unit**.

**This moves:** the unit and any child units (up to 1000), the hierarchy between them, work orders and visits, reports with revisions, report units and PDFs, readings, maintenance plans and occurrences, documents with versions, attachments/uploads, registered QR codes and comments. The root unit receives the target site. Child units lose the old site because it belonged to the previous customer and must be placed again after the move. The root unit is no longer linked to any former parent unit.

**This blocks the move:**

- **Contract lines belonging to another customer's agreement** that point to the unit or a child unit. A move cannot silently change who a financial obligation points to. Clear or end the contract lines first.
- Child units that already belong to another customer.
- Order numbers that collide with a number in the target customer's department.
- A missing or deleted target customer, or a target site that does not belong to the customer.
- More than 1000 units in the subtree.

**This does not move:** contract lines, the old customer's portal users, global documents without a customer or unit link, or the audit log itself. Contacts from the old customer are cleared on the work orders, and work orders that point to another customer's agreement lose the contract link. The preview warns you in advance. The target customer's department is used when it exists; otherwise your own department is used, and you are told.

The move is idempotent: repeating it after the unit is at the target customer reports that nothing changed. All validation runs before the first write, so a rejected move leaves no half-finished changes. The action is audit-logged. To undo it, move the unit back manually.

## Readings

1. Open the asset and find **Meter**.
2. Press **Register reading**.
3. Enter **Value**, **Date**, **Source** and optionally **Note**.
4. Press **Save reading**.

### Maintenance plans

1. Press **New maintenance plan**.
2. Enter **Title** and choose **Trigger**: **Calendar** or **Meter**.
3. For calendar: set **Interval (months)** and optionally **Lead time (days)**. For meter: set **Meter interval**.
4. Check **Next due** and save.

**Pause** stops the plan, and **Activate** starts it again; **Timeline** shows occurrences. When the plan falls due, the scheduled job creates a work order for the office to follow up.

### Service history and reports

**Service history** shows every report in which the asset was checked, with date, result, comments, checkpoints and images.

In the form, the technician chooses **Get from the customer equipment**. Each registered asset can be used only once in the same report. The report remembers the asset, serial number and answers, so the customer card and asset page show exactly what was checked.

## Calendar and dispatch

The calendar is at `/kalender` with the tabs **Resource overview**, **Month**, **Week**, **2 weeks**, **Week (time)**, **Day**, **Dispatch**, **Capacity**, **Route** and **Reports**.

| Tab | URL |
|---|---|
| **Month** | `/kalender?vis=maned` |
| **Week** | `/kalender?vis=uke` |
| **2 weeks** | `/kalender?vis=2uker` |
| **Week (time)** | `/kalender?vis=timegrid` |
| **Day** | `/kalender?vis=dag` |

At the top, **Period** shows the month or date range right next to the navigation arrows and **Today**. The calendar loads about six months back and forward, and a maximum of 1000 visits; visits outside the window may be missing.

**Filter the calendar** has **Technician**, **Status** and **No technician**. **Apply filter** and **Reset** apply to all date views.

### Visits and drag and drop

The calendar shows one event per workday with the status **Planned**, **Assigned**, **En route**, **On site**, **Done** or **Cancelled**. When the technician has recorded time, the event shows the actual start, end and duration instead of only the date. Drag a day to another date to move it. On conflict the move is not saved, and you are told about the overlapping assignment. Completed and cancelled days are locked.

**Resource overview** shows technicians, active work, unassigned work and recent completions. **Unassigned work orders** have **Assign technician**. You can drag cards between technicians to assign.

## Dispatch and capacity

**Dispatch** (`/kalender/dispatch`) assigns technicians to visits, with a separate **No technician** column. **Overdue / Unassigned** is always shown, regardless of the date. Drag a workday to a technician's column to assign it. Drop it on **No technician** to remove the source technician; other technicians are kept.

**Automatic assignment** is dry-run with **Dry run**; **Use {count} suggestions** saves only when you choose it. **Suggest technician** ranks candidates by **Skills**, **Distance**, **Travel**, **Free**, **SLA** and **Load**.

**Capacity** (`/kalender/kapasitet`) shows **Planned capacity against actual load per department, day and technician**. Choose **From** and **To**, press **Show**, and read **Capacity**, **Load** and **Utilisation**.

## Route plan

**Route** (`/kalender/rute`) suggests a driving order for one day, per technician. Choose **Date** and optionally **Technician**, then **Show route**. The page shows a map, stop numbers, **Arrival**, **Planned**, **Driving**, **Waiting**, **Overtime**, and total **Drive time**, **Distance** and **Waiting time** per technician.

The route starts at the technician's earliest located visit. The calculation uses **OpenStreetMap via OSRM** for routes and the travel matrix. If `GOOGLE_ROUTES_API_KEY` is set, **Google Routes API** is used as an optional traffic-aware source for single legs; without it, ETA is a route and speed estimate, not live traffic. If neither provider responds, local calculation is used: straight-line distance × 1.35 and average speed. **Calculation** shows which method was actually used.

**Overtime:** A visit is no longer left unassigned just because it cannot meet the time window or working hours. It is planned with overtime, and the stop is marked with `outsideTimeWindow`, `outsideWorkingHours` and `overtimeMinutes`. The route sums **Total overtime**. `ROUTE_ALLOW_OVERTIME=false` disables planning after working hours, and `ROUTE_MAX_OVERTIME_MINUTES` sets an optional cap. Only genuinely impossible visits remain unassigned: missing coordinates, missing skills, holiday/sickness, overtime limit, or more than 50 located visits.

**Manual order:** Every stop has a number field. Enter 1, 2, 3, 4 … and select **Manual order**. The numbering is normalised to 1..n, stored on the visit and moves the times with the same service layer as **Use order**. An empty field or 0 means automatic order. As long as at least one visit for the technician and day has a manual number, the route sorts by number and skips nearest-neighbour/2-opt.

**Use order** moves the visits to the suggested times in one transaction. The action requires `workorders.assign`, is audit-logged per visit, and is cancelled entirely on overlap with a fixed visit or registered holiday/sickness. Completed and cancelled visits are never moved. If the plan has changed since it was shown, you are asked to reload the page.




*Figure: Manual order is set with 1/2/3/4 in the route planner.*

Look at the number field at each stop and the summary at the bottom. The order is a suggestion until you select **Manual order** or **Use order**. The numbering does not change the date; it changes the order within the day.




*Figure: Overtime per stop and total overtime for the day.*

Notice that overtime is marked per stop, not hidden in the total. A stop can be outside the customer's time window without creating overtime if it is still within working hours. Check **Total overtime** before approving the route, especially when `ROUTE_MAX_OVERTIME_MINUTES` is set.

**Assignment rules** and **Territories** are configured under **Settings** → **Dispatch**. There you choose a **Mode**, adjust the **Weights** for **Skills**, **Distance**, **Travel time**, **Free time**, **SLA** and **Load**, and add territories with **Postal prefixes**, **Municipality codes** and **Radius (km)**. **Capacity plans** are shown in the capacity view but are not edited in **Settings** in 0.8.11. The same applies to skills, absence and working hours: dispatch and booking use them, but they have no separate form in **Settings**.

## Agreements, SLA and price lists

Agreements are at `/avtaler` and on the customer card's **Agreements** tab.

### Create

1. Open the customer card → **Agreements**.
2. Enter **Title**, optional **Agreement number**, **Department**, **Start date** and **End date**.
3. Choose **Interval** (3, 6, 12 or 24 months), **Status** (**Draft** or **Active**) and **Renewal** (**Manual** or **Automatic**).
4. Set **Notice before expiry (days)** and optionally **Price adjustment (%)**.
5. Choose **Contact**, **Checklist**, **SLA policy** and **Price list**, and add a line with **Equipment**, **Coverage**, **Visits/year**, **Response (min)** and **Resolution (min)**.
6. Press **Create service agreement**.

### Lines and coverage

**Coverage ({count})** shows the agreement lines. An agreement **without lines** covers all equipment; a line **without equipment** covers all equipment; a line **with equipment** covers only that unit. Read equipment, coverage, visits per year and response times. Only **Active** agreements within the start and end dates count. No automatic “covered” badge is shown, so check manually before promising cover.

### Interval, renewal and lifecycle

**Next service** is calculated from the start date and interval. **Renew** extends; **Pause** and **Activate** stop and start new work orders. **Automatic** renewal is run by the scheduled service agreement job. **Manual** renewal becomes **Expired** after the end date.

### SLA

An SLA policy sets **Response time (minutes)** and **Resolution time (minutes)**. It can be department-specific or global, and limited to **Count business hours only**. **Business hours** and **Holidays** decide when the clock runs; without saved business hours, weekdays 08–16 are used. Policies, business hours, holidays and price lists are maintained by an administrator under **Settings → SLA and business hours**.

SLA requires an **active service agreement** with a valid date and an active SLA policy. Customers without an active agreement get no SLA clock, and the work order shows no SLA panel. On the work order, **SLA** shows **Respond by**, **Resolve by** and **On track**, **At risk**, **Breached**, **Met** or **Cancelled**. **En route** or **On site** records response; **Done** records resolution.

### Price lists

**Price lists are reference data for display and for the hourly rate in profitability. Without `invoicing`, the system does not invoice or calculate invoice amounts; with the module on, the price list is a basis when you create an invoice.**

### Recurring work orders

At each interval a new **Planned** work order is created with the agreement's customer, contact, department and checklist, plus one workday without technicians. Expired agreements do not generate new work orders.

## Bookings

The booking queue is at `/bookinger` with **All**, **Pending**, **Confirmed** and **Closed**. Each card shows **Reference**, **Customer**, **Requested time**, **Location**, **Service**, **Technician** and status.

1. Open `/bookinger` and filter on **Pending**.
2. Fill in **Date**, **From** and **To** for the planned visit. The requested date is prefilled but can be changed.
3. Choose **Technician** in **Choose technician**. All active employees in the department can be chosen; absence, working hours and holds are checked on confirmation, while overlap is only shown as a busy marker.
4. Press **Confirm booking**, or **Cancel** to decline.

On confirmation the work order becomes **Planned**, the visit is created or updated, and the customer receives an SMS if the contact has consented. If the technician already has another visit at that time, it is shown as a **busy marker**, not a hard block. You may still confirm, but should consider capacity and move one of the days if needed. Absence and working hours remain hard requirements.

The customer chooses a site, service and requested date in the portal, and writes the desired time in the description. The customer cannot select a time block, and the booking always creates a new work order with status **Open/New**. No visit or confirmed time is created.

The office or technician plans the work order and sets the status to **Planned**. If the time does not fit, it is set to **Waiting** for replanning. The customer receives the final confirmation only then. A customer booking is already a work order; there is no separate **Convert to work order** button.

If the customer already has a confirmed appointment, **Request a new time** may use a short hold while the form is filled out. That hold applies only to rescheduling the confirmed appointment, never to new bookings.

# Inventory, spare parts and purchasing

The inventory module (`inventory`) requires the equipment module. It is used by both office and technicians, but the office owns the item register, suppliers and purchasing.

| Page | Use |
|---|---|
| `/lager` | Item list, locations, purchase suggestions and new item/location |
| `/lager/varer/<id>` | Stock per location, movements, reservations and adjustment |
| `/lager/lokasjoner/<id>` | Stock and movements for one warehouse or one service van |
| `/lager/oppdrag/<id>` | Spare parts for one work order, with reservation, release and consumption |
| `/lager/innkjop` | Purchase orders, filter and drafts from purchase suggestions |
| `/lager/innkjop/<id>` | Order lines, ordering, cancellation and goods receipt |
| `/lager/innkjop/leverandorer` | Supplier list and new supplier |

A **location** is a warehouse (`WAREHOUSE`) or a service van (`VAN`), with an optional department and responsible user. An **item** has a SKU, name, unit, reorder point and active flag. Stock is held per item and location. Movements are `IN`, `OUT`, `ADJUST`, `RESERVE`, `RELEASE` and `CONSUME`.

## Reserve parts on a work order

1. Open the work order and go to the spare-parts page (`/lager/oppdrag/<id>`).
2. Choose item and location, and enter the quantity.
3. Select **Reserve**. The reservation reduces available stock but does not move the item physically.
4. If needed, select **Release** to drop the reservation without touching stock.
5. When the part is used, select **Consume**. The quantity is then deducted from physical stock.

Consumption is deliberately done from the spare-parts page, not automatically when the visit is set to **Completed**. That avoids hidden side effects in the field flow, but requires the office or technician to record consumption.

## Suppliers and purchase orders

1. Create the supplier under **Inventory** → **Purchasing** → **Suppliers** with name, organisation number, email, phone and optional department. A supplier without a department is shared across the installation.
2. Select **New purchase order** at `/lager/innkjop`, or build a draft from the purchase suggestions at `/lager`. The suggestion is `reorder point - (on hand - active reservations)`.
3. Choose supplier, receiving location and expected delivery. Add lines with item, ordered quantity and unit cost.
4. Select **Order**. The order receives a number in the form `PO-YYYY-NNNNN`.
5. On receipt, open the order and register the received quantity. Partial receipt is supported; the remainder stays open.
6. When everything is received, the order is **Received**. Over-receipt and negative quantities are rejected.

| Status | Meaning | Can move to |
|---|---|---|
| **Draft** | Freely editable | Ordered, cancelled |
| **Ordered** | No goods received | Partially received, received, cancelled |
| **Partially received** | Some goods are in stock | Received |
| **Received** | Everything received | – |
| **Cancelled** | Cancelled before receipt | – |

Only drafts can change lines, supplier, location and note. Cancellation is possible only before the first item is received. Goods receipt increases stock and writes an `IN` movement in the same transaction as line and order status. The order is locked with `SELECT ... FOR UPDATE`, so two concurrent receipts cannot over-receive. Every mutation is audit-logged.

Unit cost is 0 until edited manually; the total is therefore 0 until cost is filled in. The purchase suggestion is a calculated read-only view and orders nothing automatically. Supplier invoices, invoice control, sending orders to the supplier, returns/credit notes and serial/lot tracking are not part of v1.




*Figure: The purchase order shows status, lines and what remains to receive.*

Look at the status and received quantity per line. **Partially received** means some goods are already in stock and the rest can still be received. The receipt button applies only to the remaining quantity; the system rejects over-receipt.

**Permissions:** view inventory with `equipment.view`, manage items, locations, suppliers and purchasing with `equipment.manage`, and reserve/release/consume with `workorders.update` or `equipment.manage`. Department scoping follows the location; an administrator sees all departments.

## Mileage, expenses and payroll basis

The expenses module (`fieldExpenses`) lets the technician record mileage and expenses, and the office approve and export the payroll basis.

- **The technician** records in `/utlegg`: date, from/to, kilometres, rate, category, gross amount, VAT and receipt.
- **The office** approves in `/utlegg/kontor`, with preview and export per period.
- **The export** is downloaded from `/utlegg/kontor/eksport?start=…&end=…`.

Mileage allowance is `km × rate`, rounded to the nearest øre on the server. VAT is `gross × rate / (100 + rate)`. The default is NOK 3.50/km and 25% VAT when the form does not state otherwise. Rates are stored per row, so historical exports are not changed by later rate changes. The client cannot submit a final amount that differs.

1. Open **Expenses → Office**.
2. Choose a period and optional department.
3. Review mileage and expenses. Approve with **Approve**, or reject with **Reject** and a reason.
4. Select **Export payroll** for the period. The file is locked and marked with the export time.
5. Download the CSV and import it into the payroll system.

Mileage and expenses start as **Pending** (or **Draft** when the client asks for it). Only approved rows are exported; rejected and unresolved rows are always excluded. Rejection requires a reason. Department scoping prevents office A from approving rows in office B. Export is idempotent per department and period: the same period gives exactly the same CSV and checksum, and overlapping periods are rejected.

The CSV is UTF-8 with BOM, semicolon-separated and uses `\r\n`, ready for Norwegian Excel. The first row is:

```
type;bruker;dato;referanse;beskrivelse;mengde;enhet;belop;mva
```

`type` is `WORK`, `TRAVEL`, `MILEAGE` or `EXPENSE`. Time and travel time come from the visit timestamps; the module does not create its own time recording. There is no integration with an external payroll provider – the CSV contract is the interface.




*Figure: The office approves or rejects rows before the payroll CSV is locked.*

Notice that approval and export are two steps. An approved row is not exported until the period is locked, and an exported row cannot be changed. Check the receipt and VAT before you approve, not afterwards.

# Profitability

**Profitability** ( `/lonnsomhet` ) gives office and administrator one view of work orders, time, first-time fix, **estimated** and **actual** revenue by department, customer and technician. The module is `bi` and requires `reports.view` ; technicians and customer portal users get `403` .

1. Choose **From** and **To**, and optionally **Department**, **Customer** and **Technician**.
2. Read **Estimated revenue**, **Actual revenue**, **First-time fix**, **Utilisation** and **Result**.
3. Select **Export CSV** for the same filter and department scope.

**Estimated revenue** is the hourly rate from the agreement's price list × work time. Travel time is not included. **Actual revenue** comes from invoices when `invoicing` is on: net invoiced amount excluding VAT, with credit notes deducted and drafts excluded. The invoice must be linked to a work order in the selection and issued in the period. The period for estimate and work orders follows the work order's planned date, while actual revenue follows the invoice date. A work order planned in one month and invoiced in the next therefore gets the estimate in the first and actual revenue in the second.

**Result** is calculated only when every work order in the selection has an hourly rate and `BI_INTERNAL_COST_RATE_NOK` is set. Without the rate, "not calculated" is shown. Actual revenue does not affect the result calculation; it is still based on the estimate. The limit is 5,000 work orders, 5,000 invoices and 366 days per report. CSV export is deterministic and audit-logged with filter, row count and checksum.

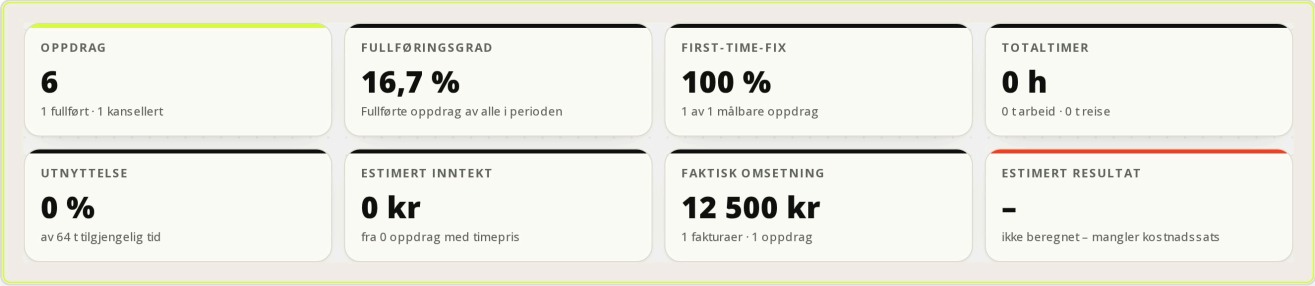



*Figure: The KPIs show estimated and actual revenue side by side.*

Look for the note on whether actual revenue is available. If `invoicing` is off, or there are no non-draft invoices in the selection, the report shows the estimate alone and explains why. The numbers are not a guarantee of profit: the cost is a global internal rate, not salary, vehicle, parts and expenses per work order.

# Reports

The report list is at `/rapporter` and shows **Service reports** – completed checklists with signatures and photos.

## List and filters

Choose **This month**, **Last month** or **All**, and filter on **Free text**, **Customer**, **Technician**, **Deviations only** and **Acknowledgement** (**All**, **Acknowledged**, **Not acknowledged**). **Sorting** is **Newest first**, **Oldest first**, **Customer (A–Z)** or **Technician (A–Z)**. Press **Filter** or **Reset filters**. The list shows **Locked**, **Open for correction** or **Unsigned**, and **Report no.** when numbered.

### Open, PDF and content

**Open form** opens the report with **Main result**, **Signed by**, **Units that were checked**, **Answers to checkpoints** and **General notes from the technician**. The PDF is shown in the app's own PDF viewer. **Open in new tab** shows the same PDF in the browser, **Download PDF** creates an attachment, and **Close PDF** goes back. If the PDF cannot be shown, for example because it contains active content that cannot be removed, the viewer shows an error with **Try again** and download. Revisions are opened the same way from **Revisions** using **Open revision (PDF)**.

A report can have a **Technician** and a **Customer** signature. Photos are added per checkpoint or unit. Deviations appear as **Deviation found**. When a report contains at least one deviation, the system automatically creates **one undated remediation work order** titled “Utbedring av avvik – …”. It inherits customer, department and contact, gets the submitter as technician, and describes the units and checkpoints. The generated title is Norwegian (“Utbedring av avvik – …”) in 0.8.11, even when the interface is English.

> **Note:** If a photo or signature is missing from the finished PDF, the PDF shows a warning with the number of images that could not be loaded. The usual causes are missing `Upload` metadata, a file removed from storage, or a file owned by another work order. Re-upload the image from the report and generate the PDF again. The operations log shows `report_pdf_images_missing` with a count, without file names or personal data.

### Status and locking

A submitted report is locked. **Unlock for editing** requires a **Reason for unlocking** and is an administrator action.

### Customer portal and equipment history

Only submitted, locked reports can be opened. **View report** opens the PDF in the app, and **Download PDF** creates an attachment. An unlocked report is unavailable until it is locked again. The customer can **Acknowledge receipt** and **Request a contact information update**; the latter does not update the customer card automatically.

The report's units are linked to the customer's equipment. **Service history** on the asset shows every check with date, result, comments, checkpoints and images.

## Documents

The document library is at `/dokumenter` and is internal documentation with folders, tags, versions and full-text search.

### Folders and upload

1. Press **New folder**, or **Import folder structure** for a whole folder of PDFs.
2. Press **Upload PDF**, choose the **PDF file**, and enter **Title**, optional **Description** and **Tags**.
3. Press **Save document**.

Only PDFs are allowed. **Move to folder** and **Add tag** work on several documents at once. **Delete selected** is an administrator action.

### Tags, search and OCR

Document tags are separate from work order tags. Search **title, description, file name or tags**, and choose **This folder** or **Whole library**; **Filter by tag:** narrows further. Text is extracted on upload, and scanned PDFs are run through OCR. **Index all PDFs for search** fills in missing content.

### Versions and sharing

**Edit → Replace file (optional)** creates a new version. Previous versions are under **Previous versions ({count})** with **Download** and **Restore**.

The library is internal and controlled by **View documents** and **Manage documents**. To share with a customer, add the file to the customer card under **More → Attachments**; it appears in the customer's **Documents**. Work order attachments are internal. Reports are shown only when submitted and locked.

## Map, search, notifications and messages

**Map** ( `/kart` ) shows active work orders grouped by location. Filter by **Technician**, **From date** and **To date**. **Open in Maps** gives directions. Markers are placed at the address when geocoded; otherwise the work order is shown at the customer's city.

**Search** ( `/sok` ) finds customers, work orders, equipment, contacts, reports, comments, employees, checklists, folders and documents. Use the top bar, **Ctrl+K** (⌘K on macOS) or ***/***. Enter at least two characters. **Best matches** appears first, **See all ({count})** opens the full list, and **Filters** narrows by type, status, customer, technician, date, tag and HMS type. Search is fuzzy and tolerates typos; document hits open the document in the library. **Save this search** stores a saved search.

**Notifications** ( `/varsler` ) shows **{unread} unread of {total} total**, **Unread only** and **Mark all as read**. The types are **Assignment**, **Date change**, **Cancellation**, **Report**, **Booking**, **Service agreement**, **Overdue** and **Contact update**. **Settings** lets you choose **In app** and **Email** per event, and turn **Notification sound** on or off.

**Messages** ( `/meldinger` ) collects SMS conversations. Filter on **All** or **Unread only**, search by customer, contact or work order, and **Start conversation** from a work order. **Quick messages** sends **Send booking confirmation**, **Send reminder**, **Send on my way**, **Send completed** or **Request review**. **Consent and opt-out** shows **Consent registered** or **Not registered**; without consent no SMS is sent. Each outgoing message shows **Status** (Queued, Sent, Delivered or Failed). Threads are scoped to your department; portal users see only their own.

An administrator can configure **Automatic reminders** under **Settings → SMS**. Rules send SMS, email or both a chosen number of hours before the visit, with quiet hours and a maximum per visit. **Recent reminders** shows what is planned, sent or skipped — without the message text.

## Customer portal from the office

The customer sees **My Company: {customerName}** and the tabs **Overview**, **Bookings**, **Reports**, **Documents**, **Equipment**, **Contracts**, **Visits**, **Messages** and **Contact**.

1. Open the customer card → **More → Portal users**.
2. Press **Create and send invitation**, and enter name, email and contact.

3. Send the invitation; the user sets a password.
4. Press **Activate** once the password is set.

**Deactivate** revokes access immediately. A portal user must be linked to a contact at the same customer.

**Bookings** shows requests and confirmed appointments. **Reports** lets the customer acknowledge. **Documents** shows files you have shared through the customer card's **Attachments**. **Equipment** shows status, warranty, latest reading and plans. **Contracts** shows service agreements and coverage lines. **Visits** shows tracking links and reviews. The tracking page shows the planned time when the technician has not shared a location, and a live ETA only when the latest position is fresh (under ten minutes). If the position is older, it says explicitly how old it is; the page never pretends an old ping is live. The ETA refreshes automatically every 30 seconds. **Messages** lets the customer reply by SMS with consent.

The customer can **Send a date request**, **Request a new time**, **Cancel request**, **Acknowledge receipt** and **Request a contact information update**. Booking and cancellation notify the office; contact updates must be made manually on the customer card.

## Daily and periodic routines

**Morning:** Check **Overdue**, **Today**, **Unassigned**. Assign anything that is without a technician today, and check double bookings and visits without a technician.

**During the day:** Follow **In progress** and **Overdue**. Move visits in the calendar, use dispatch for sickness or urgent work, and answer customers in **Messages**. Register consent before sending SMS.

**End of day:** Check that today's work orders are **Completed** or have a follow-up. Review **Unsigned reports**, check tomorrow's visits, and clear **Bookings** and **Notifications**.

**Weekly:** Review **Overdue** and **Unassigned**, next week's **Capacity**, **Expires within 60 days**, **Reports** with **Deviations only**, and customer portal users.

**Monthly:** Review expired agreements and agreement lines. Deactivate portal users who have left, and review equipment without readings or with expired warranty. SLA policies, business hours, holidays and price lists are managed by an administrator; ask for changes when needed.

## Customer transfer (administrator only)

The **Customer transfer** tab is under **Settings** and is shown only to an active administrator who also has **Manage settings**. Office staff, technicians and portal users get `403` even if they call the route directly. The tab moves one or more customers between installations; it is not a backup.

### Export

1. Open **Settings** → **Customer transfer**, search for the customer and tick it.
2. Choose whether files, revision snapshots, deleted rows and legal hold should be included. Signature images are off by default and require a documented assessment.
3. Read the personal-data notice and tick the confirmation.
4. Select **Start export**. You get a ZIP with `manifest.json`, `checksums.sha256`, data files and files. The package is available for 24 hours and can only be downloaded by you.

5. Optionally verify manually with `sha256sum -c checksums.sha256`. The checksums detect corruption but do not prove who created the package.

The package never contains passwords, password hashes, 2FA secrets, sessions or API keys. Portal users follow as metadata and are imported deactivated without a password.

**Import**

1. Upload the ZIP. The server validates structure, checksums, file types and references, and shows a preview. Nothing is written yet.
2. Review counts, warnings and errors. Errors block the import. For conflicts, choose **skip** (default), **create new** or — for customer and equipment — **update**. Reports are never updated.
3. Tick the confirmation and select **Import the package**. All rows and files are written in one transaction; if anything fails, the database is unchanged and temporary files are cleaned up.
4. After import, each portal user has a new id, is deactivated and has no password. Send a normal invitation or reset the password before sign-in. Employees and templates are never created automatically.

**Repeated attempts and errors:** export is idempotent. The same administrator, the same customer selection and the same choices within ten minutes reuse the same job instead of starting a new one, so a double click does not consume another attempt. There are at most 10 export starts per hour; exceeding the limit gives a clear message to wait. Errors are stored on the job and shown in the interface, so you can correct the cause and try again without guessing. The preview is valid for 30 minutes; if the package or selection changed since it was shown, upload and preview again. Import errors block writing, and an aborted import rolls back the whole transaction and cleans up temporary files.

**Limits:** the package is limited to 500 MiB and 50 MiB per file, and the export runs synchronously. Global documents without a customer or unit owner are not included. Signatures, SLA clocks, bookings, notifications, SMS, tracking and report drafts are deliberately excluded. See `docs/backlog-customer-export-import.md` for the full list.

## Interface with the technician and administrator parts

| The office does | The technician does | Where it is described |
|---|---|---|
| Creates work orders with customer, contact, equipment and checklists | Takes the work order, opens the form and completes checkpoints | Technician part (Part 2) |
| Plans the date and technician per workday | Starts **En route**, marks **On site**, **Done** and **Finish travel** | Technician part |
| Reads reports, deviations and signatures | Signs, attaches images, submits the report | Technician part |

The administrator part describes **Employees & Roles**, **Roles & permissions**, **Departments**, **Checklists / Templates**, **SLA and business hours**, branding, API keys, audit log, legal hold and restoration. See the administrator part (Part 3).

# Common mistakes and troubleshooting

| Symptom | Common cause | Do this |
|---|---|---|
| Cannot find the work order | Wrong department, view or filter | Reset search/filter, check department |
| Cannot assign | Missing **Assign technicians**, locked or legal hold | Request permission, or reopen as administrator |
| Drag and drop did not change the date | Drag and drop assigns a technician | Move the day in the calendar |
| Calendar is empty | No workdays, filter or outside ±6 months | Reset filter, choose **Today**, check the date range |
| Overlapping workday | Technician has several jobs at the same time | Check capacity and move one of the days if needed |
| Report is locked | Submitted report | Ask an administrator to **Unlock for editing** |
| Customer cannot see the report | Not locked, or missing portal/document access | Lock the report, check portal user |
| Customer cannot book | Missing contact/active user, or date outside the window | Check contact, user, 24 hours and requested date |
| SMS is not sent | Missing consent, phone, quota or dry run | Register consent, check number/settings |
| Agreement creates no work orders | Paused/expired, future start or job not running | Activate, check start date and job |
| SLA is not shown | No active agreement with an active policy | Activate the agreement and link a policy to it |
| Price list does not invoice | Reference data, or `invoicing` is off | Enable invoicing and create the invoice under **Invoices** |
| The payment button does not appear | Provider or webhook secret is missing | Check `PAYMENTS_PROVIDER` and the keys |
| Accounting sync is failed | Wrong key, account or VAT | Read the error code and sync again |
| Stock does not match | Reservation not consumed, or manual adjustment missing | Check reservations and movements on the item |
| Expenses cannot be exported | The period overlaps a locked export, or the row is not approved | Choose a new period and approve the row |
| Attachment is not in the portal | Work order attachments are internal | Add the file to the customer card → **More** → **Attachments** |
| Unit cannot be moved | Active contract lines, number conflict, cross-customer reference or invalid target | Read the blocker in the preview, clear the contract line, or choose the right customer/site |

| Symptom | Common cause | Do this |
| --- | --- | --- |
| Upload is rejected | File too large, user/customer quota reached, or proxy limit | Check file size, `UPLOAD_QUOTA_*`, disk and the proxy body limit |
| Customer transfer does not start | 10 starts per hour used up, an existing job is reused, or the preview expired | Wait, follow the existing job, or upload and preview the package again |

PART 2

# Technician

The field app for the **Technician** role: sign-in, My work orders, work order details, checklists, units, offline drafts and submission. Version 0.8.11 (2026-11-08).

The bottom bar has **Today** ( `/mine-oppdrag` ), **Work orders**, **Notifications** and **More**. **More** opens the same menu as the hamburger button at the top. Language, notification sound, reports, customers/equipment and settings are in that menu.




*Figure: Today is the field app's home screen.*

Notice **Next job** at the top and the **Overdue** and **Available** groups below. This is where you see whether something is assigned to you, whether you can take available work, and whether anything has been left from earlier days.

# First time: sign-in, 2FA and language

## Sign in

1. Open `/logg-inn`.
2. Enter **Email** and **Password**.
3. Select **Log in**.

You land on **Today**. A wrong password gives “Wrong email or password.” An account that has not been activated gives “Your account has not been activated yet. An administrator must activate it before you can log in.”

The session normally lasts 12 hours. Local drafts remain when the session expires, but are not sent until you sign in again.

## Two-factor authentication

2FA is optional but recommended. If enabled, you go to `/logg-inn/2fa` after your password.

1. Open your authenticator app.
2. Enter the 6-digit **One-time code** for the company.
3. Select **Verify**.

If you get “Incorrect one-time code. Try again.”, wait for the next code. If you have lost your phone, select **Sign in with a recovery code**, enter one of the codes you saved when you enabled 2FA, and select **Sign in with code**. An administrator can reset 2FA.

You cannot change your password in the field app. Sign out and use **Forgot password** on the sign-in page, or ask an administrator to send a new link. 2FA and recovery codes are managed under **Settings** → **Security**.

## Change language

1. Open **Menu** with the hamburger button at the top, or select **More** in the bottom bar.
2. Scroll to the bottom. **Change language** is there.
3. Choose **Norsk**, **English**, **Polski** or **Deutsch**.

The choice is stored in the `NEXT_LOCALE` cookie and takes effect without changing the URL. Dates and numbers follow the language. The fixed labels in the report PDF remain Norwegian in 0.8.11.

## Install the field app (PWA)

The field app is installable. It opens in its own window, starts at `/mine-oppdrag` and is locked to portrait orientation.

- **Android/Chrome:** Browser menu → **Install app** or **Add to Home screen**.
- **iPhone/iPad/Safari:** Share icon → **Add to Home Screen**.
- **Chrome/Edge on desktop:** The install icon in the address bar, or the menu → **Install**.

There is no separate “Install” button inside the app. Installation uses the manifest and requires HTTPS. The service worker is registered in production builds only. It caches all same-origin GET pages, API GET requests

and files under `/uploads/` for seven days per user.

## Native app and server address

The Android and iOS app lives in the separate `fsm-mobile` repository and is built and signed there. The platform no longer contains Capacitor projects, shells or signing files.

1. Install the app from the company distribution or app store.
2. On first start you arrive at a local setup page. Enter the **FSM server URL**, for example `https://fsm.example.no`.
3. The address is stored natively on the device and is not built into the app. Select **Connect**.
4. Later you can change the address from the gear in the app, or use **Switch server**. This clears the cache and sign-in for the old server.

If you use a browser or PWA instead, the same address and sign-in apply. The native app and the PWA use the same API and the same access control.

## Push notifications

Push complements in-app notifications. Each notification is first stored under **Notifications**, and then a copy is sent to your active mobile devices.

1. Sign in to the app and allow notifications when prompted.
2. The app registers the device with a push token after sign-in and when the token rotates.
3. Tap a notification to open the right page. Deep links always go to an internal path, for example `/oppdrag/<id>`, `/bookinger` or `/varsler`.
4. On sign-out the device is unregistered. If a device is signed in again with a different account, it moves to the new account, and old pending notifications are deleted.

Notification sound is turned on and off in the menu. The push setup itself (FCM for Android and APNs for iOS) is done by an administrator and requires credentials in the environment. If no provider is configured, the device can be registered but nothing is sent.

> **Note:** Push is not the same as SMS. Push requires the app to be installed and the device to allow notifications; SMS requires consent and a phone number.

## Scan QR and barcode

1. Select **Scan** on **Today**, in the top bar or in the menu.
2. Select **Start camera** and allow camera access the first time.
3. Hold the QR code on the equipment label steady inside the frame. The app opens the unit or work order if you have access.
4. If the camera does not work, enter the code in **Code or reference** and select **Look up**. You can paste the full link from the label.

The camera requires the app to be opened over **HTTPS** (a secure address). On a plain `http://` address, such as a LAN address during development, the browser blocks the camera and the app shows a separate HTTPS message. Use the installed app or the `https://` address in production.

- **iPhone/iPad (Safari):** Allow camera access when Safari asks. If you declined, go to **Settings** → **Safari** → **Camera** (or **Settings** → **the app**) and choose **Ask** or **Allow**. Camera scanning works in iOS Safari even without the browser's built-in barcode API; the app then uses a fallback scanner.
- **Android (Chrome):** Select **Allow** on the camera prompt. If you declined, open the lock icon by the address → **Permissions** → **Camera**, or **Settings** → **Apps** → **Chrome** → **Permissions** → **Camera**.
- **Camera is busy:** Close other apps that use the camera and select **Start camera** again.
- **The code is not read:** Hold the label steadier, avoid backlight and make sure there is enough light. The app keeps looking and shows a separate message if the image cannot be decoded.

If the browser has no camera support at all, a message says so – but on iPhone and Android the fallback scanner should be available as long as the app runs over HTTPS and you have granted camera access.

The app accepts unit QR codes, unit IDs, work order references (for example `O-261103-ABCD1234`) and order numbers. An unknown code, missing access and repeated scans give an explanation instead of opening the wrong page. Access is checked again on the server; a scan never grants more access than you already have.

If the customer already has QR labels from another system, they work too: the office registers the existing value under **Existing QR codes** on the unit. You scan the label the same way as our own, and the app opens the unit if you have access. The value is an opaque string – the app never follows a link or fetches content from the internet.

### Before you lose the network

1. Open **Today** and every work order you will perform.
2. Open every report form you need, so the page enters the offline cache.
3. Wait until the draft status shows **draft saved**.
4. Take and upload photos while online; a required photo field is not complete until the photo is uploaded.
5. Open any report PDFs you need while online.
6. Do not clear browser or app data.

The page, API and file cache lasts **seven days per user**. The caches are cleared on sign-out, an expired session and **Switch server**. The app requests persistent storage, but the operating system may clear storage if the device is full. Local drafts last seven days.

## My work orders / Today

`/mine-oppdrag` is the home screen. At the top you see your name, **On duty**, **Online** or **Offline**, today's date, the number of jobs today, the number of **Overdue** jobs, and an estimated route distance when today's work orders have coordinates. If coordinates are missing, it says **Calculating route** or nothing – the app does not invent a distance.

The search field searches title, description, order number, customer name, city, address and contact person. **Search** runs it; **Reset** clears it.

## Next job

The **Next job** card shows the first remaining job today, or the next upcoming job if today is empty. It shows the date and the status of today's session, title, customer, order number, address, contact person, **Job 1 of N**, and **Start form**, **Continue form** or **Open job**, **Call** and **Map**.

**Start form** opens the first checklist without a locked report. **Continue form** is used when a draft or unlocked report exists. If the work order has no template, the button says **Open job**.

## Overdue, available and later

- **Overdue** shows your open work orders with a scheduled date before today, oldest first.
- **Available** shows open work orders without a technician. With update permission, each card gets **Take work order**; you are assigned and see “Work order claimed and now under your work orders.” Without permission the button is hidden. Completed or cancelled work orders cannot be claimed.
- **Later today** shows the rest of the day, **Today's route list** shows the full list, and **Completed work orders / History** shows completed jobs with **Show more**.

## Job cards and status chips

The card shows title, status chip, customer, order number, contact, address, tags and date. A draft shows **Draft 14:32 · Continue**. **Call** and **Map** start the phone or Google Maps without opening the work order.

| Chip | Meaning |
|---|---|
| **Open / New** | Created, not planned or started |
| **Waiting to be planned** | The system's starting status; not selected manually |
| **Planned** | Has a planned date and workday |
| **In progress** | Work is underway |
| **Waiting for customer** | Work is waiting for the customer |
| **Completed** | Closed and locked |
| **Cancelled** | Closed; not shown in the field app |

## Notifications

The **Notifications** tab in the bottom bar opens `/varsler`, with **Unread only**, **Mark all as read**, **Mark as read** and **Open**. The bell at the top shows the unread count. In the menu you can turn notification sound on or off with **Turn on notification sound** / **Turn off notification sound**.

## No jobs?

If it says **No active work orders in your calendar right now**:

1. Select **Reset** if a search is active.
2. Select the refresh icon at the top.
3. Check **Available** – there may be work orders you can take.

4. Check that you are in the right department; department scoping hides other departments' work orders.
5. Contact the office if the job is missing a technician or has the wrong date.

# Work order details

Open a card to reach `/mine-oppdrag/<work-order-id>`. At the top is a dark job panel with status chip, title, customer, address, department, order number, planned date, assigned technicians, contact person and checklist count.

## Customer, contact, address and map

**Quick actions** give you **Call contact**, **Map**, **Customer**, **Equipment** and **Documents**. **Call contact** uses the work order contact first, then the customer contact or main phone. **Map** opens the device's map app: Apple Maps on iPhone/iPad, Google Maps or the chosen map app on Android, and Google Maps in the browser on a desktop. It uses coordinates when available, otherwise a URL-encoded address, and is hidden when both are missing.

**Copy address** puts the full address on the clipboard. **Location and contact** shows the full address, company phone, the work order contact and the customer's other contacts with phone and email. If data is missing, it says **Not provided** or **No specific contact person is registered on this work order or customer.**

## Today's session: en route, on site, done

The session card shows **Today's session** or **Next session** with date, status and technicians. Each calendar day is one session; a multi-day work order continues on the next day even when today is done. The flow is **En route → On site → Done**. Each tap records the time on the server. Tap only when you actually start driving, arrive or finish.

After **Done** the card shows **Finish travel** until the return travel is finished. Then the day is locked and it says **The day is finished.** The next day's session can be started with **En route** even while an earlier day is still waiting for **Finish travel**; the previous travel segment is then closed automatically.

The card shows today's **Travel to**, **Work**, **Travel from** and **Total** time, plus total time for the whole work order with the breakdown. **Days** lists every workday in the work order with status and time, so you can see which days are completed and which remain. A completed day that still needs **Finish travel** has the button in the day list.

With live tracking enabled, you can share your position while **En route**: **Share position** / **Stop sharing**. The position is deleted after 24 hours.

## Status, work description and comments

**Work order status** lets you choose a new status and select **Update**. **Complete** requires confirmation: “Complete this work order? The work order and report are locked for editing.” Cancellation is under **More actions** for roles with edit permission. Completed work orders are locked; reopening is done by an administrator.

**Work description / Instructions** is shown as Markdown. **Comments and updates** is a Markdown log: **Write comment → Add comment**.

### Attachments, personnel and history

**Attachments** handles photos and PDFs via **Add files**. Files upload one at a time; if one fails, its name is shown and you can try again.

**Assigned personnel and office** show who is on the work order. **Remove me** removes you, with **Undo** in the toast. **Add a colleague or office for follow-up** is used when someone else needs to see or take over.

**Activity log** shows status changes, assignments, comments, reports and locking. Service history lives on the individual unit; see “Units at the top of each checklist”.

### Checklists and reports

**Checklists for the work order** lists every template, with HMS type, **Report no.** when one exists, **Locked**/**Unlocked**, **Fill in form** →, **Edit** when unlocked for correction, and **View report (PDF)** when a report exists.

The progress ring shows answered checkpoints, deviations and draft status. **Do you need an extra form?** adds a template you have access to; otherwise the office must do it.

If it says **No checklists are linked to this work order yet**, you cannot fill in anything until the office adds a template.

## The report form

The form opens at `/mine-oppdrag/<work-order-id>/skjema/<template-id>`.




*Figure: The report form is built for one hand and an unstable network.*

Notice the step strip at the top and the unit blocks. Each checkpoint belongs to a unit, and the signature sits at the bottom. Draft status is always visible, so you know whether what you entered is stored locally or on the server.

The fixed step strip shows **Checkpoints**, **Photos** and **Sign**, plus draft status and **Save draft now**.

## Locked report and draft restore

If the report is submitted and locked, the page shows **The report is locked**. You can open **View the report (PDF)** or go back. Only an administrator can unlock it for correction.

If a saved draft exists, the form opens with the fields as you left them, and you see "📝 A saved draft has been restored with all fields as you left them." A submitted report is never overwritten by a local draft.

## Units at the top of each checklist

Each unit is a block; details sit at the top:

- **Get from the customer equipment** – choose a registered unit; designation, serial number and location are filled in.
- **New unit (not registered before)** – when the equipment is not on the customer record.
- **Designation** * – required; enter what you see, for example "North gate".
- **Serial number** and **Location** – optional, but important for history.

A linked unit shows 🔗 **Linked to a registered unit on the customer record**. A new unit with a name shows ＋ **Saved as a new unit for the customer**. On submission, the unit is created or reused on the customer.

**Each registered unit can only be used once in the same report.** A duplicate choice is rejected with "This unit is already selected in another unit row. Choose a different unit."

**Add unit** adds a row; **Remove unit** confirms and deletes its checkpoints. Use **New unit** only for new equipment, not to bypass the duplicate check.

Units provide service history, the customer's equipment list, the office overview and the basis for QR labels.

## Checkpoints

Each checkpoint comes from the template:

- **Status choice** – OK / Deviation / Correction needed / Not applicable. A follow-up choice opens a required comment: "💬 Comment for "…" – required".
- **Checkbox** – must be checked when required.
- **Free text / text area** – the text area supports Markdown.
- **Number, date and dropdown** – ordinary required fields.
- **Photo** – **Take photo** / **Choose photo**. Compressed and uploaded immediately; offline they stay **Ready to upload** or **Upload failed** and retry when online.
- **Signature** – a text field where you enter your full name.

Fields marked `*` need a real value; a checkbox must be checked and a photo field needs an uploaded photo.

**Conditional fields** appear only when another field has a specific value. If you change the source, the field may disappear. Answer every visible required field; hidden fields are not required.

Follow-up shows “One checkpoint requires follow-up” or “{count} checkpoints require follow-up”. If the unit is still **Approved / done**, the app asks you to confirm, and you can select **Set the result to “Deviation found”**.

Photos are evidence: photograph the deviation, meter, damage or completed correction.

### Signatures

**Signatures** at the bottom are optional – technician, customer or both. Draw with a finger or mouse, use **Full screen**, and **Clear** to start again. **Technician** is pre-filled with your name; **Customer** with the work order contact.

An earlier signature is kept if you do not sign again: **Kept if you do not sign again.** If you enter a name, it must be one of the work order's contact persons or technicians. Otherwise submission is rejected with “The signer name must be one of the work order's contact persons or technicians.”

### Markdown in notes and comments

Free-text fields, unit comments, general notes and work order comments use Markdown. The toolbar has bold, italic, strikethrough, highlight, heading, lists, task list, quote, callout, code, link, table, footnote and hidden comment. **Preview** shows the result.

Markdown is sanitised on the server and rendered the same way in the browser and the PDF. Use **General notes for the job** for the overall picture, and **Comment for this unit** for what applies to one unit.

### Mobile swipe-to-complete

On mobile you can approve a unit with a swipe:

1. Swipe **right** on the unit header. You see a percentage and **keep swiping**.
2. At the threshold it says **Release to complete**.
3. Release. The unit is set to **Approved / done**.

The swipe requires 84 pixels; vertical movement scrolls and cancels. The fallback **Complete unit** button is at least 48 pixels high and does the same.

After approval you get **Undo** for three seconds: “Unit N is approved. Undo is available for 3 seconds.” **Undo** restores exactly the previous status.

Swipe and button are blocked when the unit is **Approved**, set to **Deviation found**, missing **Designation**, missing a required answer, photo or checkbox, or has a status choice that requires follow-up (“Cannot complete: “…” is set to “…”. Use the Approved/Deviation options.”).

Therefore a quick gesture cannot approve an incomplete or deviating unit. The swipe uses the same validation as the form and the server.

## Submit and sign

### Review

**Sign and submit report** opens **Sign and submit the report?** with **Units: N**, **Deviations: N**, **Photos: N**, **Signatures: technician … , customer …**, and **When you submit, the report is locked and cannot be changed unless an administrator unlocks it.**

Clear **Mark the work order as completed** if the work order should stay open. It is checked by default and sets the work order to **Completed** on submission.

Select **Sign and submit report**, or **Back to the form** to correct something.

### Submission requirements

Submit when online, the template belongs to the work order, you have `reports.write` in the right department, and the report is not locked or under legal hold. When the template requires units, the report must have at least one unit, and every unit must have a **Designation** and be assessed as **Approved / done** or **Deviation found**.

Answer every visible required checkpoint; upload required photos and check required boxes. The same registered unit cannot appear in several rows. Cancelled work orders cannot be completed by technicians. Any signer names must be people linked to the work order.

A signature is optional; the report can be submitted without one. If an administrator has unlocked the report, it locks again automatically when you save.

### After submission

The server confirms with **The report has been submitted and locked**, **Open report (PDF)**, **Open work order** and, in the field app, **Next assignment**.

The report is locked, local draft and outbox entries are cleared, and the form page leaves the offline cache. The office sees the report, the customer can see the submitted report and acknowledge it, and the units update service history. If the report has deviations, one undated remediation work order is created with you as technician.

If **Mark the work order as completed** is checked, the work order is set to **Completed** in the same transaction. Correction requires an administrator to unlock the report with a reason. A new submission creates a revision and resets any customer acknowledgement.

### PDF in the app or as a download

**View report (PDF)** opens the report in the app's own PDF viewer. There you can scroll, zoom and close without leaving the work order. **Download PDF** creates a file you can send or save. A direct link to the PDF route always downloads, because only the app's own viewer asks for inline display.

The PDF is sanitised before it is shown. If it contains active content that cannot be removed safely, the viewer shows an error with **Try again** and download instead of opening anything unsafe. The report data and the lock are unchanged. If an image or signature is missing, the PDF shows a warning with the number of images that could not be loaded; re-upload the image from the report and generate the PDF again.

Open any PDFs you need while online. PDF viewing requires a connection, and a direct download should be made before you lose coverage.

## Mileage, expenses and spare parts

### Record mileage and expenses

1. Open **More** → **Expenses** ( `/utlegg` ).

2. Choose **Mileage** or **Expense**.
3. For mileage: fill in **Date**, **From**, **To**, **Kilometres** and optional **Work order**. A rate is suggested but stored per row.
4. For an expense: choose **Category**, enter the **Gross amount**, check **VAT** and upload the **Receipt**.
5. Save as **Draft**, or submit for approval.
6. Follow the status: **Pending**, **Approved**, **Rejected** or **Exported**. A rejection shows the office's reason.

You see only your own mileage and expenses. You cannot approve your own rows; the office does that. Amounts and VAT are calculated on the server, and a final amount from the client is rejected if it differs. The receipt is the evidence – upload it while you have it in front of you, not the next day.

### Spare parts on the work order

If you have inventory access, you can reserve and consume parts from `/lager/oppdrag/<id>`:

1. Choose item and location, and enter the quantity.
2. Select **Reserve** before using the part.
3. Select **Consume** when the part is fitted. It is then deducted from stock.
4. If you do not use the part, select **Release**.

Consumption does not happen automatically when you complete the visit. Make sure the reservation is consumed or released before you leave the job.

## Offline, drafts and the outbox

### Autosave

The form saves locally about 200 ms after a change, on the server after about 1.2 seconds, when the tab is hidden or closed, and on **Save draft now**.

The local draft contains units, answers, conditions, notes, names and signatures, and survives network loss, tab closure and restart. The server draft follows the user to another device. If the local draft is newer, it is restored when the form opens.

Drafts are per user, work order and template, and expire after seven days.

### The offline banner

The banner at the top is visible when you are offline or have pending drafts:

- 📴 **You are offline – drafts are stored on this device and sent when the app is open and the network returns**
- 📴 **Offline · {count} draft(s) waiting**
- 📤 **{count} draft(s) waiting to sync**
- ⚠️ **{count} draft(s) could not be sent**
- 🔄 **Syncing drafts …**
- ✓ **{count} sent**

**Open the draft list** goes to `/offline.html` with stored pages, drafts, failed sends and **Discard my drafts**.

### Outbox and background sync

Only the newest draft per report is kept. The outbox is the browser's local send queue, the payload is the content sent to the server, and backoff is the waiting time between retries.

- **Chromium** can use Background Sync: the browser wakes the service worker and sends even without an open tab.
- **Safari/Firefox** do not support Background Sync. Drafts are sent when the app is open and the network returns, when the tab becomes visible, or by the 30-second timer. Keep the app open until ✓ **sent**.

Backoff runs 15 seconds to 30 minutes, maximum 10 attempts, then terminal **could not be sent**. Payloads over 2 MB are rejected; 400/413 are terminal; 429/5xx stop the queue and back off.

### 401/403 and expired sessions

A 401/403 means the session or access is gone, not that the draft is broken. The service worker stops replay, page caches and the offline marker are cleared, and the draft/outbox entry is kept. Sign in again, then select **Send now** or save the form. A 403 can also mean lost permission; the office must restore access.

**Discard my drafts** deletes all local drafts and pending uploads for the user on the device. Use it only when sure, and always on a shared device before leaving.

## Safety and quality in the field

- **Swipe or “Complete unit” is only for fully documented, non-deviating units.** Use **Deviation found** when something is wrong, missing or needs correction.
- **Never approve something you intend to fix later.** “Approved” means checked and in order now; correction belongs in the deviation or a separate work order.
- **Describe the deviation concretely:** what, where, measurement and what must be done. Attach a photo when relevant.
- **Use the recorded timestamps.** Start driving, on site and done are recorded when you tap; do not tidy them up afterwards.
- **Check unit identity.** Correct designation, serial number and location make the history useful; one registered unit is used once per report.
- **Only sign what you have reviewed.** A signature is optional, but the name must belong to a person on the work order.
- **The report locks on submission.** Corrections require an administrator unlock and are logged as a revision.
- **Legal hold stops everything.** A frozen report cannot be submitted or changed.
- **Do not share your sign-in.** Drafts and the page cache are per user; sign out and discard drafts on a shared device.
- **Do not invent time, distance or data. Time not set** means the time is not set.

# Common field problems

| Problem | Likely cause | Fix |
|---|---|---|
| **No active work orders** | Search, wrong department, no assignment, or no jobs | Reset search, refresh, check **Available**, ask the office |
| **The report is locked** | The report is submitted and locked | Open the PDF or ask an administrator to unlock it |
| **No checklists** | The template has not been added | Ask the office to add it. If you have access, use **Do you need an extra form?** |
| **Upload failed** | Network, large file, camera/storage | Check network, **Try again**, or use **Choose photo** |
| **Draft waiting / sync stuck** | Offline, expired session, 429/5xx or terminal | Get online, keep the app open, **Send now**. On 401/403, sign in. On **could not be sent**, **Try again** or discard |
| **Wrong or duplicate unit** | Wrong row or duplicate registered unit | **Remove unit**, add the right one. For duplicates, choose another or **New unit** |
| **Cannot submit** | Missing unit, assessment, required answer, photo, checkbox or name; locked; offline; no checklist | Read the message, correct the field, try again. **Missing assessment** means **Approved** or **Deviation** per unit |
| **Cannot complete with swipe** | No name, deviation, or missing required field | Fill in the field, upload the photo, or use **Deviation found** |
| **2FA code fails** | Wrong code, clock drift, used code | Wait for a new code, check the clock, use an unused recovery code, or ask for a reset |
| **Session buttons missing / status locked** | Day or work order closed, or no permission | Contact the office; only open days and work orders can be changed |
| **PDF or offline page missing** | No network, report, access, cache, or active content that cannot be sanitised | Get online, open **View report (PDF)**, download the file, reopen the job online |
| **Image missing from the PDF** | File deleted, missing metadata, or owned by another work order | Re-upload the image from the report and generate the PDF again |
| **No push notifications** | Notifications denied, device unregistered, or provider not configured | Allow notifications in the app, sign in again, and ask an administrator to check FCM/APNs |
| **Push opens the wrong page** | Old version or invalid deep link | Update the app and open the notification again from **Notifications** |
| **Cannot record an expense** | Missing receipt, invalid amount or period | Upload the receipt, check VAT and try again |

## How technician work connects to the rest

| Area | What you do in the field | Where others see it |
|---|---|---|
| **Office** | Update status, comment, submit reports | `/oppdrag`, `/rapporter`, dashboard, notifications. The office assigns, reschedules and adds templates; an administrator unlocks reports. |
| **Customer portal** | Sign and submit a report | The customer sees submitted, locked reports in `/kundeportal/rapporter`; PDF download and **Acknowledge receipt** |
| **Equipment module** | Choose or create a unit, record serial and location | Service history, timeline, readings, QR label on `/enheter/<id>` and **Equipment** tab |
| **Administrator** | Follow template requirements, use right department and user | Manages **Checklists / Templates**, **Employees**, departments, permissions, 2FA and unlocks in **Settings** |
| **Inventory** | Reserves and consumes parts on the work order | Stock, movements and reservations at `/lager` and on the item |
| **Expenses** | Records mileage and expenses with receipts | The office approves and exports the payroll CSV |

PART 3

# Administrator

Applies to FSM Platform **0.8.11** (2026-11-08). This chapter is for system administrators who configure the platform, manage users and permissions, set up integrations, and are responsible for data, security and operations. It covers configuration and governance. Day-to-day workflows for office, technician and customer portal users are described in their own chapters.

## 1. The administrator role and delegation

Administrators always have every permission, including deletion, unlocking submitted reports, legal hold, roles, departments and API keys. The Office role is an operations role with broad access inside its own department, but without the most security-sensitive actions.

| Task | Administrator | Office | Recommendation |
|---|---|---|---|
| Roles, departments and API keys | Yes | No | Keep with administrators. |
| Administrators, deletion and moves | Yes | No | Keep with administrators. |
| 2FA, report unlocking and legal hold | Yes | No | Keep with administrators. |
| Activity log | Yes | No | Grant `audit.view` deliberately. |
| Employees in own department | Yes, all | Yes, own department | Delegate to the department lead. |
| Customers, work orders, agreements, templates and reports | Yes | Yes | Delegate to office and subject-matter owners. |

Office has `settings.manage`, `users.manage` and `dispatch.configure` by default, so it can change branding, map, SMS and dispatch settings. If that is too broad, create a separate role, for example **Service manager**, without `settings.manage`. Changes to the built-in Office role affect everyone who has it.

**Least privilege:** grant the weakest role that solves the task, use dedicated roles for leads and integrations, and keep at least two active administrators.

## 2. Departments and department scoping

Work orders, service agreements and employees belong to a department. Employees only see work orders in their own department. Customers and documents are shared; reports are department-scoped unless the user has `reports.viewAllDepartments`.

**Create a department:**

1. Go to **Settings** → **Departments**.
2. Enter **Name** and optional contact details.

3. Leave **Enable live tracking** off until privacy has been assessed.
4. Select **Create department**.

**Expected result:** the department appears with counts for employees, work orders and agreements. One department is the **Default** and is used when a work order or booking does not specify another. The default department cannot be deactivated until another department is set as default.

**Warning:** deactivating a department signs out all employees in it, clears calendar subscriptions and makes its work orders unavailable, because work-order queries require an active department. Accounts are kept and the department can be reactivated. Notify users in advance.

## Modules

The platform has a core and a catalogue of modules. The core (customers, work orders, workdays, time, revisions and settings) can run without optional modules. Modules are switched on and off under **Settings → Modules**. The module guard returns `404` for pages and APIs when the module is off, and removes navigation, settings, search hits, export/import and background jobs for that area. The data is kept.

| Module | Key | Depends on | Provides |
|---|---|---|---|
| **Quotes** | `quotes` | Customers, work orders | Quotes, versions, approval link |
| **Invoicing** | `invoicing` | Customers, work orders | Invoice, payment, credit note, accounting sync |
| **Inventory** | `inventory` | Equipment | Items, stock, reservations, purchasing |
| **Expenses** | `fieldExpenses` | Calendar | Mileage, expenses and payroll CSV |
| **Profitability** | `bi` | Reports | Estimated and actual revenue |
| **Customer portal** | `portal` | Customers | The customer's own surface |
| **API** | `api` | – | REST API under `/api/v1` |

Existing installations get every delivered module enabled. `MODULE_PRESET` can be used at startup to choose a predefined set. If you turn off a module that others depend on, validation stops and you get a clear error. Every module change is audit-logged.

> **Note:** Turning a module off hides the feature but does not delete data. An invoice, stock item or expense row remains and returns when the module is switched on again.

## 3. Users, invitations and sessions

**Create a user:** go to **Settings → Employees & Roles** or `/brukere`, select **Add new user**, enter **Full name**, **Email address** and optional **Phone**, choose **Role / Access level** and **Department**, then select **Create and generate invitation**. Employees need a department to see and create work orders.

The user is created inactive. You either get confirmation that the invitation email was sent, or an **Invitation link** you can send manually. The link is shown only in the page content, is valid for seven days, and can only

be used by an inactive account. Without `SMTP_HOST` and `SMTP_USER`, the app cannot send email. Active users must use **Forgot password**.

**Activate the account:** the user opens the link and chooses a password. The account is still inactive. Find the user under **Employees & Roles** and select **Activate**. Only then can the user sign in. Ask the user to set up 2FA under **Settings → Security**.

**Deactivate and delete: Deactivate** signs the user out, clears the calendar subscription and blocks sign-in and API access. **Activate** restores access. **Delete user** is administrator-only and cannot be used on yourself or the last active administrator.

**Passwords and 2FA:** administrators cannot see or set a password. **Forgot password** sends a one-time link valid for one hour, rate-limited per email and IP. **Reset 2FA** disables two-factor authentication, deletes recovery codes and signs out all sessions. Use it only when a device is lost or misuse is suspected.

**Sessions:** there is no session list or “sign out everywhere” button. Sessions are invalidated when the password changes, 2FA changes or is reset, the user is deactivated, or the user’s role, permissions or department change. Editing a role definition applies on the next request without signing users out.

## Change password, 2FA and recovery codes

1. To change your password, sign out and use **Forgot password** on the sign-in page. 0.8.11 has no separate change-password page. The link is valid for one hour, and all other sessions are invalidated when the new password is set.
2. Go to **Settings → Security** to turn on **Two-factor authentication**. Scan the QR code with your authenticator app and enter the 6-digit code.
3. Store the **Recovery codes** in a safe place. They are shown only once, and each code can be used once. **Create new codes** invalidates the old ones.
4. Use **Log out** on shared devices. If misuse is suspected, change the password via **Forgot password** and reset 2FA.

An administrator can reset a user’s 2FA but cannot see the password. A reset deletes the recovery codes and signs out all sessions.

## SSO and SCIM

SSO is configured with `SSO_PROVIDERS` and the `SSO_<SLUG>_*` variables, not in the interface. Entra ID, Keycloak, Authentik and Zitadel are supported with OIDC, PKCE, state/nonce and verified email. The redirect URL is `https://<NEXTAUTH_URL>/api/auth/callback/sso-<slug>`. After a restart, **Settings → SSO and SCIM** shows the provider status, issuer, default role and allowed domains. Password sign-in always remains as break-glass.

SAML 2.0 is additive and uses `SAML_PROVIDERS` and `SAML_<SLUG>_*`; OIDC stays untouched, so one installation can run both. Register the SP entity ID and ACS URL from `/api/auth/saml/<slug>/metadata` in the IdP. The server validates the signed assertion, audience, recipient and time window, and rejects assertion replay. `ALLOW_IDP_INITIATED=true` is only for IdP-initiated sign-in; signature, audience, recipient and replay checks remain active. SAML uses the same account, role, department and 2FA rules as OIDC.

SCIM 2.0 is served at `/api/scim/v2` with a dedicated Bearer token created, rotated and revoked in the same tab. `DELETE` deactivates the user without deleting it, and `ADMIN` requires both an explicit group mapping and `SCIM_ALLOW_ADMIN=true`. **SCIM Groups** at `/api/scim/v2/Groups` mirror group membership.

A group only affects role or department through `SCIM_GROUP_ROLE_MAP` / `SCIM_GROUP_DEPARTMENT_MAP`; without a mapping nothing happens, and conflicting mappings change nothing. Use one token per integration, and keep at least two local administrators with strong passwords and 2FA.

## 4. Roles and permissions

| Role | Key | Default content |
|---|---|---|
| **Administrator** | `ADMIN` | All permissions; the list cannot be changed. |
| **Office** | `OFFICE` | Work orders, customers, agreements, equipment, reports, documents, templates, employees and settings in its own department. |
| **Technician** | `TECHNICIAN` | View and create work orders, update status, view customers, agreements and equipment, fill in reports, view documents and templates. |
| **User** | `USER` | No permissions by default. |
| **Customer** | `CUSTOMER` | The customer portal, controlled by the customer link. |

Administrator and Customer are locked. Office and Technician can be edited, but system roles cannot be deleted.

**Custom role:** go to **Settings** → **Roles & permissions** → **New role**, enter **Name**, optional **Description**, select **Permissions** and choose **Create role**. The role gets a fixed key based on its name; the key does not change if the display name changes. A role in use cannot be deleted until its users are moved.

**Per user:** open **Permissions for this user**, select the list and choose **Save permissions**. The list overrides the role completely; if it matches the role's list, the override is removed. **Use the role's permissions** does the same. Customisation does not apply to administrators or portal users, and changing a role clears the list.

**Field-level access:** sensitive fields are masked on the server before they reach the API, search, exports or the UI. ADMIN and OFFICE keep full access. Technicians do not see customer contact details or invoice amounts, and only their own payroll and expense amounts. The customer portal sees only its own contact and billing data. An administrator can grant one field group explicitly with `fields.customerContact`, `fields.billingAmounts`, `fields.reportSignatures` or `fields.payrollAmounts` in a user's permissions; in v1 this is done directly in the database/API because the settings list filters unknown keys. This never grants ADMIN. See `docs/RBAC.md`.

The permission groups are **Work orders** (`workorders.*`, `dispatch.configure`), **Customers and equipment** (`customers.*`, `equipment.*`), **Agreements and SLA** (`contracts.*`, `sla.manage`), **Reports and documents** (`reports.*`, `documents.*`) and **Administration** (`templates.*`, `users.*`, `settings.manage`, `audit.view`).

**Test a role:** use a dedicated account, try a direct URL it should not have (for example `/innstillinger?tab=roller`) and one mutation, confirm both are rejected, then deactivate. Authorisation is enforced on the server; hidden buttons are not security.

**Common mistakes:** giving a broad role to everyone, forgetting a per-user override, deleting a system role or a role in use, and giving Office `settings.manage` when it is not intended.

# 5. Customers and customer portal users

Customers are created under **Customers** → **New customer**. The customer card has **Overview**, **Work orders**, **Agreements**, **Equipment** and **More**. Portal users are created under **More** → **Portal users**.

**Create a portal user:**

1. Create the customer and at least one **Contact** with an email address.
2. Go to **More** → **Portal users**.
3. Enter **Name**, **Email** and choose **Contact**.
4. Select **Create and send invitation**.
5. Send the link manually if the email fails.
6. Select **Activate** after the user has chosen a password.

The user receives the `CUSTOMER` role, a customer link and a contact link, and remains inactive until you activate the account. **New link** only works for inactive accounts.

The portal shows the customer's own overview, bookings, reports, documents, equipment, agreements, visits, messages and contact details. Reports appear only when submitted and locked; documents only when shared with the customer (`CUSTOMER`-owned).

Portal users never see other customers, internal notes, internal permissions, drafts or unlocked reports. `/uploads/...` is checked per file, and PDF responses are `no-store`.

**Security:** use separate accounts per person. **Deactivate** when someone leaves; this signs them out, clears the calendar subscription and revokes portal access. Deleting the customer deactivates all portal users, invalidates sessions and closes booking. After restoring, each portal user must be activated again. A portal user must always have a valid contact on the same customer.

**Delegation:** the portal has server-enforced capabilities for booking, documents, agreements and equipment, but 0.8.11 has no form for editing them. A contact without a delegation row gets full portal access; restricted access must be configured and tested with the actual account.

# 6. Checklists and templates

**Create a template:**

1. Go to **Settings** → **Checklists / Templates**.
2. Select **+ Create**, enter **Checklist name** and choose **Category**.
3. Select **Edit**, add fields under **Add a new checkpoint**, and set **Required field** where needed.
4. Use **Move up**/**Move down**, then select **Save template with logic**.

Field types are **Short text**, **Long text** (Markdown), **Checkbox**, **Dropdown**, **Condition + comment** (OK/Deviation/To be fixed/Not applicable), **Number**, **Signature**, **Image upload** and **Date**.

**Required and conditional:** required fields must be answered before submission. **Condition (If This Then That)** shows a field only when another field has a specific value, for example `Deviation`. **A comment is required for these options** applies to condition fields; the default is `Deviation` and `To be fixed`. Clear it to never require one.

**Units and report name: Link the report to equipment units** is on by default and gives one assessment per unit. Turn it off for reports that concern the whole work order. **Report name** appears in the PDF and report lists; an empty field uses the template name.

**Category and HMS: Service** or **HMS form**. HMS requires an **HMS type**: general, SJA, hot work, internal control, deviation, safety round or written instruction. Category and HMS type are frozen at submission.

**Linking and versioning:** templates can be linked to work orders, service agreements and maintenance plans. At submission, a snapshot of the template, report name, unit requirement, category and HMS type is stored. Later changes do not affect existing reports, even when unlocked; new fields apply only to new reports.

**Delete a template:** administrator only. A template cannot be deleted while it is used by reports, work orders or active agreements. A blocked attempt is logged.

### HMS library and QR labels

The **HMS library** is at `/hms`. It groups HMS forms by type and lists templates and completed HMS documentation. **Link HMS form to work order** chooses a form and an active work order; the form is added to the work order and completed there. HMS templates are managed under **Settings** → **Checklists / Templates** with category **HMS form**.

**QR label:** open the unit at `/enheter/<id>`, go to **QR label** and use **Rotate QR token** if the label is lost or shared. Print the label from `/enheter/<id>/etikett`. A rotated token invalidates the old label.

## 7. Work orders and reports

The statuses are **Open / New** (`OPEN`), **Waiting for planning** (`PENDING`, the system start status), **Planned** (`PLANNED`), **Waiting for customer** (`WAITING`), **In progress** (`IN_PROGRESS`), **Completed** (`COMPLETED`) and **Cancelled** (`CANCELLED`).

Office and technicians with `workorders.update` can change open statuses freely. Completed and cancelled are closed; only an administrator can reopen them, with a reason. Cancellation requires `workorders.edit`. The report has its own lock and is not affected by the work order status.

**Unlock a report:** open the work order, find the checklist, select **Unlock for correction**, enter **Reason for unlocking** and confirm. The report becomes **Unlocked**, can be edited, and a revision of the previous version is stored. The customer’s acknowledgement is withdrawn. On resubmission, the report is locked automatically and gets a new revision. Legal hold blocks unlocking.

**Revisions:** every unlock and submission creates a revision that freezes the form, units, comments, signatures, report number, category, HMS type and acknowledgement. **Revisions** shows the content and lets you download it as a PDF.

**Report numbers:** HMS reports receive an automatic `abc1234` number (three lowercase letters and four digits) on first submission. It is kept across revisions. Ordinary service reports have no automatic number.

**PDF and branding:** the PDF uses the company name, logo and primary colour from **Branding**, is stored as a `WORK_ORDER` upload and requires `reports.view`. The response is `no-store`. Branding is part of the PDF hash, so changes apply next time the PDF is generated. Report PDF labels are Norwegian in 0.8.11;

agreement PDFs support all four languages. The generated remediation work order title is also Norwegian (“Utbedring av avvik – …”) in 0.8.11.

**Legal hold and retention:** legal hold prevents hard deletion and freezes the report. On a work order, all reports inherit the flag. Releasing it requires a reason and does not release the reports’ own flags; **Report hold** must be released per report with its own reason. Soft-deleted work orders can be hard-deleted after `RETENTION_DAYS` (default 365) only when they have no reports. **Recycle bin** and **Restore** are administrator-only.

## 8. Service agreements, SLA and price lists

**SLA policy:** go to **Settings → SLA and business hours**, enter **Name**, **Department** or **All departments**, **Response time (minutes)**, **Resolution time (minutes)** and **Count business hours only**, then select **Create policy**. A clock starts only when the work order has an active, date-valid agreement whose SLA policy is active. There is no fallback to a department default or a global policy, so a customer without an active agreement gets no SLA clock. The clock starts at creation, response is recorded on `EN_ROUTE` / `ON_SITE`, resolution on `DONE`, and `CANCELLED` cancels it. Travel from (`travelFromEndedAt`) never moves the clock. States: `ON_TRACK`, `AT_RISK`, `BREACHED`, `MET`, `CANCELLED`.

**Business hours and holidays:** choose **Department**, select **Show**, set **Opens**/**Closes** or **Closed** per weekday, and select **Save business hours**. Add **Holidays** with date, name and optional department. Global holidays apply to all; department-specific rows override only that department. Without business hours, weekdays 08–16 in Europe/Oslo are used. All deadlines are calculated through this calendar.

**Service agreement:** create it from the customer card, **Agreements** tab, or via **Agreements**. Enter **Title**, **Start date**, **Interval** (3, 6, 12 or 24 months), optional **Contact**, **Checklist** and **Department**, then select **Create service agreement**. The agreement becomes **Active** (or **Draft** if selected) with a next service date. A start date today creates the first work order immediately; a past date uses the next future interval. Each interval creates a work order with customer data, a checklist and the previous report as a PDF.

**Coverage:** an agreement without lines covers all equipment. A line without a product also covers all; a line with a product covers only that product. A line can have **Coverage**, **Visits/year**, **Response (min)** and **Resolution (min)**.

**Renewal: Automatic** renews after the end date and generates a new work order. **Manual** becomes **Expired** after the end date. **Notice (days)** controls the expiry notice. Automatic renewal runs exactly once even with multiple instances.

**Price lists:** create them under **SLA and business hours → Price lists** with **Code**, **Description**, **Unit** and **Amount**. They are reference data for agreement PDFs and settings, and for the hourly rate in profitability. Invoicing happens only when `invoicing` is on, and the price list does not automatically determine an invoice amount.

## 9. Booking, SMS, tracking and reviews

**Booking:** the customer chooses a requested date (up to 60 days ahead) and writes the desired time in the description; the customer cannot select a time block. The request creates a new work order with status **Open/New** and no visit or confirmed time. The office or technician plans it and sets **Planned**, or **Waiting** if

the time does not fit. **Bookings** shows unconfirmed requests; they can be declined, but planning happens on the work order.

**SMS:** the provider is selected with `SMS_PROVIDER`, not in the UI. The default `console` is a dry run; `linkmobility`, `pswincom` and `twilio` are supported. **Sender** comes from `SMS_SENDER_ID`. Under **Settings → SMS and messaging** you can see the provider, sender, dry-run state and **Webhook signature**, and set **Default language**, **Opt-out keywords**, **Opt-in keywords**, **Monthly cap** and **Templates per event**.

SMS is sent only when the recipient has `optedIn = true` and `optedOut = false`. STOP/START is enforced from the keywords, and consent is checked when queuing and before sending. `SMS_ALLOWLIST` can restrict test and staging. The webhook at `/api/webhooks/sms/<provider>` requires an HMAC or Twilio signature; without `SMS_WEBHOOK_SECRET`, deliveries are rejected (except console in development). Outgoing SMS is a queue with three attempts; provider delivery status updates **Sent**, **Delivered** or **Failed** in the same conversation without creating a new message. Conversations appear under **Messages**.

**Reminders:** under **Settings → SMS → Automatic reminders** you create rules per department. A rule has a **Channel** (SMS, email or both), **Hours before visit** (for example 24 or 2), **SMS template**, **Quiet hours**, **Time zone** and **Maximum reminders per visit**. The worker sweeps every five minutes, skips customers without consent, respects quiet hours, and cancels the reminder if the visit is moved, cancelled or completed. **Recent reminders** shows planned, sent and skipped rows with a reason — without message text or phone numbers. Sending happens in the background, so a slow SMTP or SMS provider never blocks a page.

**Tracking:** off by default per department. Enable it under **Departments → Edit department → Enable live tracking** after a DPIA, because the technician's location is personal data. An assigned technician can share location while the visit is `EN_ROUTE`. The public link shows only first name, status, ETA, planned time and last updated — never coordinates, a map or history. Pings are kept for 24 hours and deleted on `DONE` / `CANCELLED`; the link expires two hours after the visit, and there is one active link per visit. `TRACKING_AVERAGE_SPEED_KMH` (default 50) controls the fallback estimate. With `OSRM_BASE_URL`, an external routing provider is used for ETA; without it, ETA is calculated locally. A ping older than ten minutes is shown as **last known position**, not as live. The public ETA endpoint is polled every 30 seconds and has its own per-token rate limit.

**Reviews:** internal and never published. The customer can request a review in the portal after a completed visit, or the office can send an SMS with "Request review". A review requires a completed visit, can be submitted once and is stored with a hashed token. Reviews are not shown as a separate dashboard in 0.8.11.

## 10. Branding, map, language and notifications

**Dispatch and workforce:** under **Settings → Dispatch**, configure **Assignment rules** (weights for skills, distance, travel time, availability, SLA and load) and **Territories**. Weights are relative. **Capacity plans** are shown under **Calendar → Capacity** but are not edited in **Settings** in 0.8.11. Skills, absence and working hours feed the suggestions but also have no separate form in **Settings**; keep the data updated outside the interface.

**Routing and overtime:** the route plan uses `OSRM_BASE_URL` for routes and the travel matrix, and `GOOGLE_ROUTES_API_KEY` for optional traffic-aware ETA on single legs. Without the key, ETA is a route and speed estimate, not live traffic; without either, it falls back to straight-line distance × 1.35. `ROUTE_ALLOW_OVERTIME` (default `true`) and `ROUTE_MAX_OVERTIME_MINUTES` (0 = no limit) control whether visits may be planned outside working hours. `ROUTING_PROVIDER=haversine` deliberately disables all

external route and traffic calls. Monitor Google quota and cost; cache and rate limit are per process. See `docs/ROUTING.md`.

**Branding:** under **Settings** → **Branding**, set **Company name**, **Primary colour (hex code)** and **Company logo (optional)**. The logo must be uploaded through the app; external URLs are rejected. It is used in reports, email and the dashboard header, and applies to office and customer portal, not the dark field app. Customers can have their own colour and logo in the portal.

**Map and geocoding:** under **Settings** → **Map**, choose a **Map provider**. **OpenStreetMap (no key)** is the default; **MapTiler** and **Mapbox** require an **API key**. The key is stored on the server and sent only to the map in the browser; use a domain-restricted key. Set **Default layer**, **Default zoom (2–19)**, **Center** and **Enable scroll wheel zoom**. Geocoding is on by default (`GEOCODE_ENABLED=true`) and tries OpenStreetMap/Nominatim, then Kartverket and Photon. Set `GEOCODE_USER_AGENT` to identify the installation, or disable with `GEOCODE_ENABLED=false`. The **Geocode addresses** button on the map reports whether no address was found, the service did not respond, or geocoding is disabled.

**Language:** Norwegian, English, Polish and German have complete catalogues. The top-bar selector stores the `NEXT_LOCALE` cookie and uses `router.refresh()`, without a URL prefix. Dates and numbers follow `nb-NO`, `en-GB`, `pl-PL` and `de-DE`. SMS uses the recipient contact's `locale`; email notifications use the user's `locale`. Reports have Norwegian PDF labels in 0.8.11; translations live in `messages/` and are checked with `npm run check:i18n`.

**Notifications:** in-app and email. Each user chooses events and channels under **Notifications** → **Settings**; there is no global default. Make sure SMTP is configured. The `Notification` queue retries five times with backoff, skips deactivated users and uses a lease (a time-limited lock) to prevent double sending. Types include assignment, date change, cancellation, report ready, booking, agreement, overdue, contact update, SLA and warranty.

**Messages: Messages** collects customer conversations. Threads are scoped to a department or customer; office users see their own departments, portal users only their own threads. Outgoing messages can be sent as SMS with consent; incoming SMS lands in the same thread and is idempotent per provider ID.

## 11. API keys and the REST API

The API is under `/api/v1` and uses Bearer keys prefixed `fsm_`. NextAuth cookies are never accepted.

**Create a key:** go to **Settings** → **API keys**, enter **Name** and optional **Expires (optional)**, select **Create key**, copy the key into a secret manager and select **I have saved the key**.

The key is shown only once. The server stores only a SHA-256 hash and a short display prefix. If you lose it, revoke it and create a new one.

A key inherits the owner's permissions, department and customer scope, and has explicit **scopes** in addition. A key can be marked read-only. New keys need at least one scope. A key with only read scopes gets `403` on write endpoints. The default lifetime is 90 days, maximum 365.

| Scope | Covers |
|---|---|
| `workorders:read` / `workorders:write` | Work orders and changes |
| `visits:read` / `visits:write` | Visits |

| Scope | Covers |
|---|---|
| `customers:read` | Customers |
| `contracts:read` / `contracts:write` | Service agreements and PDF |
| `sla:read` | SLA policies |
| `equipment:read` / `equipment:write` | Equipment and readings |
| `maintenance:read` / `maintenance:write` | Maintenance plans |
| `reports:read` | Reports and PDF |
| `documents:read` | Documents |
| `quotes:read` / `quotes:write` | Quotes and sending |
| `orders:read` / `orders:write` | External orders |
| `inventory:read` / `inventory:write` | Inventory and reservations |

**Quotes, external orders and inventory:** `/api/v1/quotes` creates and sends quotes; `/api/v1/orders` receives orders from an external order system; `/api/v1/inventory/**` reads items, locations and stock and creates, releases or consumes reservations. The order endpoint uses `source` + `externalId` as the idempotency key and accepts `Idempotency-Key` as a secondary key. A replay with the same content returns `200` with `Idempotent-Replay: true`; the same key with different content returns `409`. The API sends no webhooks back – the source system must poll.

**Important:** in the UI, the key is created for the signed-in administrator and therefore gets all administrator permissions. Treat it as an administrator secret. For lower privilege, create a dedicated active integration user with a narrow role and use `scripts/create-api-key.mjs` with `API_KEY_USER_EMAIL`, `API_KEY_NAME` and optionally `API_KEY_EXPIRES_AT`.

**Revoke** takes effect immediately and returns `401`. Expired keys also return `401`. The list shows **Owner**, **Created**, **Last used**, **Expires** and status **Active**, **Expired** or **Revoked**.

Each key can make **120 requests per minute**; exceeding the limit returns `429` with `Retry-After`. Pagination uses `page` and `limit` (maximum 100). OpenAPI 3.1 is at `/api/v1/openapi.json`; documentation with examples is in `docs/API.md`.

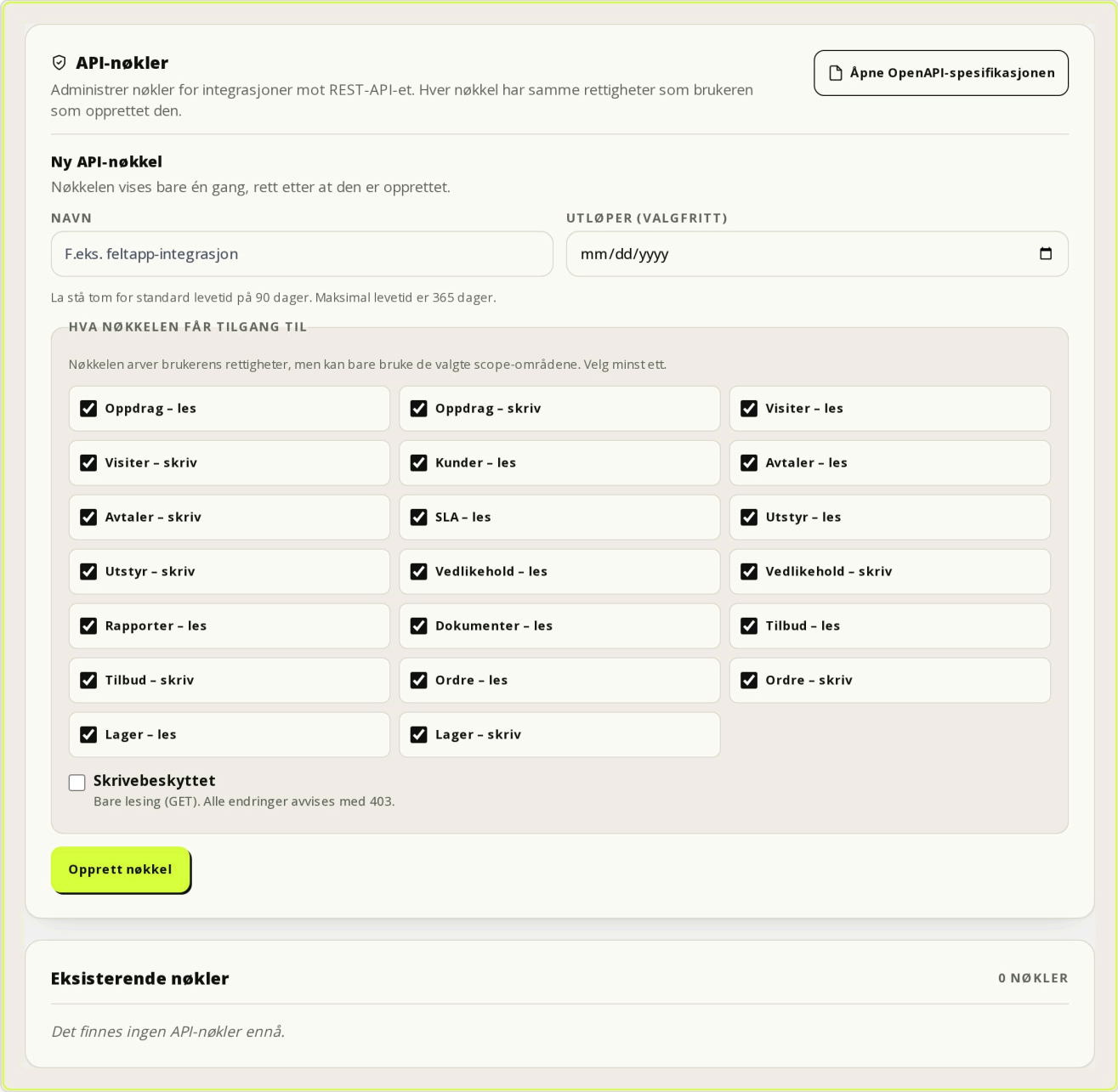



*Figure: The API overview shows keys, scopes and last used.*

Check **Scopes** and **Last used** before rotating or revoking. A key without scopes is an old key treated as "all scopes" until it is rotated. Use one key per integration, and confirm that read-only keys cannot use write endpoints.

**Safe use:** use HTTPS, one key per integration, an expiry date, a secret manager and **Last used**. To rotate: create a new key, deploy it, verify with `GET /me`, then revoke the old one. On suspected leakage, revoke first, create a new key and review the activity log.

## Automatic email and the outbox

Transactional email (quotes, invoices, payments, bookings and assignments) is sent from a durable outbox in the database, not directly from the request path. A slow or unavailable SMTP server can therefore never roll back an invoice, quote or booking, and every send has retry, history and idempotency.

| Event | Triggered by | Recipient |
|---|---|---|
| `quoteSent` | Quote sent | Customer contact |
| `invoiceSent` | Invoice sent | Customer contact |
| `paymentReceived` | Invoice paid | Customer contact |
| `bookingRequested` | Customer books | Customer |
| `bookingConfirmed` | Office confirms | Customer |
| `bookingCancelled` | Cancellation | Customer |
| `workOrderAssigned` | Assignment or move | Technician |

1. Set `SMTP_HOST`, `SMTP_PORT`, `SMTP_SECURE`, `SMTP_USER`, `SMTP_PASS` and `APP_URL`. Without username and password the outbox is a safe no-op: messages stay `PENDING` without counting as failed attempts.
2. Open **Settings** → **Automatic email**.
3. Set **Sender name**, **Sender address** and **Reply-to**. An empty sender uses `SMTP_USER`; an empty name uses the company name from branding.
4. Turn each event on or off. Off only means the email is not sent; the business event is stored as before.
5. Select **Send test email**. The test goes through the outbox but is sent immediately, and does not change the on/off settings.

The outbox retries after 60 seconds, then exponentially up to one hour, and gives up after five attempts. Only a short error code is stored, never the raw SMTP message or recipient address in the log. `FOR UPDATE SKIP LOCKED` lets several app instances share the queue. Rows left in `SENDING` by a crashed process are reset after 15 minutes. If the module for an event is off, the message is postponed rather than discarded.

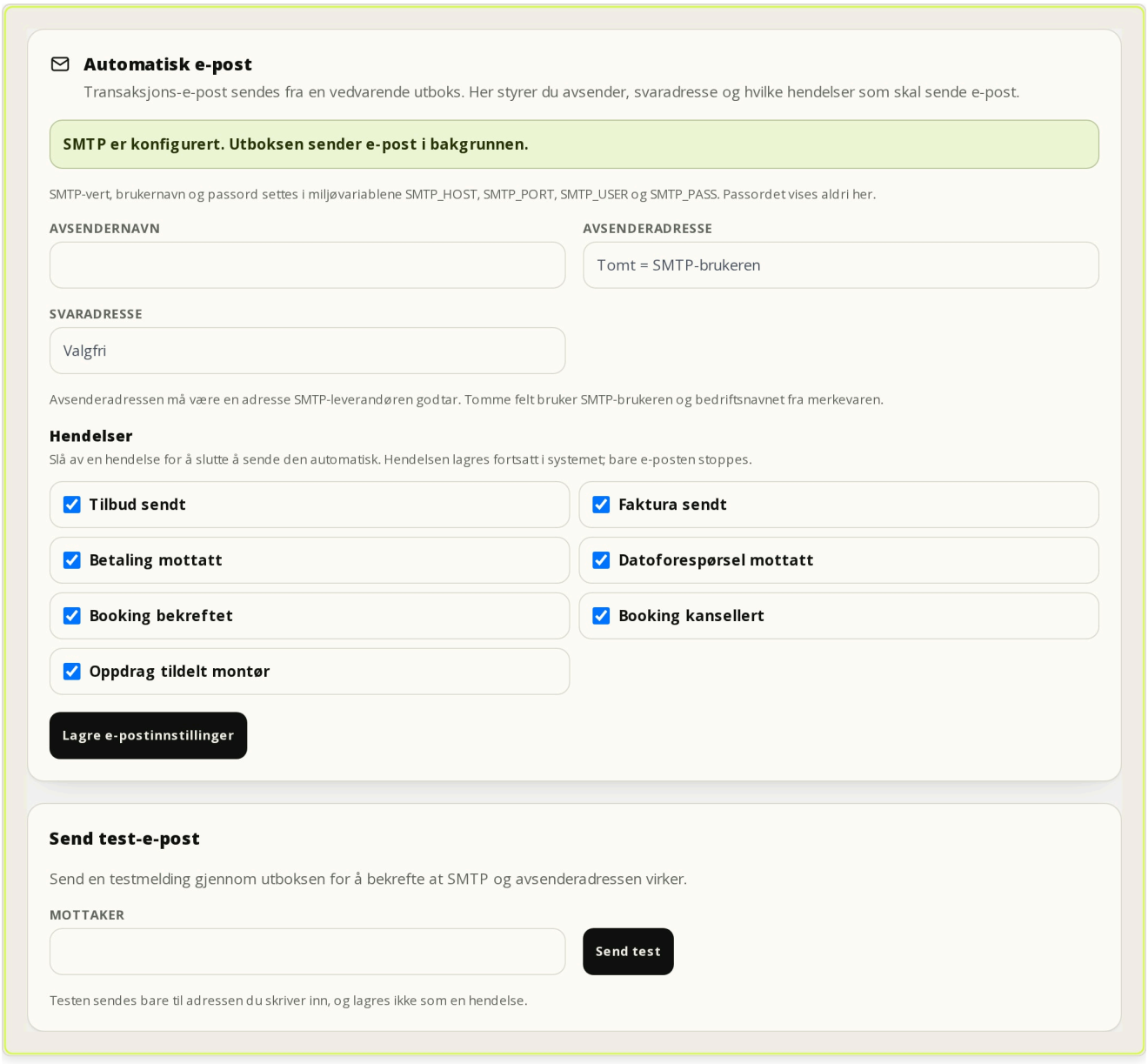



*Figure: Automatic email is controlled per event, with sender and test.*

Notice that **Send test email** is a separate form. The test proves that SMTP and the sender address work, but does not change the on/off choices. Also check **Reply-to** and the sender name before enabling customer events.

## Push notifications

Push complements in-app notifications. Each notification is first stored in `Notification`, then one row per active mobile device is added to the `PushMessage` outbox. The worker sends in the background, so a slow provider never blocks the action that triggered the notification.

- **Android:** FCM HTTP v1 with a Google service account in `FCM_SERVICE_ACCOUNT_JSON`.
- **iOS:** APNs with a `.p8` key in `APNS_KEY_P8`, `APNS_KEY_ID`, `APNS_TEAM_ID`, `APNS_BUNDLE_ID` and `APNS_ENV`.
- `PUSH_WORKER=false` disables the worker. Without a provider, the feature is a safe no-op.

The app registers the device after sign-in and on token rotation. On `UNREGISTERED` or `BadDeviceToken`, the device is deactivated. Under **Settings → Push notifications** you can see whether providers are configured, and clean up deactivated devices and devices not seen for 90 days. A test notification uses a random key and goes to the administrator's own registered devices.

Push follows the module that owns the notification type. If the module is off, no push row is created, and an already queued row is skipped. Deep links always contain an internal path starting with `/`; the app must never open external URLs. Operational metrics are in `/api/metrics` as `fsm_push_messages{status="pending"}` and `fsm_push_messages{status="dead_letter"}`.

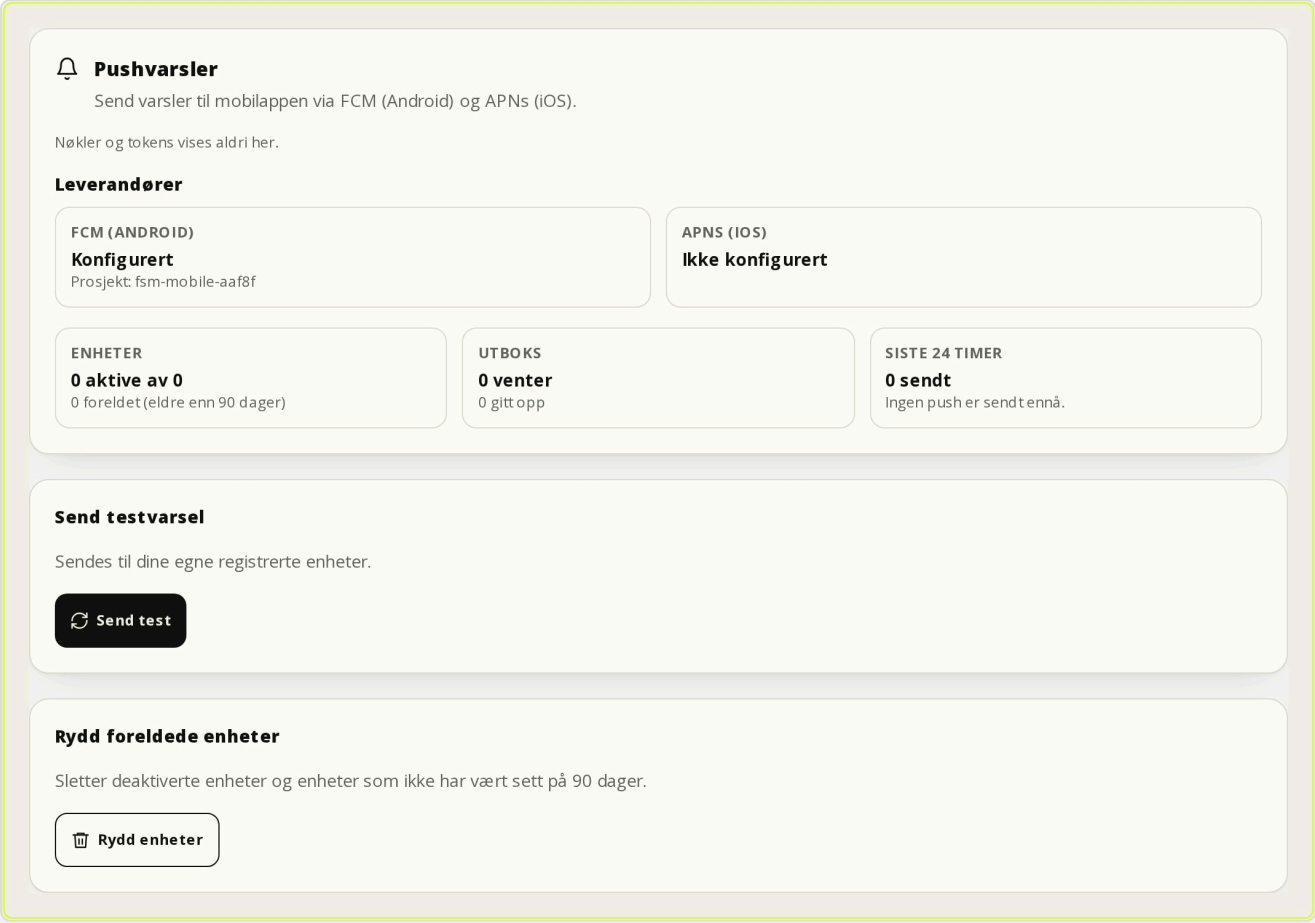



*Figure: Push settings show whether FCM and APNs are configured.*

Look at the provider status, not just the test button. If only FCM is set, Android notifications are sent while iOS devices are skipped with `not_configured`. The test returns 0 devices if the administrator has not registered a device on their own account.

## 12. Activity log, privacy and data

**Activity log: Settings → Activity log** shows system events, security events such as failed 2FA attempts, and user activity with search, **From date** and **To date**, 50 rows per page. Work orders and customers have their own activity tabs; reports have revisions. The log is written in the same transaction as the mutation, cannot be edited or deleted in the UI, and is not hash-chained in 0.8.11.

**Privacy:** collect only necessary data, use department and customer scope, and grant `reports.viewAllDepartments` only when needed. Use separate portal accounts and deactivate them on leaving. Assess DPIA before tracking in a new department; tracking is opt-in, limited to `EN_ROUTE`, shows no

route or coordinates and deletes pings after 24 hours. Obtain SMS consent and keep reviews internal. Deleting a customer deactivates portal users.

**GDPR erasure: Privacy / Erase customer data** on the customer card anonymises the customer's direct identifiers. The preview shows affected row counts and blocks when a work order, report or audit hold has a retention obligation. Enter a reason and confirm with the customer's current name. Work orders, reports, revisions, technician signatures and the audit log are preserved; the customer's own attachments and document library are deleted. Signed report PDFs are kept unchanged as service evidence and may still contain the original data. A second run is a no-op. See `docs/privacy-erasure.md` for the retention matrix and evidence.

**Export:** there is no automatic export job or self-service GDPR export. Evidence is exported through the database backup or an administrator-controlled query against `WorkOrder`, `Report`, `ReportRevision` and `AuditLog` for one work order at a time. Never export reports from the client or a public route.

**Retention:** `RETENTION_DAYS` (default 365) controls hard deletion of soft-deleted work orders without reports. Work orders with reports are kept indefinitely. Legal hold stops all automatic deletion. Files are removed only when no rows reference them.

**Backup and upgrade:** take a backup before migration and rollback. Follow `docs/OPERATIONS.md` and `docs/BACKUP-RESTORE.md`, use `scripts/backup.sh`, `scripts/restore-test.sh` and `restore-pitr.sh` for PITR. In production, set `REQUIRE_BACKUP_BEFORE_MIGRATE=true` so the app refuses to migrate without `BACKUP_CONFIRMED=true`. That only confirms a backup was taken, not that it is good; run the restore test first. Restore database and uploads from the same timestamp, stop the app before switching databases, use `prisma migrate deploy` and never `prisma db push`. Pin `APP_IMAGE` to a published version or digest, not `latest`, and never run `docker system prune --volumes`.

For PITR: choose a time **before** the failure, restore into a separate environment first, check data integrity and file references, and only then switch production. Document RPO/RTO and the result of the exercise. A backup without a tested restore is not proof that data can be recovered.

## 13. Operations and health

`GET /api/health` is liveness and returns 200 when the Node process is alive, without database, disk or queue calls; Docker uses it so a slow database does not cause a restart loop. `GET /api/ready` is readiness: unauthenticated calls get a short status and a database check, rate-limited per IP. With `READY_SECRET` or `CRON_SECRET`, the upload volume and job queue are also checked; it returns 503 when something is missing. Production Compose uses readiness but does not restart on `unhealthy`; a database outage gives 503 from the proxy. Both return `requestId`.

**Migrations:** the app runs `prisma migrate deploy` at startup. `DATABASE_MIGRATION_URL` can be used for a DDL role, `DATABASE_RUNTIME_URL` for the app. To run migrations separately, start with `SKIP_MIGRATIONS=true` and run them manually. Verify with `npx prisma migrate diff --from-config-datasource --to-schema prisma/schema.prisma --exit-code`.

**Jobs:** `RECURRING_SCHEDULER=true` runs service agreements and maintenance plans every six hours and at startup. The SLA cleanup runs in the same scheduler with `SLA_CLEANUP_INTERVAL_HOURS` (default 6) and can be disabled with `SLA_CLEANUP_ENABLED=false`. `POST /api/cron/recurring` and `/api/cron/cleanup` require `Authorization: Bearer <CRON_SECRET>`. `PDF_JOB_WORKER` controls OCR/document jobs with at

most two concurrent jobs per instance; report PDFs render in-process with a separate concurrency limit. `SMS_WORKER` is independent.

**Uploads and OCR:** files are stored in `UPLOAD_DIR` (default `./uploads`, `/app/uploads` in Docker). `/uploads/...` requires sign-in and checks the `Upload` row. Limits are 15 MB for images and 50 MB for PDFs. A rolling 24-hour quota of 2 GiB per user and 10 GiB total storage per customer also applies. Adjust it with `UPLOAD_QUOTA_USER_MB` and `UPLOAD_QUOTA_CUSTOMER_MB`; the proxy still needs its own body limit (55 MB in the Caddy example). Over quota the user gets a clear error, and the upload is not written. OCR uses `TESSDATA_DIR` and `/tmp`; large PDFs can fill `/tmp`, and the job then fails without filling the root filesystem. Check `DocumentJob`, `PDF_JOB_WORKER` and disk.

**Environment variables:** the most important are `DATABASE_URL`, `NEXTAUTH_URL`, `APP_URL`, `NEXTAUTH_SECRET`, `SMTP_*`, `SMS_*`, `SSO_*`, `SAML_*`, `SCIM_*`, `PAYMENTS_*`, `ACCOUNTING_*`, `FCM_*`, `APNS_*`, `STORAGE_DRIVER`, `S3_*`, `QUEUE_DRIVER`, `REDIS_URL`, `RATE_LIMIT_DRIVER`, `CACHE_DRIVER`, `DATABASE_POOL_MAX`, `PDF_JOB_WORKER`, `PUSH_WORKER`, `RECURRING_SCHEDULER`, `SLA_CLEANUP_ENABLED`, `SLA_CLEANUP_INTERVAL_HOURS`, `CRON_SECRET`, `READY_SECRET`, `RETENTION_DAYS`, `UPLOAD_*`, `TESSDATA_DIR`, `GEOCODE_*`, `ROUTING_*`, `GOOGLE_ROUTES_*`, `ROUTE_ALLOW_OVERTIME`, `ROUTE_MAX_OVERTIME_MINUTES`, `TRACKING_AVERAGE_SPEED_KMH`, `TRUST_PROXY` and `BACKUP_*`. `.env` must be `chmod 600` and never shared; changes require a new deploy or restart.

**Before escalating:** is `/api/health` 200 and `/api/ready` 200/503? Is `app` `healthy`, and is `RestartCount` increasing? What do the logs say? Does Postgres respond? Is there free disk and a writable upload volume? Are `DocumentJob`, `Notification` or `SmsMessage` stuck? Include `requestId`, timestamp and error message, never secrets or personal data.

## Storage, Redis and high availability

The default setup with one app, local disk and a database queue is fully supported. A separate HA profile exists for multiple replicas or external storage.

### Storage (S3/MinIO)

Uploads go through a shared `StorageDriver` interface. `STORAGE_DRIVER=local` is the default; `STORAGE_DRIVER=s3` makes files shared between replicas. URLs and stored file names are unchanged (`/uploads/<file name>`).

1. Set `S3_BUCKET`, `S3_REGION`, `S3_PREFIX` and optionally `S3_ENDPOINT`, `S3_ACCESS_KEY_ID` and `S3_SECRET_ACCESS_KEY`.
2. Run `npm run storage:migrate -- --dry-run`, then without `--dry-run` **before** switching the driver.
3. Verify with `--verify` and check with `--check`.
4. Stop the app, run the migration once more, set `STORAGE_DRIVER=s3` and start the app.
5. Enable versioning and encryption on the bucket, and block public access. `/uploads` still enforces access control.

The migration never deletes source files. Keep them until at least one successful backup/restore and an operating period on S3 have passed. Rollback means switching back to `STORAGE_DRIVER=local`, but files written while S3 was active exist only in S3. See `docs/STORAGE.md`.

### Redis and the job queue

`QUEUE_DRIVER=db` is the default and uses `DocumentJob` with `FOR UPDATE SKIP LOCKED`. `QUEUE_DRIVER=redis` also puts each job ID in a Redis sorted list. Redis is a wake-up mechanism, not a duplicate queue: the database decides whether the job runs. Redis down falls back to database claim, and an empty Redis loses no jobs. The same applies to `RATE_LIMIT_DRIVER` and `CACHE_DRIVER`; the database is always the fallback.

### HA and PITR

`docker-compose.ha.yml` has multiple app replicas, Caddy, Redis, shared S3 and Postgres with WAL archiving. Start with:

```
docker compose -f docker-compose.ha.yml up -d --scale app=3
```

Each app process opens at most `DATABASE_POOL_MAX` connections. Plan for `app_replicas × DATABASE_POOL_MAX + migrations + operations < max_connections`. Roll one replica at a time, and check `/api/ready` with `READY_SECRET` and a real upload/PDF. The WAL archive must be copied offsite; without it, PITR is local history only. Restore with `restore-pitr.sh` and the exercise checklist in `docs/BACKUP-RESTORE.md`; capacity, readiness and when **not** to enable HA are in `docs/SCALING.md`.

> **Warning:** S3, Redis and PITR are not verified in production without real credentials and an environment. Always run a restore test and a real upload/PDF generation against the HA profile before promising high availability.

## Compliance, ISO 27001 and SOC 2

The compliance package in `docs/compliance/` is a **readiness basis**, not a certification, attestation or audit report. It describes how the platform is intended to be governed, operated and evidenced, and what remains.

- **ISO/IEC 27001:** the Statement of Applicability covers all 93 Annex A controls, but only an accredited certification body can issue a certificate.
- **SOC 2:** the TSC mapping covers 51 criteria. A SOC 2 Type I or Type II requires an independent CPA firm and an examination period.
- **Status:** 29 controls are implemented in design/repo, 50 partial, 11 planned and 3 not applicable. This is design evidence, not tested operating effectiveness.
- **Evidence:** `npm run compliance:evidence` collects repository, CI and operational evidence without cloud credentials. `npm run compliance:links` checks documentation links.

Use `docs/compliance/GAP-ANALYSIS.md` as the work list. Do not describe the service as "ISO 27001 certified" or "SOC 2 compliant" in customer replies until the certificate or report actually exists and its scope matches the customer's use.

## 14. Administrator routines

**Onboard a department:** create it, enter contact details, set business hours and holidays, create a default SLA policy and optional price list, leave tracking off until DPIA is assessed, and create employees.

**Onboard a user:** create with the right role and department, send the invitation, let the user set a password, activate, ask for 2FA, and test that the user sees the right work orders and not other departments.

**Customer and portal user:** create the customer, add a contact with email, create the portal user under **More → Portal users**, send the invitation, activate and test the portal. Deactivate the test account afterwards.

**Publish a template:** build it, set the report name, unit requirement and HMS type, test on an internal work order, check the PDF, **A comment is required for these options** and conditional fields, then link it to agreements or work orders. Inform office and technicians.

**Launch an agreement and SLA:** choose or create a policy, check business hours and holidays, create any price list, create the agreement with the right interval and checklist, add lines and coverage, set renewal and notice, activate and check the next service date and SLA clock.

**Rotate an API key:** create a new key with a clear name, store it in the secret manager, deploy it, verify with `GET /api/v1/me`, revoke the old key, confirm it returns `401`, and update the documentation.

**Monthly review:** review administrators, employees, roles and departments. Review API keys, **Last used** and expiry. Review the **Activity log** for deletions, role changes, unlocks and legal hold. Check 2FA, portal users and unlocked reports. Run `scripts/restore-test.sh` and check disk, `/api/ready`, `RestartCount` and pending migrations.

## 15. Common problems

| Symptom | Cause | Action |
|---|---|---|
| User cannot sign in | Inactive, missing password or deactivated. | Activate, send a new invitation, or use **Forgot password**. |
| Customer portal is empty | Portal user inactive, customer deleted or contact missing. | Activate, restore and link a valid contact. |
| Template is missing, or changes do not apply | Wrong category/permission, or the report has a frozen template snapshot. | Check category and `templates.view`; create a new report for new fields. |
| Report cannot be unlocked | Legal hold or missing administrator. | Release the hold with a reason, or use an administrator. |
| SMS is not sent | `console`, missing consent, allowlist or cap. | Set the provider, record consent, check the cap. |
| SMS webhook rejects | `SMS_WEBHOOK_SECRET` missing or signature wrong. | Set the secret and check the signature. |
| Upload is rejected by quota | User or customer quota reached, or the proxy rejects the body. | Check `UPLOAD_QUOTA_*`, disk and the proxy body limit; remove unused attachments. |
| Unit cannot be moved | Active contract lines, number conflict or cross-customer reference. | Clear the contract lines, choose another order number, or check the preview. |
| SLA has the wrong deadline | Wrong policy, business hours or holiday. | Check the policy, department and calendar. |

| Symptom | Cause | Action |
|---|---|---|
| API returns `401` / `403` / `429` | Expired key, missing permission or rate limit. | Create a new key, grant permission, or wait for `Retry-After`. |
| `/api/ready` returns 503 | Database, upload volume or job queue. | Check Postgres, `UPLOAD_DIR`, disk and queue. |
| App restarts in a loop | Migration gate, OOM or `.env` error. | Check logs, `dmesg` and `REQUIRE_BACKUP_BEFORE_MIGRATE`. |
| Email stays `PENDING` | SMTP missing, or the event is off | Check `SMTP_*`, `APP_URL` and **Automatic email** |
| Test email fails | Wrong host, port, TLS or sender | Check the SMTP settings and sender address |
| Push returns `not_configured` | FCM/APNs missing in all app instances | Set `FCM_*` / `APNS_*` and restart all replicas |
| Push returns `invalid_token` | Device unregistered or token rotated | Let the app register the device again |
| S3 returns 404 for old files | Files were not migrated | Run `storage:migrate --check` and migrate |
| `/api/ready` returns 503 after HA start | Redis, S3 or database missing from the environment | Check `REDIS_URL`, `S3_*` and `/api/ready` with `READY_SECRET` |

## 16. Relationship to the office and technician chapters

The administrator chapter covers setup and governance; the office and technician chapters cover daily use. **Department and role** determine which work orders and actions staff see; the customer portal is controlled by the customer link. **Templates, SLA, branding and SMS** provide the forms, deadlines and messages staff work with. **Legal hold** freezes data for all parties.

See `10-office.md` for customer cards, work orders, agreements, the booking queue and reports. See `20-technician.md` for the field app, offline drafts, forms, submission and tracking. See `90-reference.md` for concepts, error codes and reference material.

# SSO and SCIM

This chapter is for administrators who will connect FSM to an external identity provider (IdP) and automate user creation and deactivation. Configuration is done with environment variables and requires an app restart. Password sign-in always remains as the break-glass path. It applies to FSM Platform 0.8.11 (8 November 2026).

## 1. Before you begin

Agree these points with the customer before changing production:

- Which IdP is used: Microsoft Entra ID, Keycloak, Authentik or Zitadel?
- Which email domains may sign in?
- Should unknown users be created automatically, or must an administrator activate them first?
- Which groups grant `OFFICE`, `TECHNICIAN` and optionally `ADMIN`?
- Who is the local break-glass administrator if the IdP is down?

**Least privilege:** start with `DEFAULT_ROLE=USER` and auto-provisioning off. Add only a small, named group to `ADMIN`.

## 2. Register the application in the IdP

The redirect URL is:

```
https://<your-fsm-address>/api/auth/callback/sso—<slug>
```

`<slug>` is the name you choose in `SSO_PROVIDERS`, for example `entra`. The URL must be exactly the same in the IdP. FSM uses authorization code flow with PKCE, state and nonce.

- **Entra ID:** Web application, redirect URL above, API access `openid`, `email`, `profile`. Add `email` and `groups` as optional claims in the ID token. Entra uses `xms_edov` instead of `email_verified`; both are accepted.
- **Keycloak:** Confidential client with standard flow and PKCE. Add the redirect URL under *Valid redirect URIs*, and a group-membership mapper that emits the `groups` claim.
- **Authentik:** OAuth2/OpenID provider with the redirect URL. Scope mapping for groups to `groups`.
- **Zitadel:** OIDC application of type Web. Request `urn:zitadel:iam:org:project:roles` and use the same name as `GROUPS_CLAIM`.

## 3. Set environment variables

Example for Entra ID:

```
SSO_PROVIDERS=entra
SSO_ENTRA_DISPLAY_NAME=Microsoft Entra ID
SSO_ENTRA_ISSUER=https://login.microsoftonline.com/<tenant-id>/v2.0
SSO_ENTRA_CLIENT_ID=<client-id>
SSO_ENTRA_CLIENT_SECRET=<client-secret>
SSO_ENTRA_ALLOWED_EMAIL_DOMAINS=example.no
SSO_ENTRA_AUTO_PROVISION=false
SSO_ENTRA_AUTO_PROVISION_ACTIVE=false
SSO_ENTRA_DEFAULT_ROLE=USER
SSO_ENTRA_DEFAULT_DEPARTMENT=Oslo
SSO_ENTRA_GROUPS_CLAIM=groups
SSO_ENTRA_GROUP_ROLE_MAP={"FSM-Office":"OFFICE","FSM-Technicians":"TECHNICIAN"}
SSO_ENTRA_SYNC_ROLES=false
```

## SAML 2.0

SAML is configured with `SAML_PROVIDERS` and `SAML_<SLUG>_`. The OIDC variables are untouched, so an installation can run both. Example:

```
SAML_PROVIDERS=entra-saml
SAML_ENTRA_SAML_DISPLAY_NAME=Entra SAML
SAML_ENTRA_SAML_ENTRY_POINT=https://login.microsoftonline.com/<tenant>/saml2
SAML_ENTRA_SAML_IDP_CERT=<certificate>
SAML_ENTRA_SAML_ALLOWED_EMAIL_DOMAINS=example.no
SAML_ENTRA_SAML_GROUPS_ATTRIBUTE=groups
SAML_ENTRA_SAML_GROUP_ROLE_MAP={"FSM-Office":"OFFICE"}
SAML_ENTRA_SAML_GROUP_DEPARTMENT_MAP={"FSM-Oslo":"Oslo"}
SAML_ENTRA_SAML_AUTO_PROVISION=false
SAML_ENTRA_SAML_DEFAULT_ROLE=USER
```

Register the **SP entity ID** and **ACS URL** from `/api/auth/saml/<slug>/metadata` in the IdP. The app validates the signed assertion, audience, recipient and time window, and rejects replay of the same assertion. `ALLOW_IDP_INITIATED=true` is only for IdP-initiated sign-in; signature, audience, recipient and replay checks remain active. SAML uses the same account, role, department and 2FA rules as OIDC.

Restart the app. Open **Settings → SSO and SCIM**. The provider should appear as **Active**, with issuer, default role and allowed domains. Red configuration errors mean a required variable is missing or invalid.




*Figure: SAML is configured per provider and shown with status in settings.*

Look at **Status**, **Issuer** and the **Redirect/ACS** details. SAML and OIDC appear side by side, and the OIDC setup is untouched when you add SAML. If the status is red, a required variable is missing or the certificate is invalid – fix it before testing sign-in.

**Expected result:** the sign-in page shows a button with the provider's name above the password form.

## 4. Test sign-in

1. Sign out, or use a private window.
2. Select the SSO button and sign in at the IdP.
3. An existing, active user with the same verified email should get in.
4. An unknown user should get a clear error when auto-provisioning is off.
5. A deactivated user should be rejected.
6. If 2FA is required for the role, you should get the code step or 2FA setup before access.

**Check:** the role should not change just because the email matched. A role change happens only with `SYNC_ROLES=true` and an explicit group mapping. Look for the audit row "SSO: linked account …" in **Activity log**.

## 5. Auto-provisioning

With `AUTO_PROVISION=true`, unknown users are created without a password. The default is `AUTO_PROVISION_ACTIVE=false`, which gives an inactive account. Go to **Employees & Roles**, check role and department, and activate the account.

Enable `AUTO_PROVISION_ACTIVE=true` only when the IdP is authoritative for the whole user population and the department is set correctly. A user who is a member of several groups with different roles is rejected – clean up the mapping instead of granting the highest role automatically.

## 6. SCIM token

1. Go to **Settings → SSO and SCIM**.
2. Enter a descriptive name, for example "Entra provisioning".
3. Leave the expiry date empty for the default lifetime, or choose a date.
4. Select **Create token**.
5. Copy the token immediately. It is shown only this time.

Use one token per integration. If leakage is suspected, select **Rotate token**. The old token is revoked in the same operation, and the integration must be updated immediately. **Revoke** blocks the token without deleting history.

## 7. Set up SCIM in the IdP

- Tenant URL: `https://<your-fsm-address>/api/scim/v2`
- Authentication: Bearer token from the previous step.
- Test connection, and enable provisioning for users.

Field mapping:

| SCIM | FSM |
|---|---|
| `userName` / `emails` | Email (unique) |
| `externalId` | External ID (unique when set) |
| `active` | Active/inactive |
| `name` | Name |
| enterprise `department` | Department |
| `roles` / `groups` | Role through explicit mapping |

`DELETE` deactivates the user; it does not delete them. Passwords are never set. Changes to role, department, email or active status sign out existing sessions. `ADMIN` requires `SCIM_ALLOW_ADMIN=true` in addition to an explicit mapping.

## SCIM groups

Groups are served at `/api/scim/v2/Groups` with the same token and error format as users. A group has `displayName`, optional `externalId` and `members` with the local user ID in `value`. `PATCH` supports `displayName`, `externalId`, `members` and `members[value eq "…"]`.

Groups grant access **only** through explicit mapping:

```
SCIM_GROUP_ROLE_MAP={"FSM-Office":"OFFICE"}
SCIM_GROUP_DEPARTMENT_MAP={"FSM-Oslo":"Oslo"}
```

Without a mapping nothing happens. Two groups pointing to different roles or departments change nothing, only an audit row. `ADMIN` still requires `SCIM_ALLOW_ADMIN=true`. `DELETE` deletes the group and memberships but keeps the users.

## Field-level visibility

Field-level RBAC is enforced on the server, not only in the interface. Every read service that returns a sensitive field masks it before the response reaches the API, search, export or UI. Masking is the default: email and phone are masked, amounts and signature URLs are set to `null`, and `redactedFields` explains why.

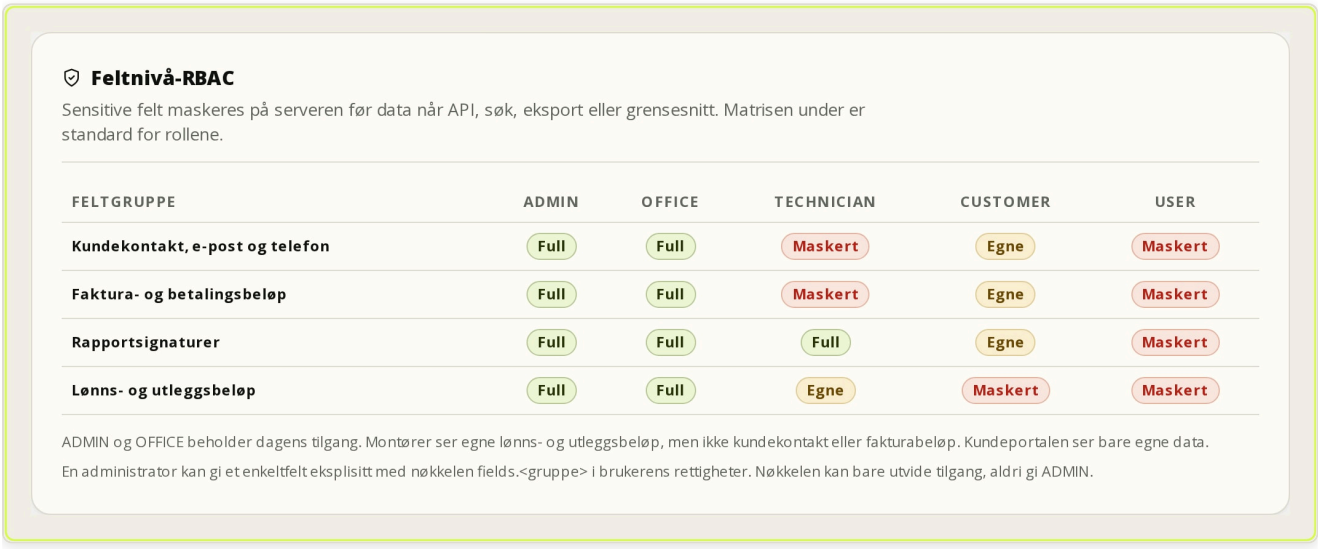
| Field group | ADMIN | OFFICE | TECHNICIAN | CUSTOMER | USER |
|---|---|---|---|---|---|
| `customerContact` – customer contact, email, phone | Full | Full | Masked | Own | Masked |
| `billingAmounts` – invoice and payment amounts | Full | Full | Masked | Own | Masked |
| `reportSignatures` – report signature images | Full | Full | Full | Own | Masked |
| `payrollAmounts` – payroll, mileage and expenses | Full | Full | Own | Masked | Masked |

"Own" means `ownerId` or `customerId` must match the actor. A technician sees their own mileage claims and expenses, but not colleagues'. The customer portal sees its own invoice amounts and contact details, but no internal amounts or signatures. The contact's name and the customer's address are not masked in v1, because they are needed to do the job. The report's `technicianName` and `customerName` are kept for the same reason, while the signature image itself is hidden.

An administrator can grant one field to one user with the key in the user's `permissions` list:

```
fields.customerContact
fields.billingAmounts
fields.reportSignatures
fields.payrollAmounts
```

The key extends access to the whole field group for that user, is not inherited from the role and cannot grant ADMIN. In v1 it is set through direct database/API access, because the settings permission list filters out unknown keys. An editable field matrix in the UI is a residual.



**Feltnivå-RBAC**

Sensitive felt maskeres på serveren før data når API, søk, eksport eller grensesnitt. Matrisen under er standard for rollene.

| FELTGRUPPE | ADMIN | OFFICE | TECHNICIAN | CUSTOMER | USER |
|---|---|---|---|---|---|
| Kundekontakt, e-post og telefon | Full | Full | Maskert | Egne | Maskert |
| Faktura- og betalingsbeløp | Full | Full | Maskert | Egne | Maskert |
| Rapportsignaturer | Full | Full | Full | Egne | Maskert |
| Lønns- og utleggsbeløp | Full | Full | Egne | Maskert | Maskert |

ADMIN og OFFICE beholder dagens tilgang. Montører ser egne lønns- og utleggsbeløp, men ikke kundekontakt eller fakturabeløp. Kundeportalen ser bare egne data.

En administrator kan gi et enkeltfelt eksplisitt med nøkkelen fields.<gruppe> i brukerens rettigheter. Nøkkelen kan bare utvide tilgang, aldri gi ADMIN.

*Figure: The field policy shows which roles see which field groups.*

Notice that the matrix is role-based and read-only in v1. It shows which groups are masked, not which individual fields are visible. To give a technician access to invoice amounts, the field key must be set explicitly; that is a deliberate action, not a role change. See `docs/RBAC.md`.

## 8. Break-glass and troubleshooting

Keep at least two local administrators with strong passwords and 2FA. The password form at `/logg-inn` does not disappear when SSO is on.

| Symptom | Check |
|---|---|
| No SSO button | `SSO_PROVIDERS`, `ENABLED`, required variables, and that the app has restarted. |
| `redirect_uri` error | The redirect URL in the IdP must be exactly `…/api/auth/callback/sso—<slug>`. |
| "not created in FSM" | Auto-provisioning is off, or the email does not exist. Create the user or enable provisioning. |

| Symptom | Check |
|---|---|
| "email not verified" | The IdP is missing `email_verified` / `xms_edov`. Add the claim to the ID token. |
| Role conflict | The user is in groups that map to different roles. Remove one group. |
| SCIM 401 | The token is wrong, expired or revoked. Rotate and update the IdP. |
| SCIM 403 | Customer portal user, or the administrator who created the token is inactive. |
| SCIM 409 | Email or `externalId` is already in use. Clean up the duplicate in FSM. |
| SAML assertion rejected | Check signature certificate, ACS URL, audience, recipient and clock. |
| Field missing or masked | Check the field group and the user's explicit `fields.*` key. |

# Quotes, invoicing and accounting

This chapter describes the optional **quotes** (`quotes`), **invoicing** (`invoicing`), **payments** and **accounting sync** modules. An administrator must enable them under **Settings** → **Modules** before `/tilbud`, `/faktura` and the invoice tab in the customer portal are available. When a module is off, both the page and the API return `404`, even if you enter the address directly.

The platform core works without these modules. Quotes, invoices, payments and accounting records are separate data with their own permissions and audit trail; they do not change work order locking or the report. All amounts are calculated on the server. The client cannot submit totals that differ from the lines.




*Figure: The quote list collects drafts, sent and accepted quotes.*

Notice the status filter at the top and the validity column: a quote that has been sent but not accepted is still an open commitment. Use the list to find quotes nearing expiry, and open a row to resend or convert it to a work order.

## Quotes

A quote can be edited only while it is a **Draft**. When it is sent, the lines are frozen in a version and the customer receives a one-time link.

### Create and send

1. Choose **Quotes** → **New quote** (`/tilbud/ny`).
2. Select **Customer**, optional **Contact**, **Work order**, **Contract** and **Equipment**. A customer name can be entered without a customer card, but invoicing requires a customer card link.
3. Add lines with **Description**, **Quantity**, **Unit**, **Price** and **VAT**. Totals, VAT and rounding are calculated per line on the server.
4. Save the draft. You can edit, clone and create a new version.
5. Select **Send quote**. The system assigns a quote number, freezes the lines, creates the approval link and queues the email in the automatic outbox.
6. Share the link with the customer. The customer accepts or rejects it in a browser without signing in.

An approval link can be used only once and has a short lifetime. If it expires, send the quote again; this creates a new version and a new link. **Print** (`/tilbud/<id>/skriv`) gives a print-friendly page. PDF is produced with the browser's **Print** → **Save as PDF**; there is no server-generated quote PDF.

### Status and versions

| Status | Meaning | Editable |
|---|---|---|
| **Draft** | Being prepared | Yes |
| **Sent** | The customer has the link | No |
| **Accepted** | The customer accepted | No |
| **Rejected** | The customer rejected | No |
| **Expired** | Validity has passed | No |

Every send and change to a sent quote creates a new version. The version keeps the lines, prices, VAT and totals, so a later price change does not alter a quote the customer has already seen. A scheduled job marks expired quotes; it is idempotent and runs once per quote.

### Convert to a work order

An **accepted** quote can be converted with **Convert to work order**. The action requires `customers.manage` and `workorders.create`, runs in one transaction and reuses or creates the customer. Equipment on the quote is linked to the work order. The report and invoice remain separate objects; conversion does not create an invoice automatically.

## Invoicing

Invoices can be created manually, from an accepted quote, from a completed work order or from a submitted report. Work orders and reports must be **Completed** before they can be used as a basis.

### Create and send

1. Choose **Invoices** → **New invoice** ( `/faktura/ny` ).
2. Select **Customer** and optionally a **Work order**, **Report** or **Accepted quote**.
3. Add lines with quantity, unit, price and VAT. The invoice is saved as a draft and can be edited.
4. Select **Send invoice**. The invoice receives a unique number, the lines are frozen, and the customer receives an email with a link to the customer portal.
5. Register payment manually with **Amount**, **Date**, **Method** and **Reference**. Partial payments are supported and update the outstanding amount.
6. If needed, register a **Credit note**. It cannot exceed the outstanding amount.

**Print** ( `/faktura/<id>/skriv` ) gives a print-friendly invoice using the same print-to-PDF flow as quotes. The invoice number is assigned on sending. Concurrent sends can leave gaps in the number sequence; the gaps do not affect uniqueness.

### Status and payment

| Status | Meaning |
|---|---|
| **Draft** | Not sent; editable |
| **Sent** | Number assigned, lines frozen |
| **Partially paid** | Payment registered, outstanding remains |
| **Paid** | Outstanding is zero |
| **Overdue** | Due date passed with an outstanding amount |
| **Credited** | A credit note has been registered |

Overdue invoices are updated manually with **Update overdue**, or by the scheduled invoicing job. A payment carries an idempotency key, so double clicks and repeated webhooks never produce a double amount. Refunds are not created from the payment provider; use a credit note.

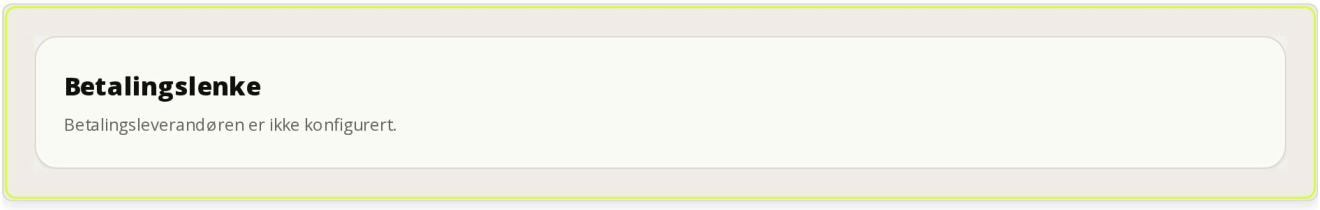



*Figure: The invoice shows lines, payments and outstanding amount together.*

Check **Outstanding** and **Payment status** before creating a new link. The payment button is active only when the provider is configured and the invoice still has an outstanding amount. Partial payments appear as separate rows, and credit notes are kept apart from payments.

## Payment links (Stripe and Vipps)

Payment is optional. Without configuration, every payment action is a silent no-op: no network call, no database mutation, and no button that promises more than the system can deliver. Card numbers, CVC, BankID and the confirmation itself stay with the payment provider. FSM stores only the provider's id, the link and the status.

### Enable

1. Set the provider with `PAYMENTS_PROVIDER=stripe` or `PAYMENTS_PROVIDER=vipps`.
2. Set the keys and webhook secret in the environment. A provider counts as configured only when both the API key and the webhook secret exist.
3. Register the webhook URL in the provider's console: `https://<app-url>/api/webhooks/payments/stripe` or `https://<app-url>/api/webhooks/payments/vipps`.
4. Restart the app and confirm that **Pay** appears on a sent invoice with an outstanding amount.

Stripe uses **Checkout** as the hosted payment page. Vipps uses ePayment with `WEB_REDIRECT` and wallet. Only NOK is supported in v1, and each link covers one amount. If the outstanding amount changes after the

link is created, create a new link.

### Flow and security

1. Office staff or an administrator open the invoice and select **Create payment link**.
2. The customer pays at Stripe or Vipps.
3. The provider calls the webhook. The signature is verified against the raw body, the event is idempotent, and amount and currency are checked.
4. The payment is registered through the same service as a manual payment, and the invoice becomes **Partially paid** or **Paid**.

The webhook returns `200` also for duplicates and events the system deliberately ignores, so the provider stops retrying. Unknown providers return `404`, and missing keys return `503`. Failed or expired links set the payment status but do not change the invoice's accounting status. Refund and capture are not done in FSM; handle them with a credit note and, if needed, in the provider's portal.

> **Warning:** The payment link is a secret on par with the invoice. Share it only with the customer. Never display webhook secrets in the interface or put them in git.

## Accounting sync (Fiken, Tripletex, PowerOffice Go and CSV)

Accounting sync is manual and per invoice. It creates or reuses the customer, invoice and positive payments at the provider, stores external IDs and a retryable state. It never pretends an external call succeeded.




*Figure: Accounting settings show the selected provider and whether the integration is active.*

Notice that the integration is **off** until an administrator enables it, even when `ACCOUNTING_PROVIDER` is set. **Test connection** stores the result, and **Last sync** shows whether anything was actually sent. CSV is always available but is marked **Exported**, never **Synced**.

### Enable and test

1. Set `ACCOUNTING_PROVIDER` to `fiken`, `tripletex`, `poweroffice` or `csv`, and fill in the provider's keys and account codes.
2. Open **Settings** → **Accounting**.
3. Enable sync and select **Test connection**. A green result means the provider responded; it does not prove every account code is correct.
4. Open a sent invoice and select **Sync to accounting**.
5. Check that the external ID and status appear. On failure, fix the cause and try again; the customer is reused and the invoice has a stable key.

### Provider differences

| Provider | Customer | Invoice | Payment |
|---|---|---|---|
| **Fiken** | Contact | Invoice | Free journal entry: debit bank, credit accounts receivable |
| **Tripletex** | Customer | Order + invoice | Payment on the invoice |
| **PowerOffice Go** | Customer | Outgoing invoice | Bank journal |
| **CSV** | Local row | Local row | Local row, no external write |

Fiken requires a bank account for invoices and account codes for payments. Tripletex requires a VAT type and payment type; without the payment type, the payment sync fails with `not_configured` instead of guessing. PowerOffice paths can be overridden because resource names vary between API versions. The CSV file contains customer names and invoice lines and must be treated as business data.

### Status and limits

| Status | Meaning |
|---|---|
| **Synced** | The provider acknowledged customer, invoice and payments |
| **Exported** | CSV was written; no external state changed |
| **Failed** | The last attempt failed; the error code is in `lastError` |
| **Not synced** | No sync attempted |

Sync is neither automatic nor two-way. There is no background job retrying failed syncs, no update of an already sent invoice, and no credit note, refund or e-invoice sync. VAT mapping is simple and must be checked manually for special rates. Resetting a sync does not delete the audit trail. Always verify the Fiken, Tripletex and PowerOffice connectors against the provider's sandbox before production.

## Customer portal

When both the customer portal and invoicing are enabled, the customer sees **Invoices** in the portal (`/kundeportal/faktura`). The customer sees their own sent invoices, lines, payment status and an optional

**Pay** button. They cannot see other customers' invoices, internal notes or drafts. Payment status is updated only after the webhook has been received and processed.

The customer cannot pay by card outside the payment link. If the provider is not configured, a short explanation appears instead of a button. Acknowledging a report and paying an invoice are two separate actions.

## Permissions, audit and routines

| Action | Permission |
|---|---|
| View quotes and invoices | `customers.view`, `customers.manage` or `workorders.view` |
| Create, edit and send quotes | `customers.manage` |
| Create and send invoices | `customers.manage` |
| Register payment and credit note | `customers.manage` or administrator |
| Create a payment link | `customers.manage` or administrator |
| Enable, test and reset accounting sync | `settings.manage` |
| Sync one invoice | `customers.manage`, and the invoice must be visible to the actor |

Every mutation, send, payment, credit note, payment event and sync attempt writes an audit row in the same transaction. The audit log contains provider, external ID, status and a short error code — never tokens, raw payloads or personal data.

**Daily:** send drafts, follow **Overdue**, register received payments and check that webhook payments have the right status.

**Weekly:** sync sent invoices, review **Failed** and **Outstanding**, and reconcile against the accounting system.

**Monthly:** check the number series, VAT, credit notes, the payment provider's reconciliation and that no invoice is stuck in **Draft**.

## Common problems

| Symptom | Cause | Action |
|---|---|---|
| `/tilbud` or `/faktura` returns 404 | The module is off | Enable `quotes` or `invoicing` under **Settings → Modules** |
| The customer does not receive the approval link | Email missing, or SMTP not configured | Copy the link manually or configure automatic email |
| The link has expired | Short lifetime or already used | Send the quote again |
| Cannot convert the quote | It is not accepted, or permission is missing | Accept the quote and check `customers.manage` and `workorders.create` |

| Symptom | Cause | Action |
|---|---|---|
| The invoice cannot be sent | The draft lacks lines or a customer | Add a customer and at least one line |
| The payment button does not appear | Provider or webhook secret is missing, or the amount is paid | Check `PAYMENTS_PROVIDER` and keys, and create a new link if needed |
| The payment is not registered | Webhook rejected, or currency/amount mismatch | Check signature, `APP_URL` and the audit log; reconcile manually |
| Overpayment is rejected | The outstanding amount changed after the link was created | Create a new link after the partial payment |
| Accounting sync is **Failed** | Wrong key, account code or VAT setup | Read `lastError`, fix the cause and sync again |
| CSV appears as **Exported** | CSV is an export, not a sync | Import the file into accounting; no external state changed |
| The credit note is rejected | The amount exceeds the outstanding amount | Check payments and outstanding first |

## Customer and contact

The contact field is shown only after a customer has been selected. The contacts are loaded for that customer and contain only people who actually belong to the customer. Changing the customer clears the contact and reloads the list, so a contact from another customer can neither be selected nor submitted.

The server validates the same rule on save. If a contact does not belong to the selected customer, the quote or invoice is rejected even if the form is bypassed.

# Reference, glossary and troubleshooting

*Applies to FSM Platform version 0.8.11, 8 November 2026.*

This chapter is the reference section. It explains permissions, statuses and terms across roles, shows how features connect, and helps you find where to look when something is missing.

## Roles and permissions in practice

Permissions have more levels than "yes" and "no". **Full** means view, create, edit and administer. **Own department** means the role acts within its own department for areas that are department-scoped. **Own customer** means the portal user only sees their own customer. **No** means the area is not available.

| Area | Administrator | Office | Technician | Customer portal user |
|---|---|---|---|---|
| **Work orders and visits** | Full, all departments | View, create, edit, assign and change status in own department | View own department, create their own work orders, claim available work orders, change status and add/remove people | View their own visits |
| **Customers and contacts** | Full | Administer; customers are shared | View | View their own customer and contacts |
| **Equipment and meter readings** | Full | Administer; equipment is shared | View | View their own equipment if delegated |
| **Service contracts** | Full | View and administer in own department | View in own department | View their own contracts if delegated |
| **Administer SLA and business hours** | Full | No | No | No |
| **Bookings** | Full | Confirm, reschedule and cancel in own department | View in own department | Book and cancel their own if delegated |
| **Reports and checklists** | Full, including unlocking and legal hold | View and complete in own department; administer templates | View and fill in reports in own department | View submitted reports and acknowledge if they have document access |

| Area | Administrator | Office | Technician | Customer portal user |
|---|---|---|---|---|
| **Documents** | Full | Administer; documents are shared | View | View their own shared documents if delegated |
| **Employees and roles** | Full | Administer users in own department; can assign roles whose permissions are a subset of their own, never **Administrator**; role definitions and department moves are administrator-only | No | No |
| **Settings and branding** | Full | Administer branding, map, templates and dispatch; departments, roles and API are administrator-only | No | No |
| **Audit log** | Global | Per work order and customer | Per work order | No |
| **Quotes and invoices** | Full | View and administer in own department; send, pay and credit | No | View own invoices and pay |
| **Payments and accounting sync** | Full | Create payment link and sync invoice | No | View and pay own invoice |
| **Inventory and purchasing** | Full, all departments | View and administer in own department; reserve and consume | View and reserve/consume if granted | No |
| **Mileage and expenses** | Full | Approve and export in own department | Record their own | No |
| **Profitability** | Full | View own department | No | No |
| **API keys** | Full | No | No | No |
| **Deletion, restore, unlocking and legal hold** | Full | No | No | No |

> **Note:** An administrator can create custom roles and give individual users tailored permissions. The table shows the standard roles. Check the user's actual permissions in "Settings" → "Employees & Roles" if something looks wrong.

Deletion is deliberately not a normal permission. An administrator can delete customers, files and other records. Work orders can be deleted by an administrator, or by a custom role explicitly granted `workorders.delete`, and only while the work order is open (not completed or cancelled). Reports and attachments are kept on soft deletion. Many deletions are soft, so the record can be restored.

# Status glossary

## Work order status

| Status | Displayed as | Meaning | Set by |
|---|---|---|---|
| `OPEN` | Open / New | Created but not planned. Used for the customer's date request before the office plans it. | System, office |
| `PENDING` | Waiting to be planned | Starting status for new and cloned work orders. Not selectable manually. | System |
| `PLANNED` | Planned | Has a date or visit and is ready for assignment or execution. | Office, system |
| `WAITING` | Waiting for customer | Work is on hold for the customer, access or clarification. | Office, technician |
| `IN_PROGRESS` | In progress | Work has started. | Technician, office |
| `COMPLETED` | Completed | Closed and treated as history. | Technician, office |
| `CANCELLED` | Cancelled | Closed without being carried out. | Office |

All open statuses can be changed freely, including to `COMPLETED` or `CANCELLED`. `COMPLETED` and `CANCELLED` are closed. Only an administrator can reopen a closed work order, and must give a reason.

## Visit status

| Status | Displayed as | Meaning |
|---|---|---|
| `PLANNED` | Planned | Planned, but no technician is assigned. |
| `ASSIGNED` | Assigned | At least one technician is assigned, but work has not started. |
| `EN_ROUTE` | On the way | The technician has started driving. Records the first response in any SLA. |
| `ON_SITE` | On site | The technician has arrived and is working on site. |
| `DONE` | Completed | The visit is finished. Tracking links are revoked. |
| `CANCELLED` | Cancelled | Cancelled before completion. |

The flow is `PLANNED → ASSIGNED → EN_ROUTE → ON_SITE → DONE`. `CANCELLED` can be used before the visit is completed. `DONE` and `CANCELLED` are closed.

> **Tip:** "On the way" starts the SLA response clock and makes it possible to share position and give the customer an estimated arrival.

## Report, unit and acknowledgement

| Term | Meaning |
|---|---|
| **Draft** | Completed fields saved locally or as a `ReportDraft`, but not submitted. A draft is not a report and is not shown to the customer. |
| **Submitted and locked** | The form has been submitted. The report is locked, has a PDF and a revision, and cannot be changed without an administrator. |
| **Open for correction** | An administrator has unlocked it with a reason. Changes are saved as a new revision, and the report locks automatically again. |
| **Acknowledged** | The customer has confirmed receipt. The acknowledgement is visible to the office and can be withdrawn on resubmission or unlocking. |
| **Revision** | An immutable snapshot before a change. Preserves names, units, answers, signatures and acknowledgement. |

| Unit result | Displayed as | Meaning |
|---|---|---|
| `GODKJENT` | Approved / done | The unit has been checked and is in order. |
| `AVVIK` | Deviation found | The unit has a deviation that needs follow-up. |
| Not assessed | Not assessed | The unit has no result. The report cannot be submitted until every unit is assessed. |

The overall report result is derived from the units: at least one deviation gives an overall deviation, all approved gives approved, and otherwise "not assessed" is shown. If the report contains deviations, one automatic remediation work order is created.

## Service contract and SLA

| Contract status | Displayed as | Meaning |
|---|---|---|
| `DRAFT` | Draft | Being set up; does not generate work orders. |
| `ACTIVE` | Active | Valid and can generate work orders when the next service is due. |
| `PAUSED` | Paused | Paused; does not generate work orders. |
| `EXPIRED` | Expired | Passed its end date without renewal. |
| `TERMINATED` | Terminated | Ended before its planned end. |

The renewal mode is `AUTO` (renews automatically) or `MANUAL` (must be renewed by the office). `Next service` is the date the system uses to decide whether to create a new work order.

| SLA state | Displayed as | Meaning |
|---|---|---|
| `ON_TRACK` | On track | Both response and resolution deadlines are within limits. |
| `AT_RISK` | At risk | The deadline is approaching; a notification is sent on the transition. |
| `BREACHED` | Breached | The deadline has passed; a notification is sent on the transition. |
| `MET` | Met | Response and resolution were recorded within the deadline. |
| `CANCELLED` | Cancelled | The work order was cancelled and the clock stopped. |

SLA deadlines are calculated using the department's business hours and holidays. The first "En route" or "On site" records the response. "Done" on the workday records the resolution; "Finish travel" does not move the clock. An SLA clock exists only when the work order has an active, date-valid agreement with an active SLA policy. Without that agreement, no SLA is shown.

> **Note:** Without the `quotes` and `invoicing` modules, the system does not calculate prices or invoices, and price lists are reference data. With the modules on, quotes, invoices, VAT, payments and credit notes are calculated on the server.

## Booking status

| Status | Displayed as | Meaning |
|---|---|---|
| `PENDING` | Pending | In the office's booking queue as a date-only request. Time and technician are not planned. |
| `CONFIRMED` | Confirmed | The office has planned time and technician, and the work order and visit have been created. |
| `CANCELLED` | Cancelled | The booking was cancelled. |
| `EXPIRED` | Expired | A pending request expired without confirmation. |

New bookings are always date-only requests: the customer chooses a date, not a time block, and the office plans time and technician. No new booking becomes `CONFIRMED` directly. Every pending request can be confirmed; older rows with a stored time window may use it, otherwise the office supplies date, start and end. Overlapping visits are a busy marker, not a hard block. A short hold ( `BookingHold` ) is used only when an already confirmed appointment is moved. The event log uses `CREATED` , `CONFIRMED` , `RESCHEDULED` , `CANCELLED` and `EXPIRED` .

## Quotes, invoices and payments

| Quote status | Displayed as | Meaning |
|---|---|---|
| `DRAFT` | Draft | Freely editable |
| `SENT` | Sent | Link issued; lines frozen |

| Quote status | Displayed as | Meaning |
|---|---|---|
| `ACCEPTED` | Accepted | The customer accepted |
| `REJECTED` | Rejected | The customer rejected |
| `EXPIRED` | Expired | Validity has passed |

| Invoice status | Displayed as | Meaning |
|---|---|---|
| `DRAFT` | Draft | Not sent; editable |
| `SENT` | Sent | Number assigned; lines frozen |
| `PARTIALLY_PAID` | Partially paid | Outstanding remains |
| `PAID` | Paid | Outstanding is zero |
| `OVERDUE` | Overdue | Due date passed |
| `CREDITED` | Credited | Credit note registered |

| Payment status | Meaning |
|---|---|
| **Pending** | Link created, but payment not confirmed |
| **Paid** | The webhook registered the payment |
| **Failed** | The provider rejected or the payment failed |
| **Expired** | The link expired without payment |

A payment link can be used once and covers one amount. Refunds use a credit note, not the payment provider. Accounting sync has the statuses **Synced**, **Exported** (CSV) and **Failed**.

## Inventory, purchasing and expenses

| Purchase status | Displayed as | Meaning |
|---|---|---|
| `DRAFT` | Draft | Editable |
| `ORDERED` | Ordered | No goods received |
| `PARTIALLY_RECEIVED` | Partially received | Some goods in stock |
| `RECEIVED` | Received | Everything received |
| `CANCELLED` | Cancelled | Cancelled before receipt |

Inventory movements are `IN`, `OUT`, `ADJUST`, `RESERVE`, `RELEASE` and `CONSUME`. A reservation reduces available stock but does not move the item until it is consumed. Only `ADJUST` can produce negative stock.

| Expense status | Displayed as | Meaning |
|---|---|---|
| `DRAFT` | Draft | Not submitted for approval |
| `PENDING` | Pending | Submitted, not processed |
| `APPROVED` | Approved | Can be exported |
| `REJECTED` | Rejected | Requires a reason; not exported |
| `EXPORTED` | Exported | Locked in a payroll CSV |

## Notifications and audit log

A notification is a message to one user about an event. It appears in the bell and under "Notifications", and can also be sent by email. The user chooses which types appear in the app and which are sent by email.

| Notification type | Typical recipient | Triggered by |
|---|---|---|
| `ASSIGNMENT` | Technician | The work order is assigned |
| `DATE_CHANGED` | Technician | The work order date changes |
| `CANCELLED` | Technician | The work order is cancelled |
| `REPORT_READY` | Technician, office and customer portal | The report is submitted and ready |
| `BOOKING` | Office and assigned technician | A new or confirmed booking |
| `CONTRACT_DUE` | Office | A service contract is due |
| `OVERDUE` | Technician and office | The work order is overdue |
| `CONTACT_UPDATE` | Office | The customer requests updated contact information |
| `SLA_AT_RISK` / `SLA_BREACHED` | Technician and office | The SLA deadline is approaching or has been breached |
| `WARRANTY` | Office | Equipment warranty is about to expire |
| `MESSAGE` | Office or customer | A new message in a conversation |
| `INVOICE_PAID` | Office and customer | The invoice has been paid |

The audit log is a permanent, non-editable history of actions: who created, changed, moved, cancelled, deleted, unlocked or applied a legal hold, with time and metadata. You find it on the work order, on the customer card and under "Settings" → "Activity log" for administrators.

# How features talk together

> **Note:** `WorkOrder`, `VisitTechnician`, `BookingRequest` and `MaintenancePlanOccurrence` are internal model names. In the interface they are called work order, technician assignment, booking and maintenance occurrence.

| Action | Data created or updated | Who sees it |
|---|---|---|
| Customer sends a date request | `BookingRequest(PENDING)`, `WorkOrder(OPEN)`, booking event, `BOOKING` | Office in "Bookings" |
| Office confirms a pending request | `BookingRequest(CONFIRMED)`, `WorkOrder(PLANNED)`, `Visit(PLANNED)`, optional SMS | Customer, office, technician |
| Customer moves a confirmed appointment | New time window, `Visit(PLANNED)`, `RESCHEDULED`; short `BookingHold` only in this flow | Customer, office, technician |
| Office creates a work order | `WorkOrder(PENDING)`, `Visit(PLANNED)` if dated, audit log | Office, calendar |
| Office assigns a technician | `VisitTechnician`, technician list, `ASSIGNMENT`, audit log | Technician, office |
| Technician starts driving | `Visit(EN_ROUTE)`, `enRouteAt`, SLA response, tracking | Office, customer |
| Technician marks "On site" | `Visit(ON_SITE)`, `onSiteAt` | Office, customer |
| Technician completes the visit | `Visit(DONE)`, `doneAt`, tracking revoked | Office, customer |
| Technician submits the report | `Report` locked, revision 0, PDF, units, equipment history, `REPORT_READY`, optional remediation work order | Office, customer, equipment history |
| Customer acknowledges | `customerAcknowledgedAt`, acknowledgement comment | Office, report |
| Service contract becomes due | `WorkOrder(PLANNED)`, `Visit`, previous report as PDF, `CONTRACT_DUE` | Office, calendar |
| A meter reading passes the threshold | `MaintenancePlanOccurrence`, new maintenance work order | Office, customer |
| A document is uploaded | `Document`, `Upload`, OCR job, search index | Staff with document access |
| Role or permissions change | `User.sessionVersion` increases, API keys inherit new permissions | User, administrator |
| Quote sent | `Quote(SENT)`, version, one-time token, `EmailMessage` | Customer, office |
| Customer accepts quote | `Quote(ACCEPTED)`, audit row | Office, customer |
| Quote converted | `WorkOrder`, customer link, equipment | Office, calendar |

| Action | Data created or updated | Who sees it |
|---|---|---|
| Invoice sent | `Invoice(SENT)`, number, frozen lines, `EmailMessage` | Customer, office |
| Payment received | `Payment`, `Invoice.paidTotal`, status, audit row | Office, customer |
| Payment link created | `Payment` metadata, external ID, audit row | Customer, office |
| Invoice synced | `AccountingSync`, external ID, status | Office, accounting |
| Item reserved | `InventoryReservation`, `StockMovement(RESERVE)` | Inventory, technician |
| Item consumed | `StockMovement(CONSUME)`, reduced stock | Inventory, technician |
| Purchase order received | `PurchaseOrder`, `PurchaseOrderLine`, `StockMovement(IN)` | Inventory, office |
| Expense approved | `Expense` / `MileageTrip`, approver, audit row | Office, technician |
| Payroll CSV exported | `PayrollExport`, locked lines, checksum | Office, payroll |

> **Tip:** When troubleshooting, start with the action and follow the row in the table. You will see which side is missing data, and which role should see it.

## Common cross-role problems

| Problem | Check first | Where to look |
|---|---|---|
| **The customer portal user cannot sign in or see data** | Is the user active and linked to a contact belonging to the right customer? Is the delegation active with the right sub-permissions? Has the invitation been completed and a password set? | "Customers" → customer card → **More** → **Portal users**; "Settings" → "Employees & Roles" |
| **The customer cannot see documents, contracts or equipment** | Does the contact have `canViewDocuments`, `canViewContracts` or `canViewEquipment`? Is the document owned by the customer and shared? Is the contract in the right department? | Customer card → **More** → **Portal users** and the portal's "Documents", "Contracts" and "Equipment" tabs |
| **The work order is missing from the calendar** | Does the work order have a date and a visit? Undated work orders have no visit. Is the visit cancelled or completed? Is the date outside the ±6-month window? Is a filter hiding it? | "Work orders" → visit section; "Calendar"; "Dispatch" |
| **The report is not visible** | Was the form submitted, or is it only a draft? Is the report locked and submitted? Do the date, customer and department filters match? Is it deleted? Has the PDF been generated? | "Work orders" → "Checklists"; "Reports"; customer portal → "Reports" |

| Problem | Check first | Where to look |
| --- | --- | --- |
| **The service contract is not generating a work order** | Is the contract `ACTIVE`, not `PAUSED` or `EXPIRED`? Is `Next service` due? Has the end date passed? Is the customer deleted? Has the work order already been generated? Is the scheduled job running? | "Contracts" → contract; customer card; operations log for recurring jobs |
| **The notification does not arrive** | Is the type enabled for the app or email? Is the user active and actually a recipient? Is the email in spam? Was the notification deduplicated? Is the email queue being processed? | "Notifications" → "Settings"; "Notifications"; operations log for email |
| **The offline draft is not syncing** | Are you on the same device, in the same browser and signed in as the same user? Has the app been reopened with a connection? Is the draft in the outbox? Is it older than seven days, too large, or rejected because the report is locked? | Field surface → offline banner and draft list; browser storage |
| **The technician cannot see the work order** | Is the work order in the technician's department? Is it assigned, or available? Is it completed or cancelled? Is the date correct? Does the technician have "Take work order" access? | "Today"; "Work orders"; "Calendar" |
| **An action is rejected with "no access"** | Does the user have the right role and permission? Is the user in the right department? Is the account active? Is the work order locked or under legal hold? | "Settings" → "Employees & Roles"; the work order's audit log |
| **Sites or units from the wrong customer appear** | Wrong customer selected, or the page was not reloaded after switching customer | Switch customer again, reload the page and check the saved selection |
| **A unit cannot be moved** | Active contract lines, number conflict, cross-customer reference or invalid target | Read the blocker in the preview, clear the contract line, or choose the right customer/site |
| **An upload is rejected** | File too large, user/customer quota reached, or proxy limit | Check file size, `UPLOAD_QUOTA_*`, disk and the proxy body limit |
| **Customer transfer does not start** | 10 starts per hour used up, an existing job is reused, or the preview expired | Wait, follow the existing job, or upload and preview the package again |
| **Quote or invoice page returns 404** | The `quotes` / `invoicing` module is off | Enable the module under **Settings** → **Modules** |
| **Payment button does not appear** | Provider or webhook secret missing, or the amount is paid | Check `PAYMENTS_PROVIDER` and keys, and create a new link |
| **Payment is not recorded** | Webhook rejected, or currency/amount mismatch | Check signature and audit log; reconcile manually |
| **Accounting sync fails** | Wrong key, account or VAT setup | Read `lastError`, fix the cause and sync again |

| Problem | Check first | Where to look |
|---|---|---|
| **Stock does not match** | Reservation not consumed, or adjustment missing | Check movements and reservations on the item |
| **Expenses cannot be exported** | Period overlaps, or the row is not approved | Choose a new period and approve the row |
| **Email stays queued** | SMTP missing, or the event is off | Check `SMTP_*` and **Automatic email** |
| **Push does not arrive** | Device unregistered, notifications denied, or provider not configured | Register the device again and check FCM/APNs |

## Customer portal: access is more than signing in

A portal user is always linked to a contact at a customer. The contact can have a delegation with separate choices for booking, documents, contracts and equipment. An inactive delegation closes the portal completely for that contact. A contact without a delegation row has full portal access for backward compatibility.

Check therefore that the user is active and belongs to the right customer, that the contact belongs to the same customer, that the delegation is active, and that the sub-permission for the missing area is enabled.

> **Note:** In 0.8.11 there is no dedicated form in the interface for editing the delegation. Restricted portal access must be configured and tested with the actual account.

## Calendar: date, visit and window

The calendar shows visits, not work orders themselves. A work order without a date has no visit and cannot appear. The calendar loads a limited window around the period you are viewing, and the list may be truncated. If a work order is missing, it is almost always because the work order is undated, the visit is deleted, cancelled or completed, the date is outside the window, a filter hides it, or it belongs to another department.

## Reports: draft, submitted and published

A report does not become visible to the customer just because the technician has started filling it in. The sequence is: draft → submit → the server locks and generates a PDF → the report becomes available under "Reports" and in the customer portal → the customer can acknowledge it. If the report is missing, find out where in this chain it stopped. A draft that only exists locally is not a report.

## Service contracts: only active and due contracts generate work orders

A service contract does not generate work orders just because it exists. It must be `ACTIVE`, `Next service` must be due, and the contract must not have passed its end date. The system also does not create a new work order for the same contract and date twice. If the contract is `PAUSED`, `EXPIRED` or `TERMINATED`, the office must activate or renew it first.

### Notifications: app, email and recipients

Notifications are both in-app and email. A missing notification in the inbox does not necessarily mean it was not created. Check whether the user has disabled the type, whether the user is active and actually a recipient, whether the notification is under "Notifications" without an email, whether the email is delayed or in spam, and whether the same event was already notified with a deduplication key.

### Offline: a draft is not the same as sent

The field surface saves drafts locally when the connection disappears. The draft is sent when the app is open and the connection returns. It cannot be sent if the report is locked or under legal hold, if the draft is older than seven days or larger than the limit, if the device has lost its storage, or if a different user or browser opens the app. Submission requires a connection; signature is optional.

## Norwegian/English glossary

| Norwegian | English | Meaning |
|---|---|---|
| Avdeling | Department | Organisational unit that scopes access |
| Oppdrag | Work order | The service job that is planned and followed up |
| Visitt | Visit | Time block within a work order, with technicians |
| Montør | Technician | The person carrying out field work |
| Kontor | Office | The planning and administration role |
| Kunde | Customer | The company receiving service |
| Kontaktperson | Contact | The person at the customer |
| Sted | Site | Physical address or facility |
| Utstyr / enhet | Equipment / unit | The machine or system being serviced |
| Rapportenhet | Report unit | The unit as it appears in a report |
| Sjekkliste | Checklist | The points being checked |
| Mal | Template | Reusable checklist or HMS form |
| Rapport | Report | Completed and submitted checklist |
| Avvik | Deviation | A finding that needs follow-up |
| Godkjent | Approved | The unit has been checked and is in order |
| Revisjon | Revision | Immutable snapshot before a change |
| Kvittering | Acknowledgement | Customer confirmation of receipt |
| Låst | Locked | Cannot be changed without an administrator |

| Norwegian | English | Meaning |
|---|---|---|
| Serviceavtale | Service contract | Agreement for recurring service |
| Fornyelse | Renewal | Automatic or manual extension |
| SLA | Service level agreement | Agreed response and resolution time |
| Åpningstider | Business hours | The hours within which SLA deadlines are calculated |
| Bestilling | Booking | The customer's request or confirmed appointment |
| Ledig time | Available slot | Suggested time in the slot picker; used only when moving a confirmed appointment, not for new bookings |
| Reservasjon | Hold | Short-term reservation of an available slot in the legacy reschedule flow |
| Varsel | Notification | Message about an event |
| Aktivitetslogg | Audit log | Traceable history of actions |
| Rettighet | Permission | What a role or user can do |
| Bevaringsplikt | Legal hold | Freezes data against deletion and change |
| Dokumentbibliotek | Document library | Shared file area with folders and tags |
| HMS | HMS | Health, safety and environment |
| Måler | Meter | Equipment reading, for example hours |
| Vedlikeholdsplan | Maintenance plan | Calendar- or meter-based maintenance |
| Sporing | Tracking | Temporary sharing of status and arrival |
| Omtale | Review | Internal customer feedback after a visit |
| Kundeportal | Customer portal | The customer's own surface |
| Kladd | Draft | Locally saved, unsubmitted form |
| Kundeadskillelse | Customer scoping | Site and unit choices are limited to the selected customer |
| Enhetsflytting | Equipment move | Moving a unit with its service history to a new customer |
| Kundeoverføring | Customer transfer | The administrator's export/import of a customer package |
| Kontorinnboks | Office inbox | Collected SMS conversations for the office |
| Tilbud | Quote | Price proposal with lines, versions and approval |
| Faktura | Invoice | Invoiced amount with lines, payment and credit note |
| Kreditnota | Credit note | Reduction of outstanding amount, never above it |

| Norwegian | English | Meaning |
|---|---|---|
| Betalingslenke | Payment link | One-time link to Stripe Checkout or Vipps |
| Regnskapssynk | Accounting sync | Manual transfer of customer, invoice and payment |
| Lager | Inventory | Items, stock and locations |
| Innkjøpsordre | Purchase order | Supplier order with goods receipt |
| Varemottak | Goods receipt | Registration of received goods against an order |
| Reservasjon | Reservation | Part held for a work order |
| Forbruk | Consumption | Physical deduction of a reserved item |
| Kjøretur | Mileage trip | Mileage allowance with kilometres and rate |
| Utlegg | Expense | Expense with receipt and VAT |
| Lønns-CSV | Payroll CSV | Locked export of approved time, mileage and expenses |
| Lønnsomhet | Profitability | Estimated and actual revenue by department, customer and technician |
| Utboks | Outbox | Durable queue for email and push |
| Pushvarsel | Push notification | Notification to mobile via FCM or APNs |
| SAML | SAML | Federated sign-in with signed assertion |
| SCIM-gruppe | SCIM group | Group granting role/department through explicit mapping |
| Feltmaskering | Field redaction | Server-side hiding of sensitive fields |
| S3 | S3 | Object storage for uploads |
| Redis | Redis | Wake-up channel and cache, with database fallback |
| PITR | PITR | Restore to a chosen point in time |
| RPO/RTO | RPO/RTO | Maximum data loss and maximum downtime |

## Support information to collect

When you report a problem, it helps to collect the information below. Write briefly what you did, what you expected, and what actually happened.

- **Version:** FSM Platform 0.8.11, and where you saw the problem (office surface, field surface, customer portal, PDF or API).
- **Time:** Date and time with time zone, preferably Europe/Oslo.
- **User:** Email, role, department and any customer. Never provide passwords or one-time codes.
- **Case:** Work order number or ID, title, customer and date. For a report: report number or ID. For a booking: reference. For a contract: contract number.

- **Environment:** Browser and version, operating system, phone or PC, and whether the field surface is installed as an app.
- **Network:** Were you online or offline? Did it happen in a draft, during submission or afterwards?
- **Error message:** Reproduce the text exactly, and take a screenshot showing the time, user and error message.
- **Reproduction:** Write down the steps you took, and whether the problem happens every time or only sometimes.
- **Notification:** For notification problems, provide the type, recipient, app or email, and whether the notification exists under "Notifications".
- **API:** Provide the key prefix (never the full key), endpoint, time, HTTP status, `error.code` and any `requestId`.
- **PDF and email:** State whether it concerns generation, display, download or sending, and whether it concerns a report or a contract.
- **Offline:** State whether the draft is in the outbox, whether it is older than seven days, and whether the same user and device were used.
- **Payments and accounting:** Provide invoice number, provider, amount, currency and whether the failure occurred at the link, webhook or sync. Never full keys.
- **Email and push:** Provide event type, recipient, whether it concerns app or email, and whether the row is `PENDING`, `FAILED` or `DEAD_LETTER`.

> **Warning:** Never send passwords, one-time codes, full API keys or complete personal data in a support case. Mask screenshots where possible.

# About this manual

This manual describes FSM Platform 0.8.11 as of 2026-11-08. It is divided into an introduction, one chapter for each role and a reference chapter.

| | |
|---|---|
| **Document** | FSM User Manual |
| **Version** | 0.8.11 |
| **Date** | 2026-11-08 |
| **Audience** | Office · Technician · Administrator |
| **Source** | `docs/manual/en/` (Markdown) |

The source is split by role so a single chapter can be updated without rewriting the whole document. Build the PDF with `node scripts/build-manual.mjs`. The generated PDF is a snapshot of the product configuration described in this edition; local settings, permissions and integrations may differ.